OFFICIAL DOCUMENTATION · 2026 EDITION



Tchoua.com

Solidarity, digitized.
Complete functional manual for the platform.

21 chapters | 39 real screenshots | 10 diagrams | Tontines · Loans · Solidarity

OFFICIAL DOCUMENTATION

Document record

This manual describes how the Tchoua platform actually works: its modules, workflows, calculation rules and administration. It is intended for associations, boards, treasurers, administrators and partners.

Purpose	Complete functional documentation for tchoua.com
Version	2.0 — November 2026 — anonymized edition
Scope	Public site, member area, association console, system console, tontines, loans, solidarity, meetings, reports, administration
Audience	Association boards, members, treasurers, auditors, administrators, partners
Language	English (the platform is multilingual: FR, EN, ES, DE and local languages)
Sources	Production application, ARP reference of 13 November 2026, FEV / SLD rules, ETS HORIZON data, repository specifications
Status	Reference document — every product change must be reflected here

> **NOTICE**
>
> Amounts, rates and caps are those actually configured for the anonymized associations at the time of writing. They remain fully configurable: no rule is hard-coded. Association and people names have been replaced by fictional identifiers; mechanisms, figures and screenshots remain concrete without identifying anyone.

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NAVIGATION

Each chapter starts on a new page. Page numbers are computed from the final layout.

CHAPTER 1

Vision, positioning and promise

Tchoua is not just a tontine calculator. It is a trust infrastructure for every form of tontine-type association: friendly societies, sections, federations, cooperatives, mutuals, women's groups, hometown associations.

1.1 The problem

In Cameroon and across much of Africa, the tontine is a savings and credit mechanism built on a spoken word, a notebook, a treasurer and collective memory. This model works as long as the group is small and members know one another. It weakens as soon as the association grows: records get lost, contributions are not reconciled, loans are not tracked, aid is granted without an objective basis, and distant sections do not know what is being decided at national level.

Existing tools are either spreadsheets or tontine apps limited to the rotating turn. None covers the reality of a friendly society such as the FEV: bylaws, a steering committee, sections, a national fund, illness and bereavement aid, distinctions, sanctions, quorums, terms of office and documents to be preserved.

1.2 The promise

Tchoua brings three inseparable things:

- **Transparency** — every member sees their contributions, loans, aid, interest shares and balance; the board sees the whole picture; every movement is time-stamped and attributed.
- **Configurability** — no rule is hard-coded: amounts, frequencies, quorums, caps, sanctions, cycles and allocation methods are configured per association and per activity.
- **Complete coverage** — from creating the association to closing the cycle, including membership, meetings, tontines, loans, solidarity, reports and administration.

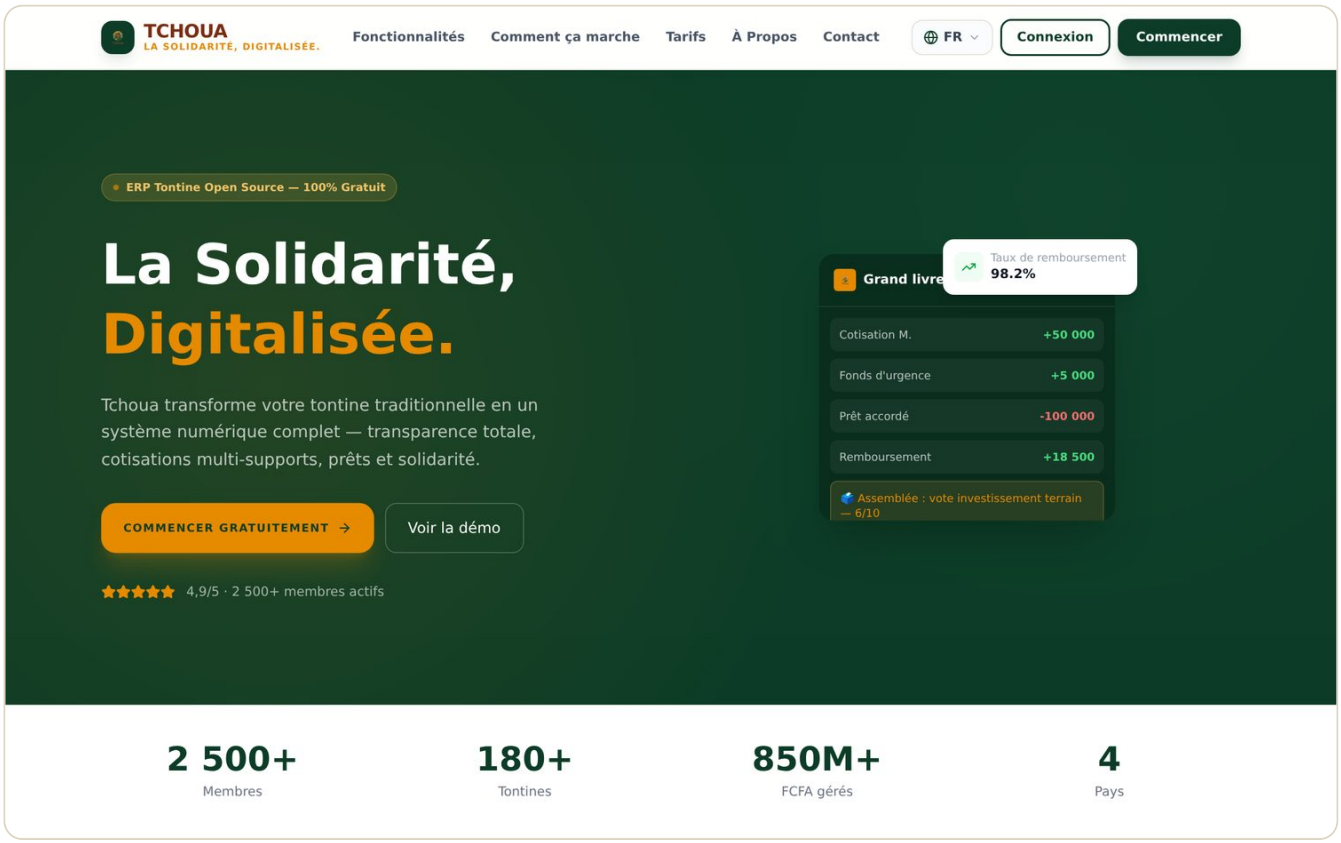


Figure 1.1 — Public home page: promise, social proof and access to account creation. The navigation is deliberately short: features, method, pricing, about, contact.

1.3 The three product pillars

Pillar	What it provides	Where to see it
Association	Identity, bylaws, internal rules, board, sections, documents, lifecycle, governance	Association console, FEV and SLD pages
Financial activities	Tontines, savings, investment, loans, catering, group purchases, in-kind contributions	Activities, sessions, loans, history
Social protection	Solidarity fund, illness, bereavement, marriage, birth, education and emergency aid	Solidarity, social aid, reports

1.4 The personas

Persona	Needs	Tchoua answer
Member	Know what they owe, what they have received, what remains; take part in decisions	Member area, launchpad, personal history, notifications, voting
Treasurer	Collect, reconcile, track cash boxes, produce reports	Sessions, contributions, loans, cash boxes, PDF/CSV exports
President / board	Steer, arbitrate, convene, control, represent	Association console, meetings, governance, reports, audit
Auditor	Verify movements, supporting documents, balances	Audit log, reports, history, exports
System administrator	Create templates, manage roles, configure the platform	System console, templates, users, settings
Local section	Apply national bylaws while managing its own funds	Parent/child hierarchy, inherited and overridden configurations

1.5 What sets Tchoua apart

Business depth. A tontine is not just a rotating turn: it is a contract with shares, fractions, sessions, a pot, auctions, beneficiaries, a balance and a closing. Tchoua implements these objects and their rules, including edge cases (insufficient pot, exhausted shares, defaulting member, multiple beneficiaries).

Governance. Bylaws and internal rules are not decorative: they become parameters (quorums, majorities, terms of office, sanctions) and enforceable documents, with version history and approvals.

Federation. A parent association can group sections together; each section keeps its own funds and local rules. This is the FEV / SLD model, transferable to all hometown associations.

> **POSITIONING**
>
> The reference platform for managing every form of tontine-type association in Cameroon and Africa: web, mobile, multilingual, multi-currency, configurable without code.

CHAPTER 2

Architecture, data and security

Tchoua is a modern web application, deployed in production and available at tchoua.com and www.tchoua.com. It is designed to serve several associations, several hierarchical levels and several currencies without duplicating code.

2.1 Overview

The application is built on Next.js (App Router), React, TypeScript, Prisma and PostgreSQL. Public pages and the member area share the same foundation; the APIs systematically verify association membership and role before any read or write. Files (bylaws, minutes, supporting documents) are stored on the server and referenced in the database.

Next.js
server rendering and API routes

Prisma
typed access to PostgreSQL

JWT
signed sessions, secure cookie

PM2
supervised process, automatic restart

2.2 Technical components

Layer	Component	Role
Interface	React + Tailwind	Public pages, member area, association console, system console, reusable components
Application	Next.js App Router	Server rendering, REST API routes, authentication middleware, exports
Authentication	NextAuth (JWT)	Credential login, signed session, role checks
Domain	TypeScript modules	Frequency engine, tontine engine, loan rules, interest allocation, solidarity, internal rules
Data	PostgreSQL + Prisma	Relational model, constraints, transactions, migrations
Files	Local storage /uploads	Bylaws, internal rules, minutes, supporting documents, exports
Operations	PM2 + HAProxy + Certbot	Processes, reverse proxy, TLS, certificate renewal

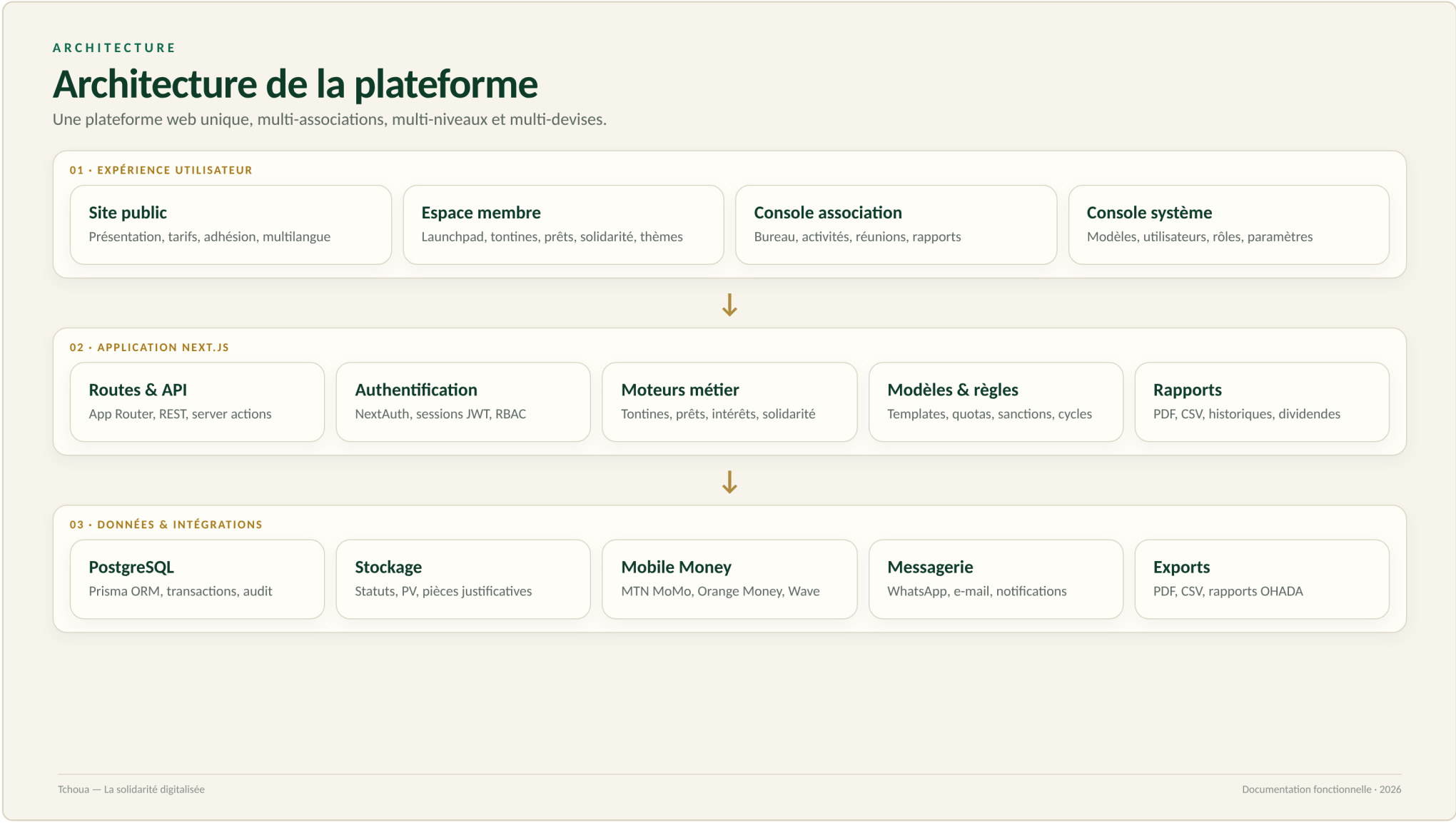


Figure 2.1 — Three-layer architecture: user experience, Next.js application, data and integrations.

2.3 Data model

The model is deliberately centred on the association. A user can belong to several associations with different roles. An association owns activities; each activity owns sessions, contributions, beneficiaries, loans and repayments. Documents and rule articles are attached to the association.

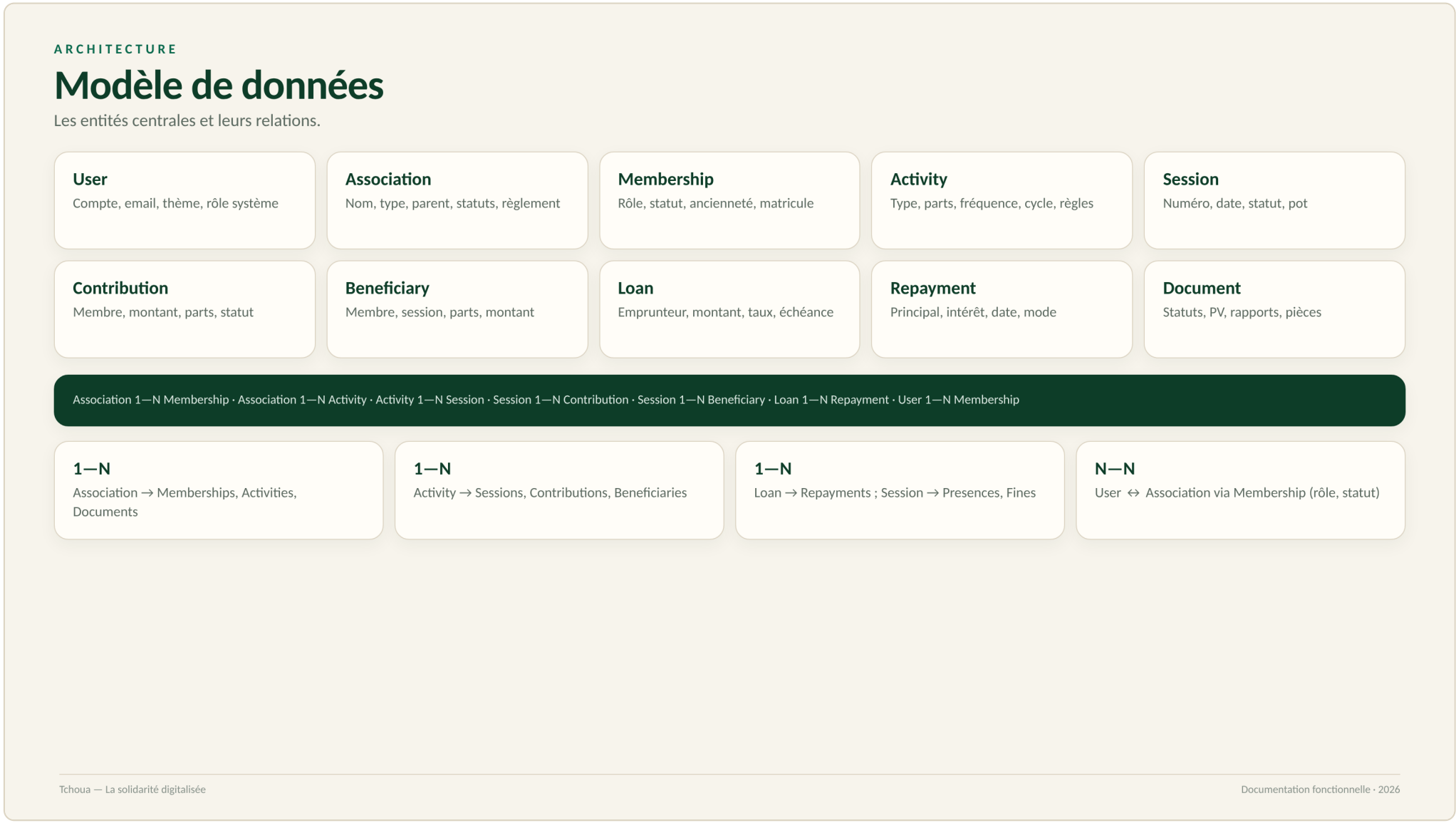


Figure 2.2 — Core entities and relationships. Full cardinalities are described in the repository's Prisma schema.

2.4 Security

- **Authentication** — hashed passwords (bcrypt), signed JWT sessions, `__Secure-next-auth.session-token` cookie, expiry and renewal.
- **Authorization** — two-level control: association membership and role (member, board, system administrator). Sensitive routes use a shared guard.
- **Least privilege** — a regular member sees only their own operations; the board sees the whole picture; the system administrator does not stand in for the board on business decisions.
- **Audit** — administrative actions and status changes are logged with author, date and reason. Financial movements are never deleted: corrections create reversing entries.
- **Transport** — HTTPS mandatory, HTTP → HTTPS redirect, Let's Encrypt certificates renewed automatically.
- **Files** — controlled types and sizes (PDF, JPEG, PNG, WebP; 10 MB), access by role.

> **WARNING**
>
> Credentials, session secrets, API keys and passwords must never appear in the interface, screenshots or documents. Demo accounts are disabled in production.

2.5 Multi-association and hierarchy

Each association is a sealed space: its members, activities, cash boxes, documents and reports. An association can have a parent (federation, national friendly society) and children (sections). Roles are evaluated at the level of the association concerned; a member can be a regular member in one section and a national board member.

This architecture can serve both a neighbourhood tontine of ten members and a national federation of several thousand members, without changing code or database.

2.6 Operations and continuity

Production is supervised: PM2 processes, HAProxy reverse proxy, TLS certificates, database backups and logging. Deployments are automated by a script that aligns the repository, regenerates the Prisma client, builds the application and restarts the process. SQL migrations are idempotent and applied before the restart.

Risk	Measure
Process unavailability	PM2 restarts automatically; health check on key pages
Data loss	PostgreSQL backups, regular exports, audit log
Access leak	Session control, roles, secure cookies, HTTPS
Deployment error	Verified build, controlled restart, rollback by commit
Malicious file	Allowed types, size limit, storage outside the codebase

CHAPTER 3

Accounts, roles and permissions

Trust rests on a simple rule: everyone sees what concerns them, the board sees what it administers, the system administrator sees the platform — but never makes business decisions in place of the board.

3.1 Two families of roles

Tchoua distinguishes **association roles** (president, treasurer, secretary, member, etc.) from **system roles** (chief administrator, operator, auditor). A user can hold both an association role and a system role, but the two scopes remain separate.

3.2 Association roles

Role	Scope	Typical actions
Founder	Association creation	Initial setup, appointment of the board, publication
President	Whole association	Convene, arbitrate, approve, represent, close
Vice-president	Whole association	Stand-in duties, delegations, oversight
General secretary	Memberships, meetings, documents	Minutes, convening notices, files, archives
Assistant secretary	Memberships, contributions, fines	Data entry, follow-up, payment receipts
Treasurer	Cash boxes, banks, reports	Receipts, disbursements, reconciliations
Assistant treasurer	Treasury stand-in	Oversight, data entry, replacement
Auditor	Oversight	Audit of movements, quarterly and annual reports
Censor	Discipline	Applying sanctions, keeping the register
Adviser	Advisory	Opinions, committees, mediation
Member	Own data	Consult, contribute, request, vote

Table 3.1 — Roles available within an association. Labels and powers are configurable; custom roles can be created and combined with a base role.

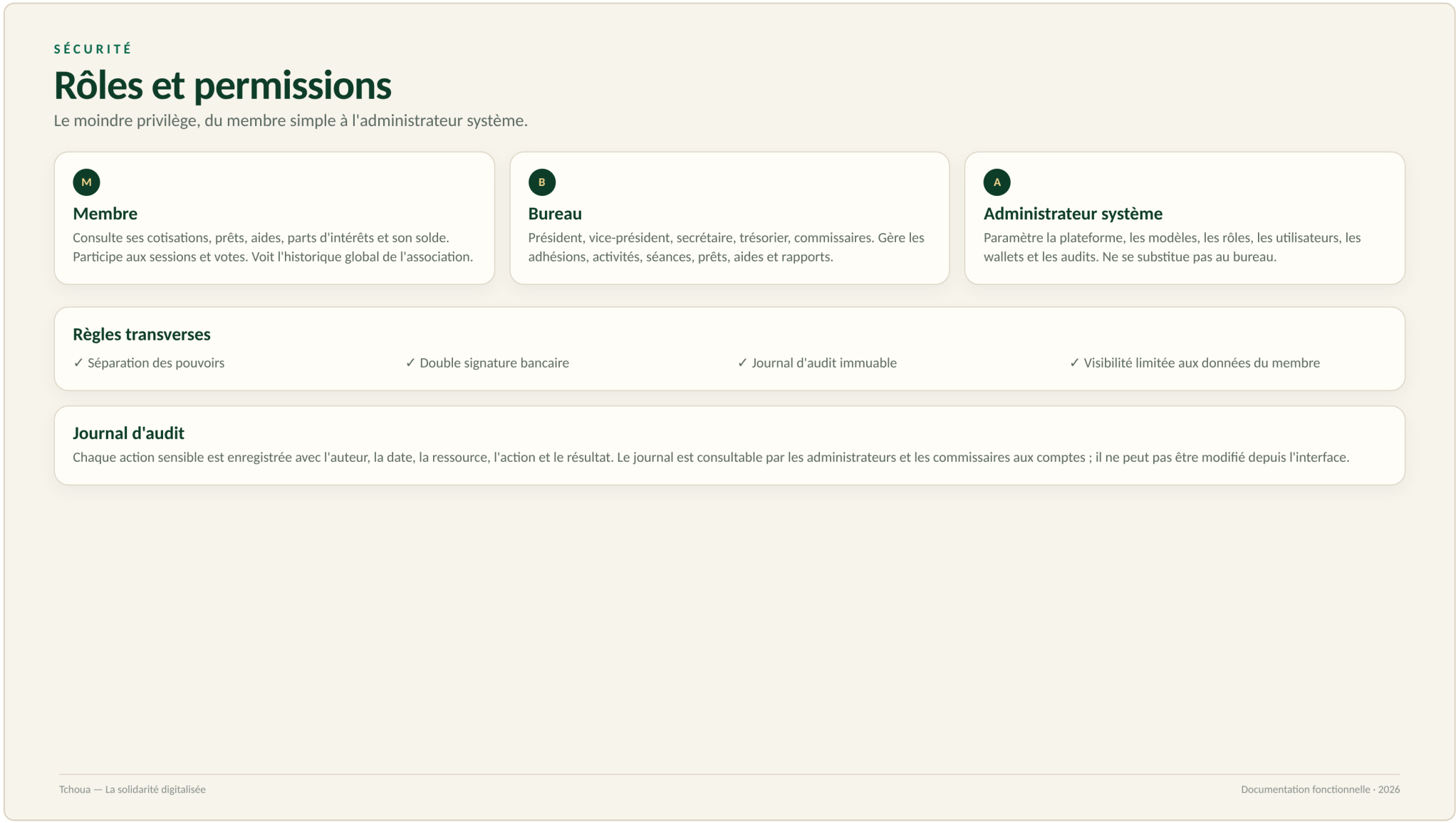


Figure 3.1 — Three levels of privilege and four cross-cutting security rules.

3.3 Visibility matrix

Data	Member	Board	System admin
Profile and contact details	Their own	All	System account
Contributions	Their own	All	Depending on role
Loans and repayments	Their own	All	Depending on role
Interest shares	Their own	All	Depending on role
Social aid	Their own	All	Depending on role
Overall history	View	Full	Full
Association settings	Read	Edit	Supervision
System roles and templates	No	No	Yes

3.4 Account lifecycle

1. **Registration** — name, e-mail, password, acceptance of terms; or invitation by the board.
2. **Verification** — the account is active after e-mail validation or activation by an administrator.
3. **Membership** — the member joins an association via a code, an invitation or a request approved by the board.
4. **Activation** — the board assigns the role and member number; the join date is recorded.
5. **Account life** — contributions, loans, aid, votes, documents, notifications.
6. **Exit** — resignation, removal or death; the status changes to `LEFT`, entitlements are calculated and the history is retained.

> **INFORMATION**
>
> A member can belong to several associations with different roles. Their dashboard aggregates their commitments, but each association keeps its own accounting and rules.

3.5 Profile settings

Each user manages their profile, password, language, member-area theme and notifications. Appearance preferences are stored server-side and applied as soon as the page loads, without flicker.

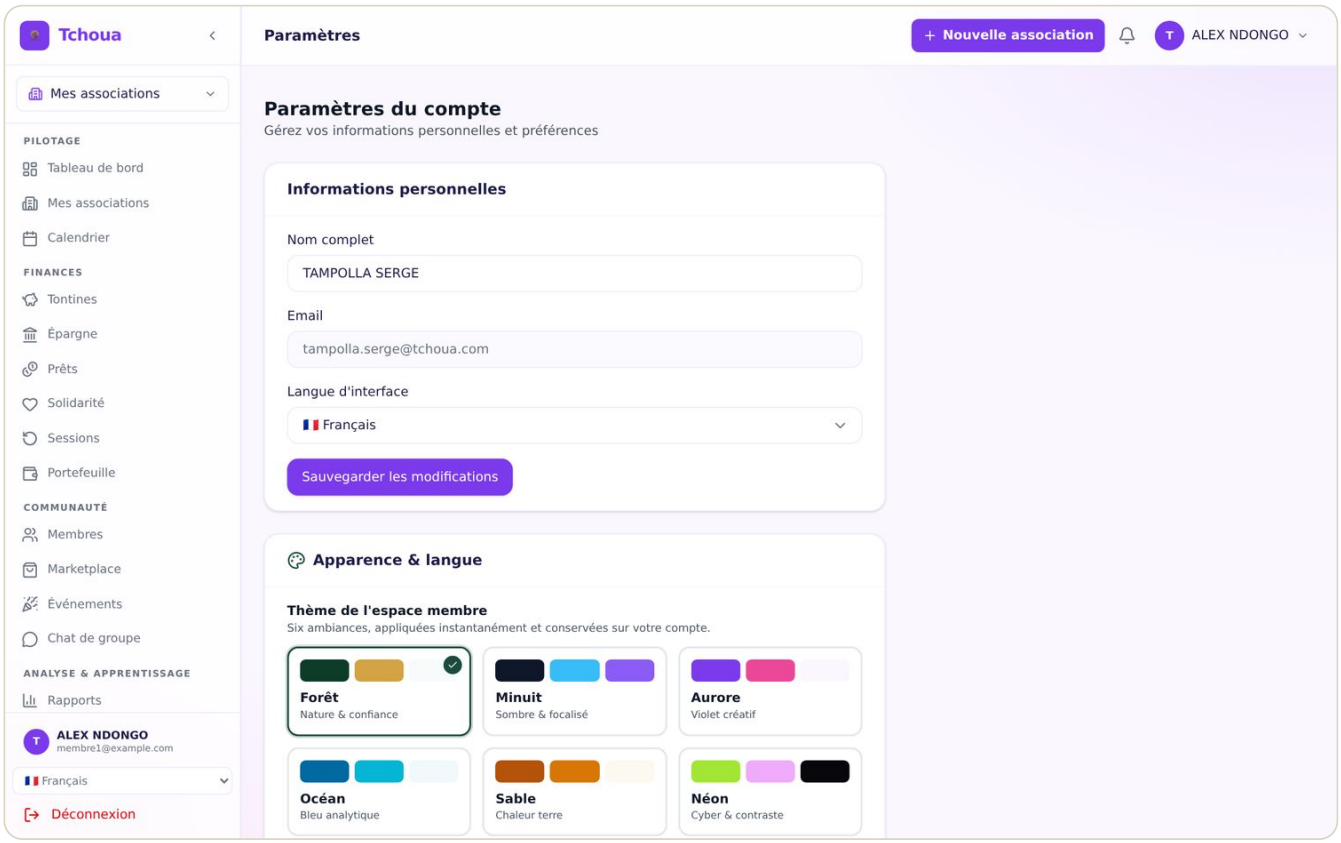


Figure 3.2 — Account settings: personal information, language and theme selector (Forest, Midnight, Dawn, Ocean, Sand, Neon).

3.6 Good practice

- ✓ One account per person, never a shared account.
- ✓ The board never shares its password; rights are granted by role.
- ✓ Sensitive functions (membership approval, disbursement, closing) are traced with the author.
- ✓ Inactive or departing accounts are disabled, not deleted, to preserve the history.
- ✓ System administrators are few, and their actions are audited.

CHAPTER 4

Association lifecycle

Creating an association is not just entering a name. You have to define its identity, its affiliation, its board, its membership conditions, its meetings, its aid, its sanctions and its internal rules — then bring it to life.

4.1 The six steps

1. Creation

The creator enters the name, the type (friendly society, section, federation, tontine, cooperative, mutual…), the region, the colour, the logo, the description, the possible parent association and the starting template. A unique invitation code is generated.

2. Configuration

The board configures how the association operates: bodies, quorums, majorities, terms of office, membership conditions, joining fee, annual contribution, meeting frequency, aid catalogue, sanctions, bank accounts. These settings are stored as JSON and remain editable.

3. Membership

Candidates complete an application, provide their information and are sponsored where the rule requires it. The board approves or rejects; activation records the join date, the member number and the role.

4. Activities

The association creates its activities: rotating tontine, ASCA tontine, auction tontine, savings, investment, loan fund, solidarity fund, catering, group purchases, in-kind contributions.

5. Sessions

Each activity schedules its sessions according to its frequency. Members contribute, attendance is recorded, fines are applied, beneficiaries are designated, loans are disbursed and repaid.

6. Closing

At the end of the cycle, the association produces its statement, calculates dividends, archives the documents and switches to read-only. A new cycle can be created without erasing the history.

4.2 Statuses and transitions

Status	Meaning	Allowed transitions
DRAFT	Configuration in progress, not visible	ACTIVE, DISSOLVED
PENDING	Awaiting approval	ACTIVE, DRAFT
ACTIVE	Normal operation	SUSPENDED, DISSOLVED
SUSPENDED	Activities frozen, consultation possible	ACTIVE, DISSOLVED
DISSOLVED	End of life, read-only	—

Table 4.1 — Each transition is logged with author, date and reason (lifecycle event).

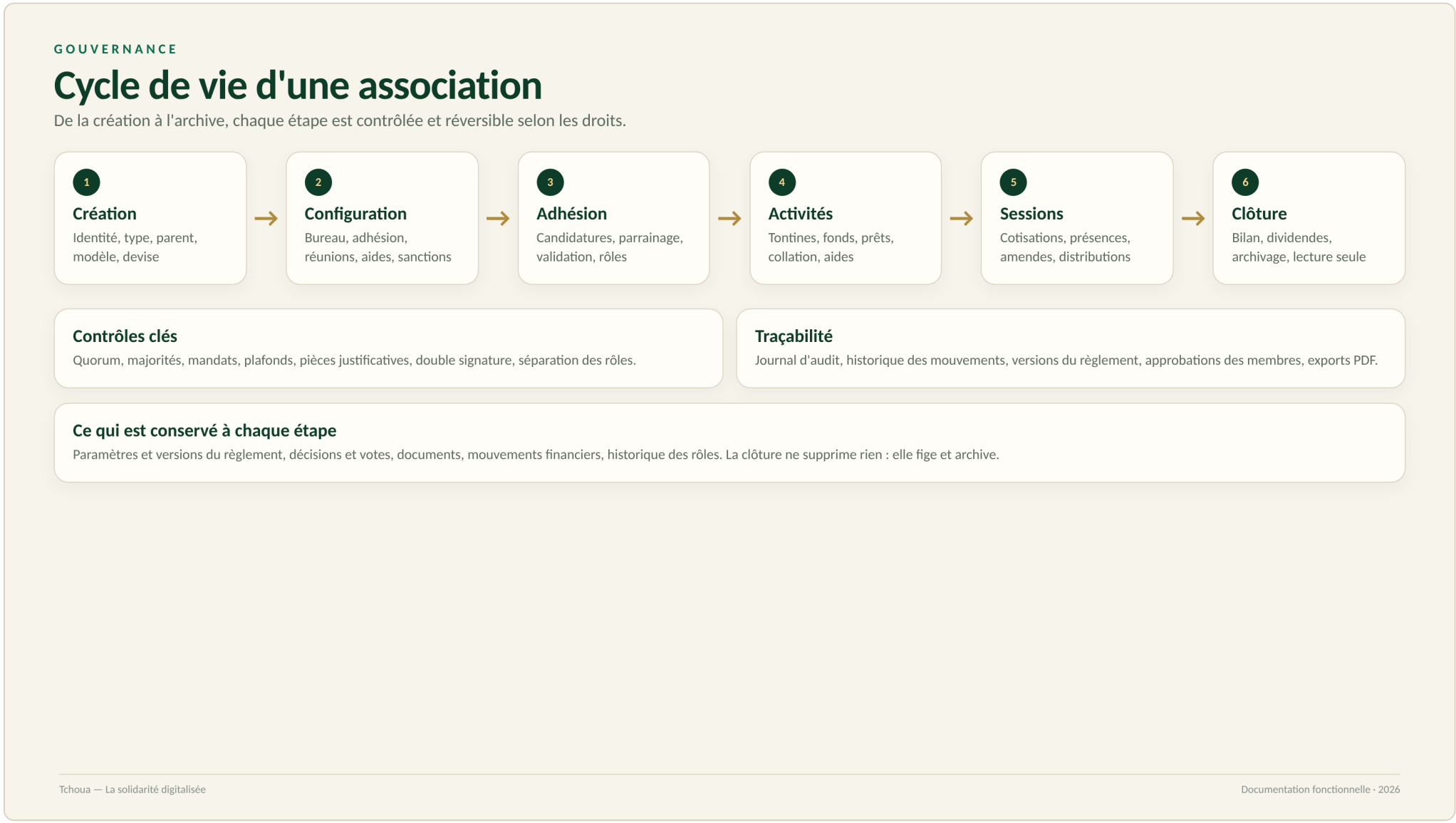


Figure 4.1 — From creation to archiving: six steps, with checks at every transition.

4.3 Two-level governance

The FEV / SLD model shows the power of hierarchy: the FEV is a national friendly society with a general assembly, a steering committee, an executive board, committees and sections. The SLD is the Douala section, with its own committee, its own rules and its own funds.

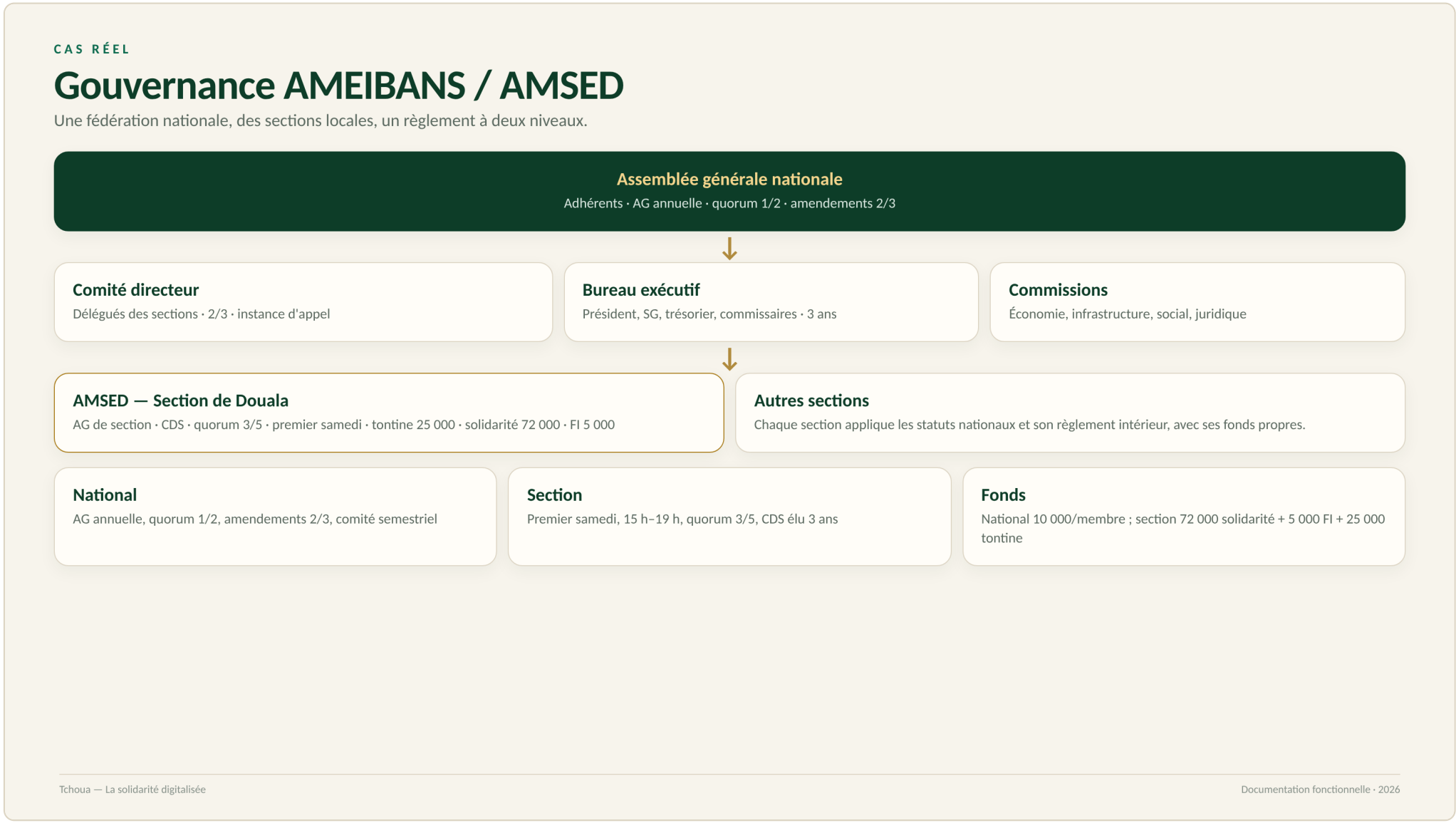


Figure 4.2 — Actual FEV / SLD organisation: national GA, steering committee, board, committees, Douala section.

Each level has its own rules: the national GA meets once a year with a quorum of half the members; the steering committee decides by a 2/3 majority; the Douala section meets on the first Saturday after the 5th, with a quorum of 3/5. Tchoua stores these parameters and surfaces them in the governance screens.

4.4 Bylaws, internal rules and documents

The bylaws and the internal rules are structured into numbered articles that members can consult and approve. The original documents (PDF, DOCX) can be downloaded from the association record. Versions are retained and changes are traced.

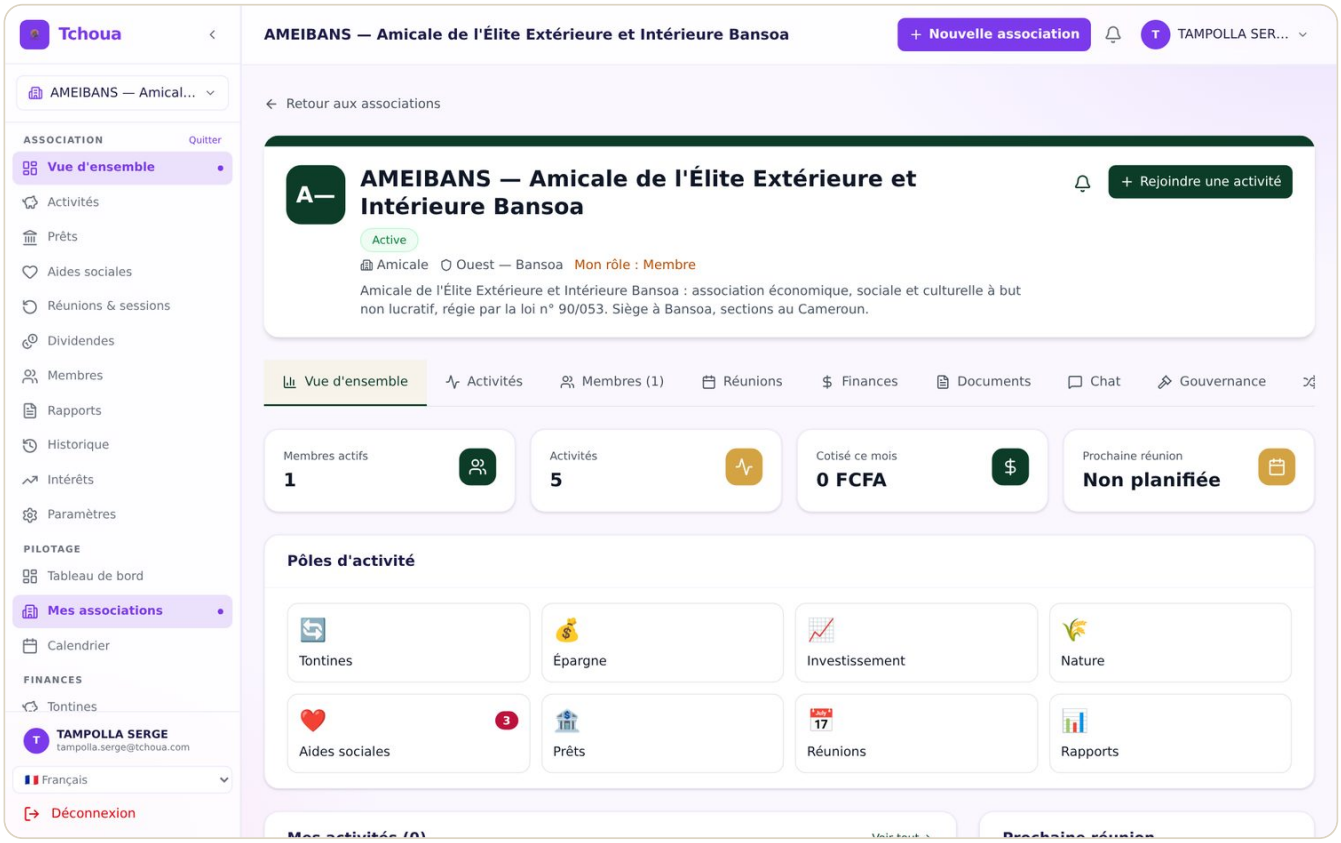


Figure 4.3 — FEV record: 1 member, 5 activities, 14 rule articles, 3 documents, national affiliation.

4.5 Special cases

- **Section without legal personality** — it inherits the parent's bylaws and manages its own funds. Tchoua models it as a child association.
- **Federation** — it groups autonomous associations; consolidation of headcounts and funds is described as a future development.
- **Dormant association** — SUSPENDED status, consultation possible, entries frozen.
- **Dissolution** — remaining assets are allocated according to the bylaws; the history remains consultable.

> **GOLDEN RULE**
>
> An association with movements is never deleted: it is closed. The history is the association's memory and the proof of its transparency.

CHAPTER 5

Module overview

Eight domains cover the entire lifecycle of an association. They share the same data model and the same logging, which avoids duplicate data entry and inconsistencies. Since November 2026, a versioned **ARP framework** governs these modules and conditions new subscriptions; it is described in chapters 19 to 21.

5.1 Module descriptions

Module	Functions	Main screens
Associations & sections	Creation, identity, hierarchy, bylaws, internal rules, documents, lifecycle	My associations, association record, settings, governance
Membership & members	Applications, sponsorship, approval, roles, member numbers, removals, directory	Members, applications, member record
Tontines & activities	Activity types, shares, fractions, cycles, auctions, beneficiaries, balances	Activities, cycle, sessions, tontines
Contributions & sessions	Calls, collections, statuses, arrears, attendance, fines, catering	Sessions, contributions, meetings
Loans & interest	Applications, guarantees, approval, disbursement, repayments, allocation	Loans, loan record, interest
Solidarity & aid	Catalogue, caps, applications, approval, disbursement, follow-up	Solidarity, social aid
Accounting & reports	Cash boxes, journals, statements, PDF/CSV exports, dividends, audit	Reports, dividends, history
Administration & templates	Templates, system users, roles, settings, maintenance	System console

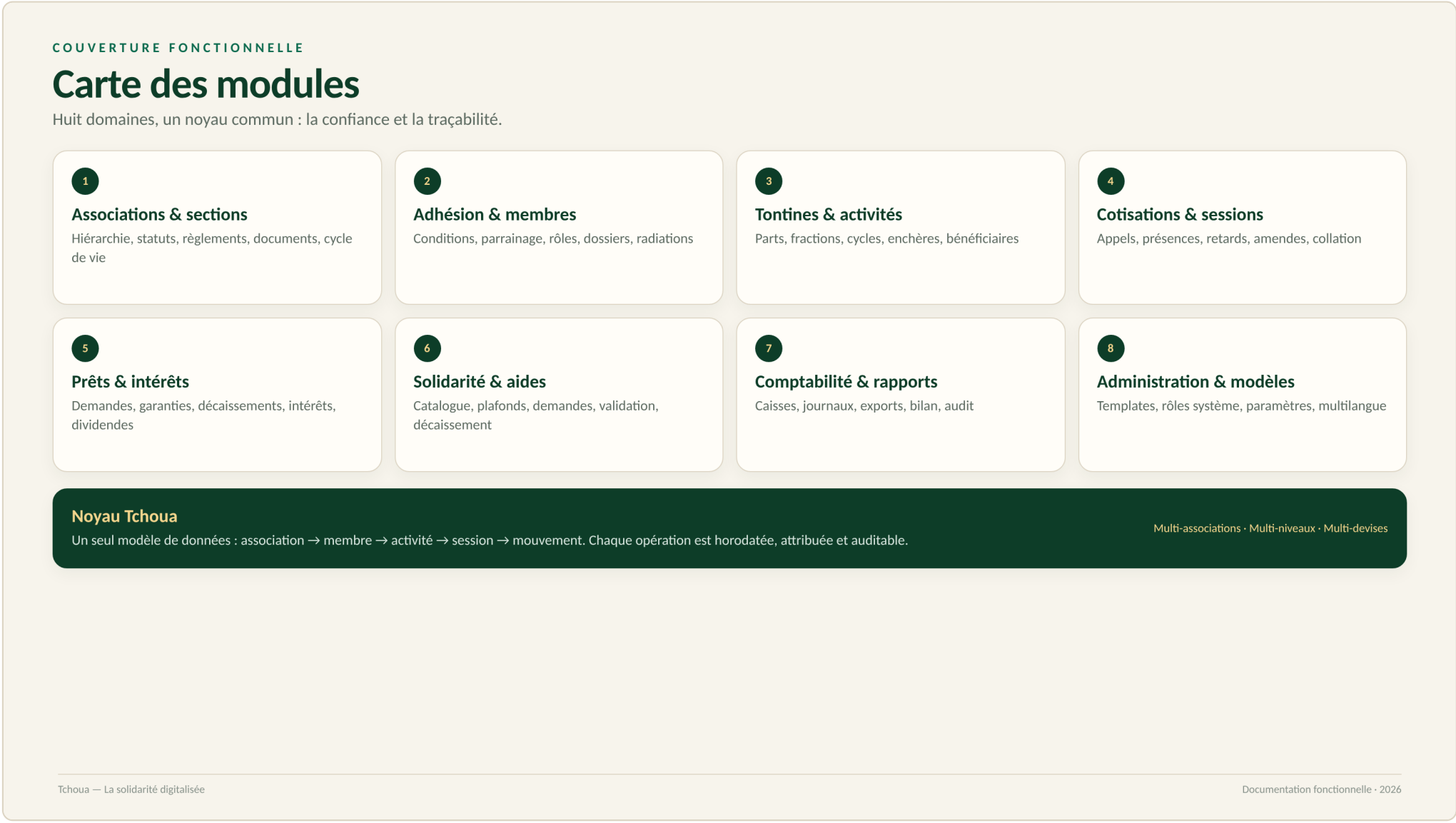


Figure 5.1 — The eight modules and the shared core: association → member → activity → session → movement.

5.2 Member area navigation

The side navigation is grouped by intent: **Steering** (dashboard, associations, calendar), **Finances** (tontines, savings, loans, solidarity, sessions, wallet), **Community** (members, marketplace, events, chat), **Analysis & learning** (reports, AI advice, gamification, academy). The association's contextual section adds overview, activities, loans, aid, meetings, dividends, members, reports, history, interest and settings.

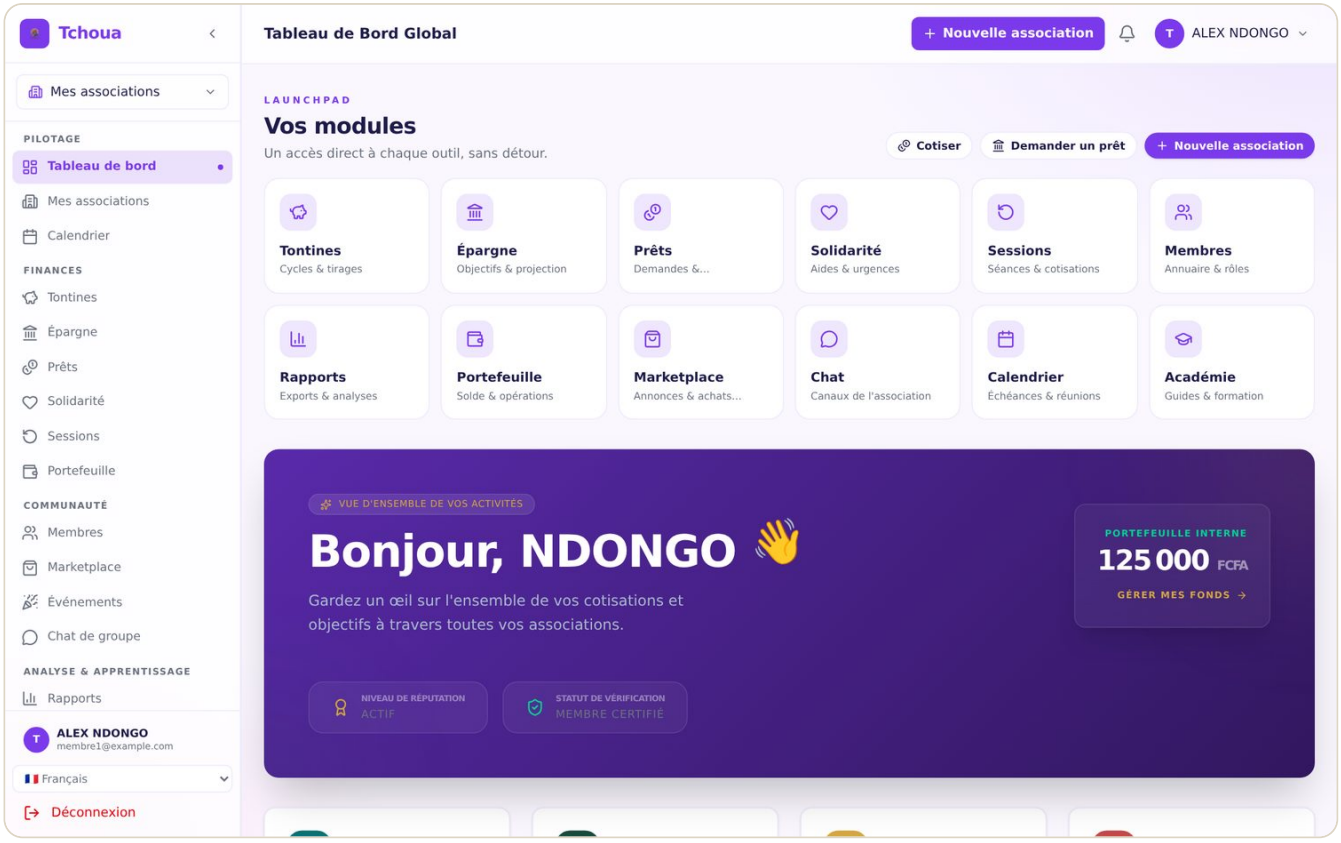


Figure 5.2 — Member dashboard: personalised hero, indicators, launchpad and shortcuts. The theme and language are those of the profile.

5.3 Launchpad

The launchpad gives direct access to the twelve most-used modules: tontines, savings, loans, solidarity, sessions, members, reports, wallet, marketplace, chat, academy, events. It reduces the number of clicks and makes the member area usable on mobile.

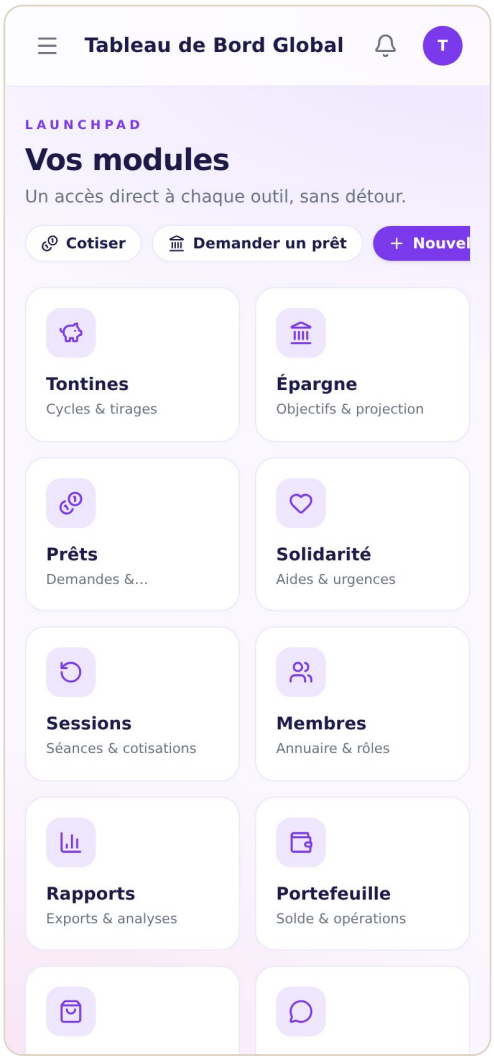


Figure 5.3 — Mobile launchpad: module grid, quick actions (contribute, request a loan, new association) and thumb-friendly navigation.

5.4 Interactions between modules

- An **approved membership** creates the member, their role and their member number; it feeds the contribution calls.
- A **session** generates the expected contributions, attendance, fines and catering.
- A **tontine** consumes contributions, calculates the pot, designates beneficiaries and produces a cycle balance.
- A **loan** debits the cash box, generates a repayment schedule and interest; repayments feed the history and the dividends.
- An **aid payment** debits the solidarity fund and creates a traceable application.
- **Reports** aggregate all movements; the **history** surfaces them at member, activity or association level.

> INFORMATION
>
> Each module can be enabled or disabled per association according to its configuration. A pure tontine does not need the loans module; a national friendly society will mainly use membership, solidarity, meetings and reports.

5.5 Shared principles

All modules share four guarantees: the same member identity, the same clock, the same currency and the same journal. An operation entered in one module is immediately visible in the others, with no manual synchronisation.

Guarantee	Concrete effect
Single identity	One member, one account, one member number per association
Time-stamping	Every movement carries a date, an author and a reference
Single currency	Amounts are stored as values and displayed in the association's currency
Shared journal	Reports and history aggregate all modules

CHAPTER 6

Activities and the tontine engine

The tontine is the historic heart of Tchoua. The engine goes far beyond the rotating turn: shares, fractions, sessions, pot, auctions, multiple beneficiaries, fixed cycles or share exhaustion, balance and closing.

6.1 Definitions

Term	Definition
Share	Unit of participation; a full share is worth the amount defined by the activity.
Fraction	1, 1/2 or 1/4 of a share depending on the configuration and whether fractions are allowed.
Subscription	A member's commitment to pay a number of shares at each session.
Contribution	The actual payment for a session, equal to subscribed shares × share amount.
Pot	Sum of the contributions collected for the session, plus any carry-over and adjustments.
Beneficiary share	Share allocated to a beneficiary; it entitles the holder to the amount per beneficiary share.
Session	A dated meeting during which members contribute and distributions are made.
Cycle	The complete period between the opening and the closing of the activity.

6.2 Activity types

Type	Principle	Typical use
Rotating tontine	Each member receives the pot in turn	Neighbourhood group, family, friends
ASCA tontine	Internal savings and credit, loans with interest	Cooperative, women's group
Auction tontine	The pot goes to the highest bidder; the discount feeds the interest	SLD, elite tontines
Savings / investment	Capitalised contributions, interest or dividends	Investment fund, projects
Loan	Loan fund for members	HORIZON, section fund
Solidarity	Non-repayable contributions and aid	Friendly societies, mutuals
Catering	Contribution to reception costs	Statutory meetings
In-kind	Contributions in goods, converted into value	Agricultural cooperatives
Group purchases	Collective orders	Groupings, consumer tontines

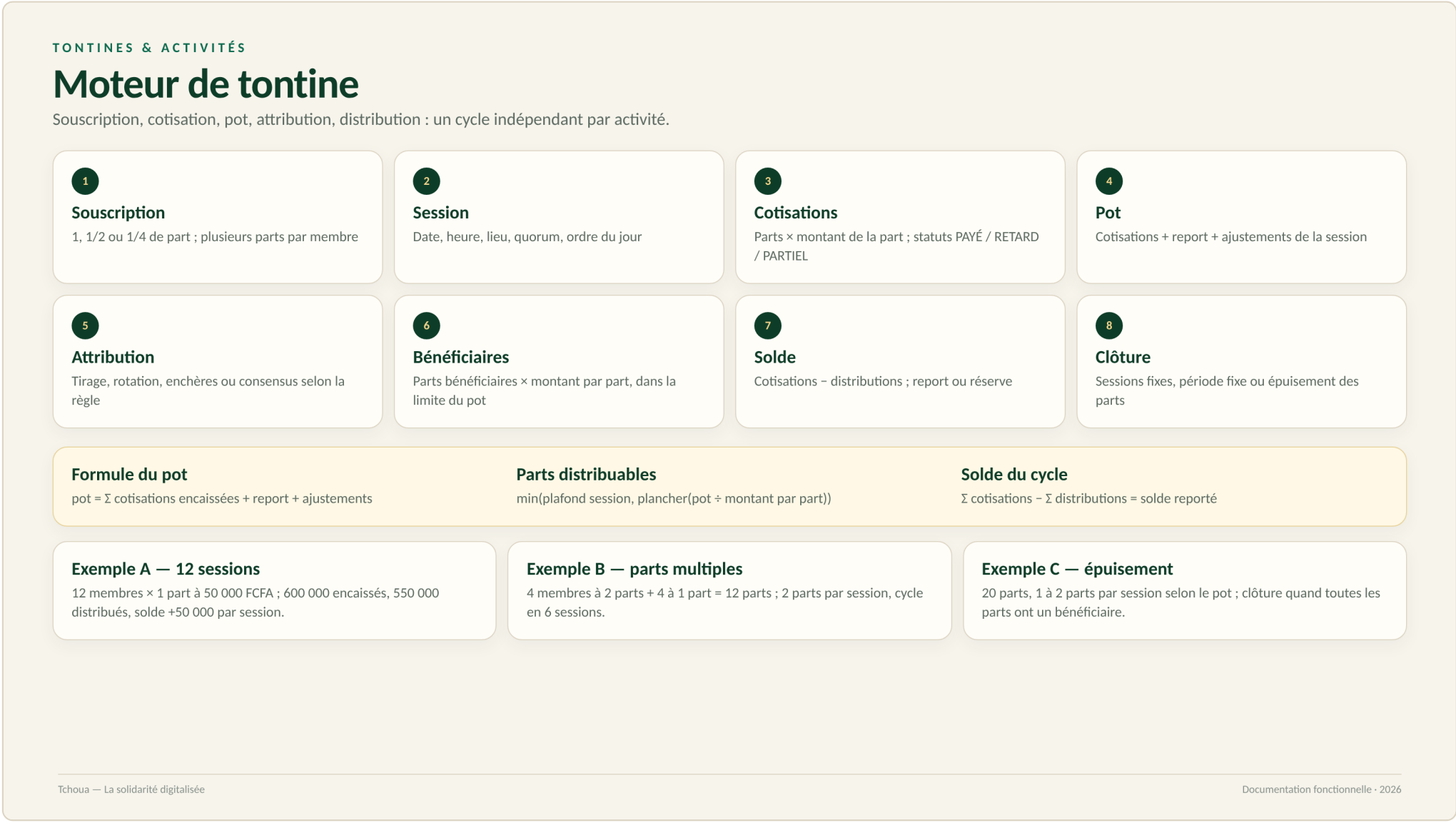


Figure 6.1 — The engine's eight steps, from choosing shares to closing the cycle, with the three calculation formulas.

6.3 Subscription and shares

An active member subscribes between the minimum fraction and any applicable cap. Their expected contribution at each session is their shares multiplied by the share amount. If they subscribe several shares, their contribution is multiplied accordingly; their beneficiary entitlement follows the same proportion.

```
expected contribution = subscribed shares × share amount
beneficiary entitlement = beneficiary shares × amount per beneficiary share
```

Parameter	Effect	Example
Share amount	Basis of the contribution	25 000 FCFA
Minimum fraction	1, 1/2 or 1/4	Half share = 12 500 FCFA
Max shares per member	Caps the commitment	4 shares maximum
Amount per beneficiary share	What a beneficiary receives	300 000 FCFA
Max beneficiary shares / session	Caps the distribution	2 shares per session
Same beneficiary several times	Allows accumulation	Yes / no

6.4 The pot and distributable shares

A session's pot is the sum of the contributions collected, any carry-over and approved adjustments. The number of distributable shares is the smaller of two numbers: the cap on shares per session, and the number of shares the pot can finance.

```
pot = Σ contributions collected + carry-over + adjustments
distributable shares = min(session cap, floor(pot ÷ amount per share))
cycle balance = Σ contributions − Σ distributions
```

> **WARNING**
>
> If the pot is insufficient, the distribution may be blocked, partial or deferred according to the activity's rule. No distribution may make the balance negative without explicit authorisation of a cash advance.

6.5 The three cycle modes

Mode	End of cycle	Consequence
Fixed sessions	After N sessions (e.g. 12 monthly)	If shares remain, they are carried over or refunded according to the rule
Fixed period	On the end date	The number of sessions depends on the frequency
Share exhaustion	When all subscribed shares have a beneficiary	The number of sessions varies; shares per session may increase with the pot

Progress is calculated differently depending on the mode: sessions held / sessions planned for a fixed-number cycle, shares distributed / shares subscribed for an exhaustion cycle.

6.6 Allocating beneficiaries

Mode	Rule	Control
Rotation	Fixed order of members	A member only comes round once per round
Monthly draw	Random draw among eligible members	Exclusion of previous beneficiaries if the rule requires it
Single draw	Full schedule set in advance	Check of remaining shares
Auctions	The pot goes to the highest bidder; the discount feeds the interest	Minimum starting bid, number of bidders
Merit / needs	The board designates with justification	Traceability of the decision
Hybrid	Combination of rules	Explicit configuration

A beneficiary's eligibility takes into account their active status, contributions up to date, remaining shares and any previous receipts. A member can receive several shares in the same session if the activity allows it and within the cap.

6.7 Auctions at the SLD

The SLD runs an annual tontine of 25 000 FCFA and its multiples, allocated by public auction with a starting bid of 5% of the contribution rate. The member who wins the auction receives the pot minus the discount; the interest generated is distributed in lots or carried over. Defaulting members incur a 15% surcharge if they have not yet benefited and 20% if they have already benefited.

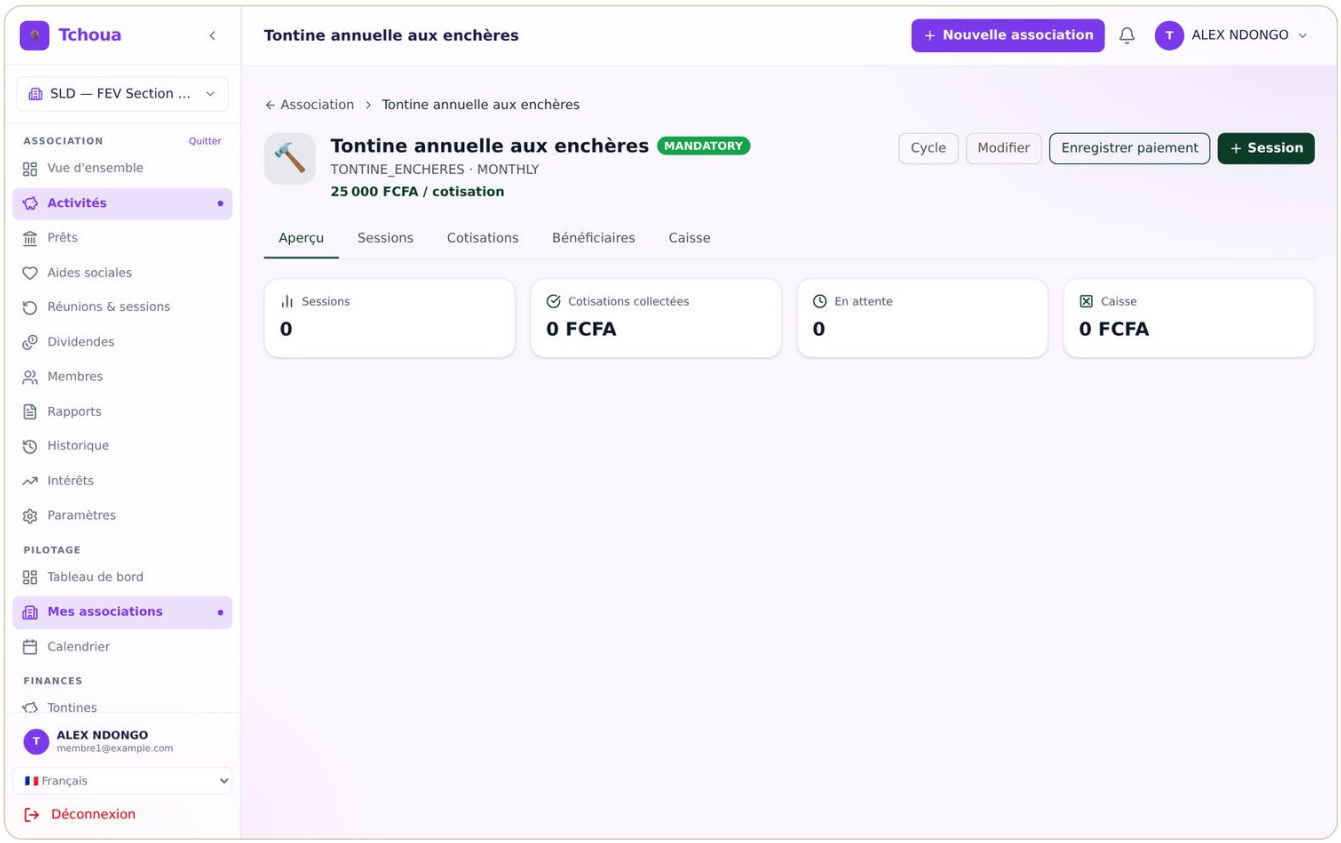


Figure 6.2 — The SLD's "Annual auction tontine" activity: 25 000 FCFA per share, 12 sessions, auction mode. The cycle and share parameters are editable.

6.8 Worked examples

Example A — 12 sessions, 1 share, 12 members

Share at 50 000 FCFA, amount per beneficiary share 550 000 FCFA, one beneficiary per session. Each session collects 12 × 50 000 = 600 000 FCFA, distributes 550 000 FCFA and leaves a balance of 50 000 FCFA. After 12 sessions, all 12 members have benefited and the cumulative balance is 600 000 FCFA.

Example B — multiple shares

Four members with 2 shares and four members with 1 share represent 12 shares. With two beneficiary shares per session, the cycle ends in six sessions.

Example C — share exhaustion

20 shares, one distributable share per session, variable pot. Session 1 distributes one share, session 2 may distribute two if the pot allows; the cycle stops when all 20 shares have a beneficiary.

Example D — half share

Share at 100 000 FCFA, member with 0.5 share: contribution of 50 000 FCFA and beneficiary entitlement of 0.5 × the amount per share.

6.9 Actual cycle and balance

The cycle screen shows contributions, distributions, balance, subscribed and distributed shares, progress and each member's position. The following figure shows a demo cycle with four members and two sessions: 800 000 FCFA contributed, 800 000 FCFA distributed, zero balance, two shares distributed out of four.

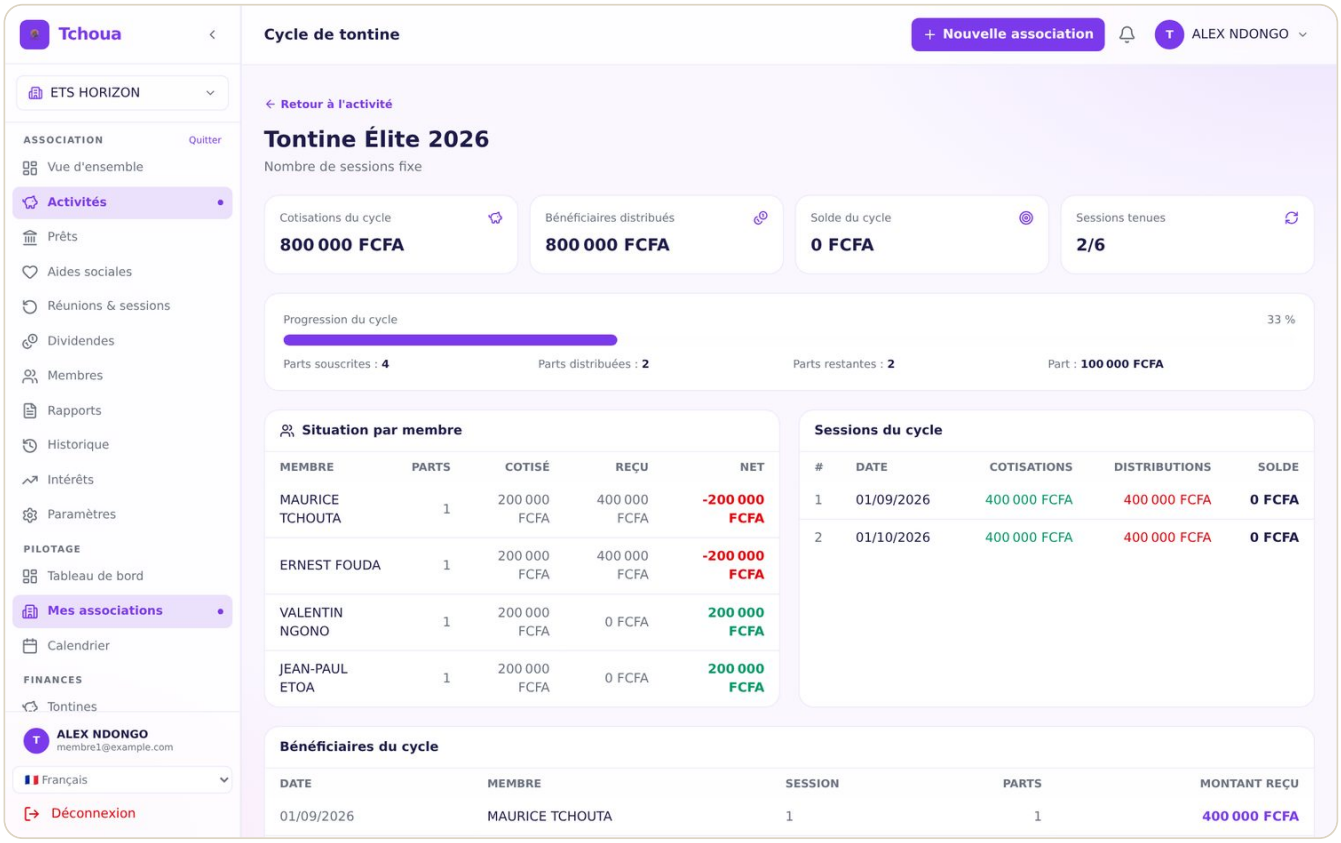


Figure 6.3 — Tontine cycle: contributions, distributions, balance, progress, each member's position, sessions and beneficiaries. Demo data.

6.10 Edge cases and management rules

Case	Expected behaviour
Absent member	Contribution due; fine according to the internal rules; eligibility suspended if the rule requires it
Partial contribution	PARTIAL status; remainder due; beneficiary share reduced pro rata if configured
Late contribution	LATE status; penalty; defaulters ranked last
Insufficient pot	Blocking, partial distribution or carry-over according to the rule
Excess pot	Carry-over, an extra share within the cap, or cash box
Departing member	Contributions vested; shares not received recalculated or refunded
Double distribution	Forbidden: uniqueness (session, member, share rank)
Change of share during the cycle	Forbidden unless by amendment with an effective date
Early closing	Allowed if the rule permits; remaining shares handled according to the rule

6.11 Configuration in the interface

The activity form lets you define the type, the share amount, the frequency, the allocation mode, the amount per beneficiary share, the minimum fraction, the caps, whether fractions are allowed and the cycle mode. Advanced parameters (surplus rule, late-payment tolerance, early closing) are stored in the cycle configuration and documented in the tontine specification.

6.12 Other activities

The engine is not limited to financial tontines. Complementary activities use the same sessions, contributions and reports, with adapted rules.

Activity	Principle	Control points
Savings	Regular capitalised contributions, withdrawal under conditions	Balance, cap, notice period, interest
Investment	Pooled fund for projects or productive loans	Shares, project, profitability, dividend
In-kind	Agricultural produce or materials valued	Unit, quality, conversion, storage
Group purchases	Collective order, pro rata allocation	Quote, delivery, quality, payment
Catering	Session contribution and flat fee to the recipient	Rate, arrears, handover, journal
Services / time	Time bank, skills exchange	Hours, valuation, assessment

Each activity keeps its own cash box, frequency and rules. An association can therefore combine a tontine, an investment fund, a loan fund and a catering fund without mixing the flows.

> **KEY TAKEAWAY**
>
> The engine calculates and exposes balances, shares and distributions. Automatic designation of beneficiaries based on eligibility and auctions remains a workflow to be completed; configuration and calculation are operational.

CHAPTER 7

Contributions, sessions and attendance

The session is the unit of time in association life: members meet, contribute, check attendance, apply fines, serve the catering and make decisions.

7.1 Anatomy of a session

Element	Description
Number	Sequence within the activity cycle
Date and time	Planned date, actual start and end
Venue	Head office, a member's home, community hall, rented room
Type	Ordinary, extraordinary, GA, board meeting
Quorum	Number or percentage required to deliberate
Agenda	Items to be addressed, attached documents
Attendance	Present, late, absent, excused
Contributions	Amounts expected and collected per member
Fines	Sanctions applied according to the internal rules
Decisions	Resolutions, votes, owners, deadlines

7.2 Lifecycle of a contribution

1. **Call** — the open session generates the expected contributions from the subscribed shares.
2. **Collection** — the treasurer enters the amount, the method (cash, Mobile Money, transfer, in-kind) and the reference.
3. **Status** — PAID, PARTIAL, LATE, EXEMPT. The status determines eligibility and reminders.
4. **Reconciliation** — the amount is assigned to the activity's cash box and to the session pot.
5. **Reminder** — unpaid amounts trigger a notification and, if configured, a fine.
6. **Closing** — the session is frozen; later corrections create an adjustment.

```
expected = shares × share amount
collected = Σ validated payments
outstanding = expected − collected
```

7.3 Statuses and consequences

Status	Definition	Consequence
PAID	Full amount collected	Eligibility maintained; full beneficiary share
PARTIAL	Incomplete collection	Outstanding balance; reduced eligibility if configured
LATE	Deadline passed	Penalty; reminder; defaulters ranking
EXEMPT	Waiver (special member, GA decision)	No penalty; rights maintained
CANCELLED	Session cancelled or error	Reversing entry, no deletion

7.4 Attendance and fines

Attendance is recorded for each session. The SLD's internal rules provide a complete scale: late 500 FCFA, absence 1 000 FCFA, two consecutive absences 3 000 FCFA, three absences 5 000 FCFA, leaving before 19:00 500 FCFA, chattering 500 FCFA, indiscipline 3 000 FCFA, unauthorised commitment 10 000 FCFA, missed committee meeting 5 000 FCFA, absence from the national GA 20 000 FCFA, absence from a funeral 25 000 FCFA.

Breach	SLD sanction	Tchoua effect
Lateness	500 FCFA	Fine attached to the session
Single absence	1 000 FCFA	Fine + ABSENT status
Two consecutive absences	3 000 FCFA	Cumulative fine
Three consecutive absences	5 000 FCFA	Automatic warning
Early departure	500 FCFA	PARTI_TOT status
Indiscipline	3 000 FCFA	Fine + report to the censor
Missed committee meeting	5 000 FCFA	Fine + mission follow-up
Absence from national GA	20 000 FCFA	Exceptional fine

Table 7.1 — SLD scale. All amounts are configurable; no value is hard-coded.

7.5 Catering

Catering is a reception fund: each member contributes a fixed amount per session, and the member hosting the meeting receives a flat fee, after deduction of their arrears. At the SLD, the contribution is 3 000 FCFA and the flat fee 150 000 FCFA.

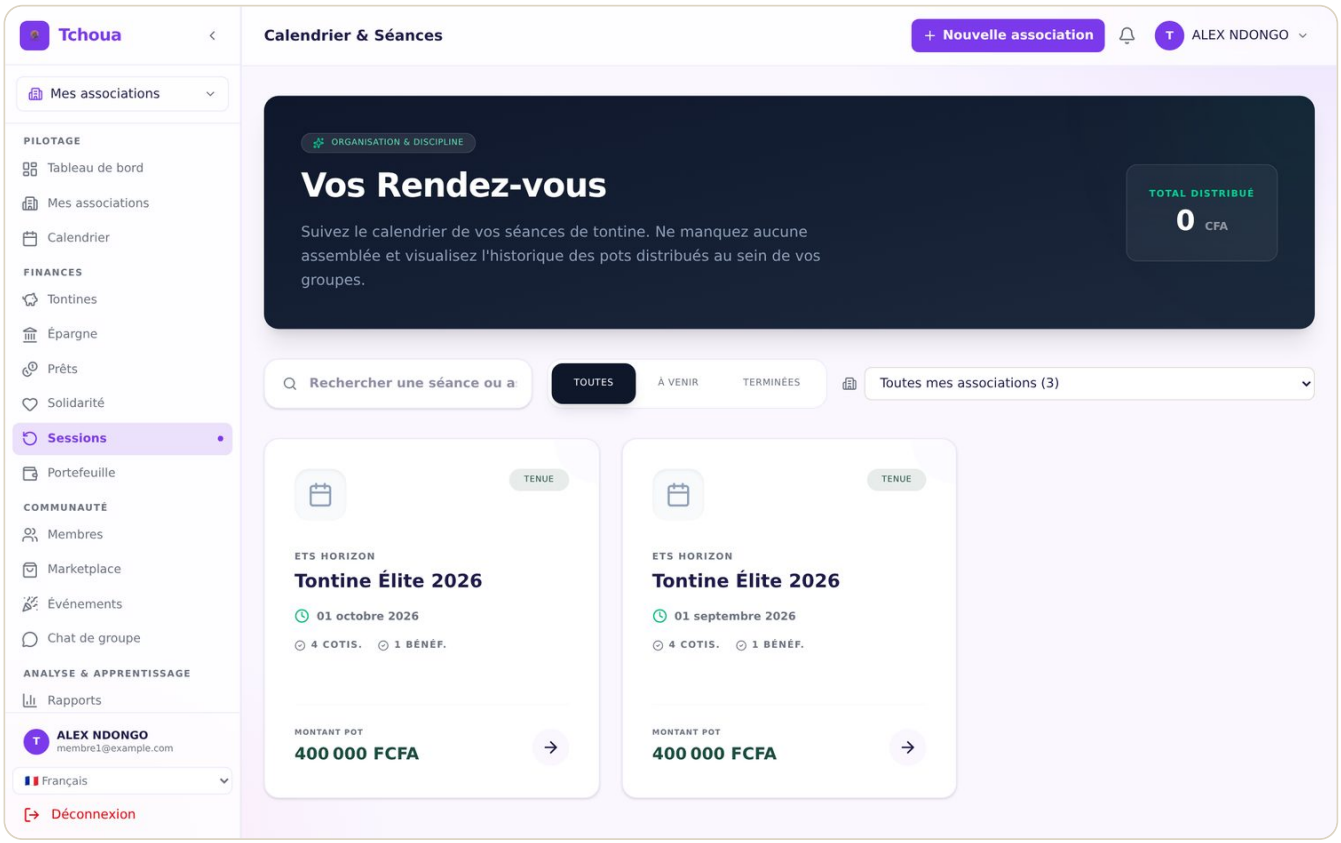


Figure 7.1 — Example of a session calendar (demo data): dates, activities, statuses and direct access to the session. Sessions come from the configured frequency.

7.6 Activities and contribution calls

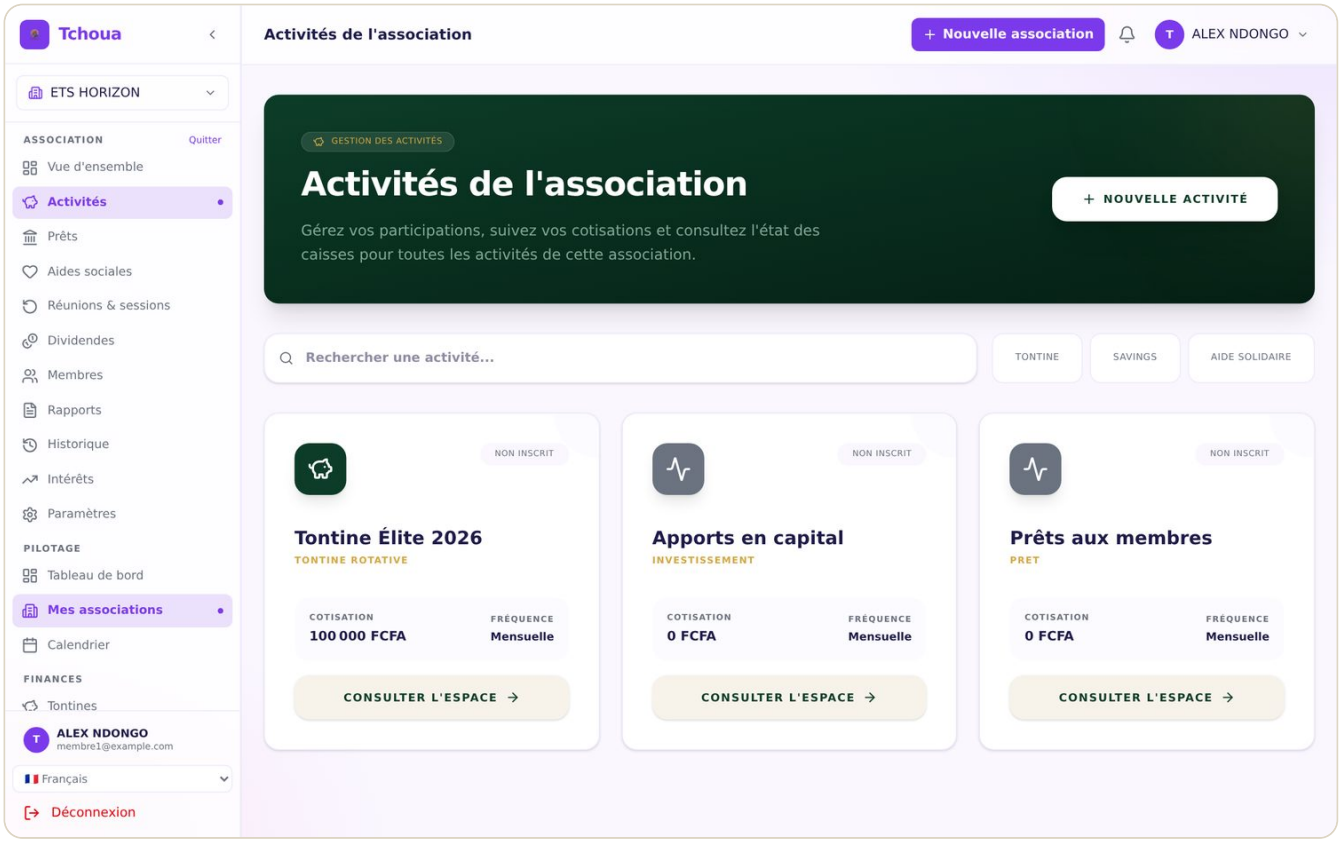


Figure 7.2 — Association activities: cards per activity, amount, participation, cash box, subscriptions and access to details.

Each activity card shows the contribution amount, the frequency, the available cash box, the number of subscribers and the actions (view, contribute, manage). Mandatory activities appear in the member area with the expected amount and the due date.

7.7 Reminders and automation

A scheduled job identifies late contributions, loan deadlines and upcoming sessions, then sends notifications. The job can be run in simulation mode to check the recipients before the actual send. Reminders are logged to avoid duplicates.

> **INFORMATION**
>
> Notification channels (e-mail, WhatsApp, SMS) are configured at platform level. The content and language follow the member's preferences.

7.8 Treasury good practice

- ✓ Enter contributions the same day, never at the end of the month.
- ✓ Attach the Mobile Money reference or the receipt to each payment.
- ✓ Have corrections approved by a second person.
- ✓ Close the session before opening the next one.

- ✓ Export the cash journal at the end of every month.
- ✓ Reconcile the Tchoua balance and the bank balance before the GA.

7.9 Session closing checks

Closing a session is a control point, not a formality. It verifies that collections, attendance, fines and catering are consistent before freezing the pot and producing the minutes.

Check	Question to ask	Trace
Cash box	Does the balance match the cash, Mobile Money and bank?	Cash journal
Unpaid amounts	Are all expected contributions paid, partial or documented?	Status list
Fines	Are lateness, absences and breaches qualified and quantified?	Censor's register
Catering	Is the recipient's flat fee calculated after deducting arrears?	Receipt and journal
Minutes	Are decisions, votes and owners recorded?	Signed minutes

> **INFORMATION**
>
> A closed session can no longer be modified directly: any later correction creates an adjustment visible in the history and the report.

CHAPTER 8

Loans, repayments and interest

The loans module turns collective savings into controlled credit: application, review, guarantees, disbursement, repayment schedule, repayments and transparent allocation of interest.

8.1 Loan types

Type	Source	Typical rules
Member loan	Activity cash box	Monthly rate, term, cap per member, guarantees
Exceptional loan	Investment fund	Committee, specific conditions, information to the GA
Illness loan	Solidarity fund	Interest-free, capped, repayment after hospitalisation
Cash advance	Treasury	Short term, board approval

8.2 Loan rules

- **Cap** — fixed amount, percentage of the cash box or multiple of the member's stake.
- **Rate** — monthly rate, possibly decreasing after a few months.
- **Term** — number of months and repayment schedule.
- **Guarantees** — savings, surety, guarantor, cheque, acknowledgement of debt.
- **Approval** — board, secretariat or automatic depending on the amount and the configuration.
- **Simultaneous active loans** — maximum number per member.
- **Arrears** — penalty, surcharge, suspension, recovery.

```
interest = outstanding principal × monthly rate × number of months
instalment = principal ÷ term + interest for the period
cap = min(available cash box, member cap, multiple of the stake)
```

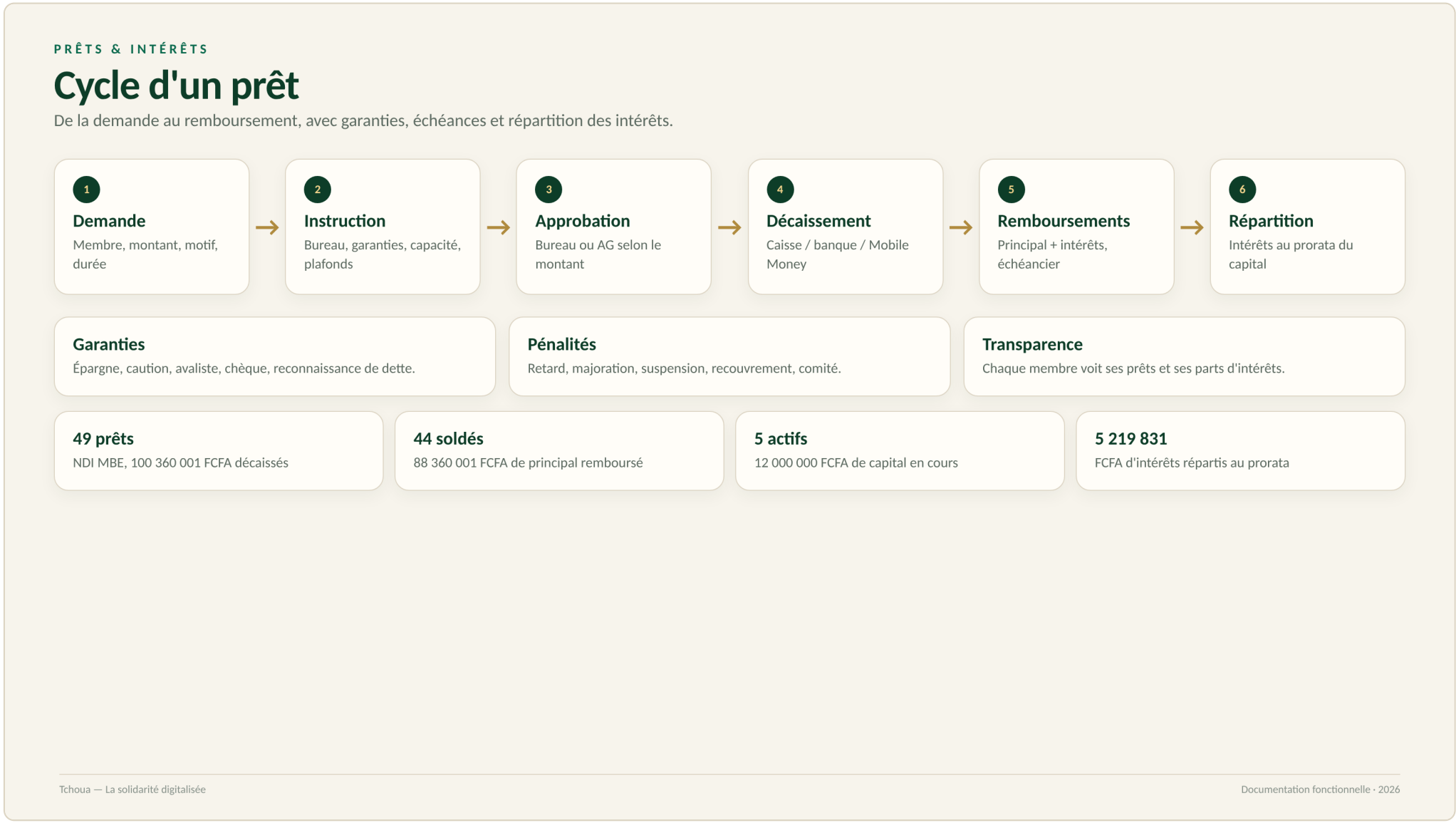


Figure 8.1 — The six steps of a loan, from application to interest allocation, with guarantees and penalties.

8.3 Loan cycle

1. The member creates an application: amount, purpose, term, proposed guarantees.
2. The board reviews it: check of the cash box, the cap, active loans and repayment capacity.
3. The application is approved or rejected with reasons; the decision is notified.
4. The disbursement is recorded (cash, Mobile Money, transfer) and the cash box is debited.
5. Repayments are entered with principal and interest separated.
6. The loan is settled; the interest feeds the allocation and the dividends.

8.4 Interest allocation

Each repayment includes a principal portion and an interest portion. Interest is allocated among members in proportion to the capital they held on the loan's disbursement date. This key prevents a member who joined after the loan from receiving interest they did not help to generate.

```
share = member's capital on the loan date ÷ total capital on the loan date
interest share = interest repaid × share
```

The "Interest" screen details, for each repayment: the date, the borrower, the activity, the total interest, the member's capital, the overall capital, the share and the amount due to them. The board can expand the allocation across all members.

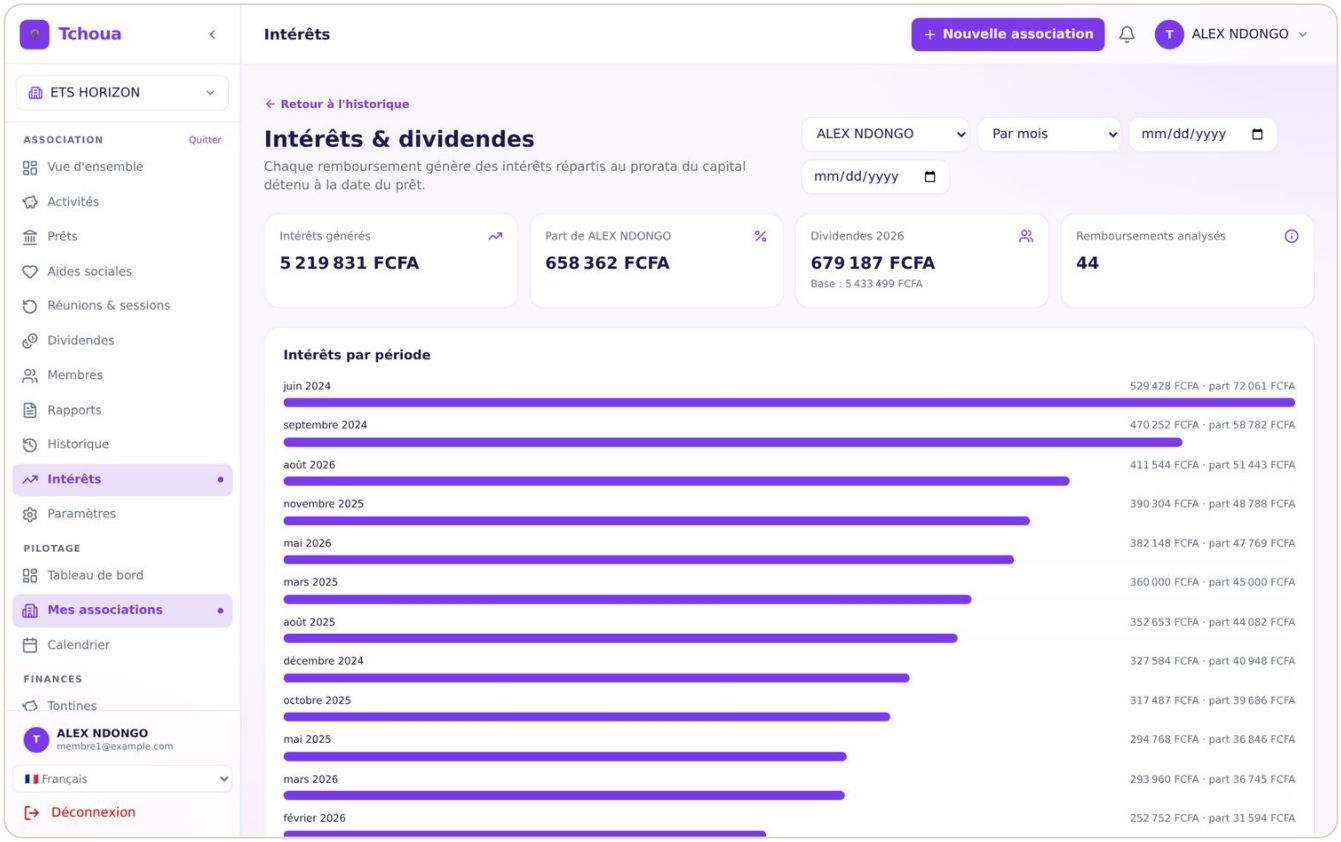


Figure 8.2 — Interest and dividends: 5 219 831 FCFA of interest generated, the member's share, 2026 dividends and the detail by repayment.

8.5 The ETS HORIZON case

HORIZON is an investment and lending association with seven members. The imported data make it possible to verify the module on a real volume: 49 loans, 100 360 001 FCFA disbursed, 44 loans repaid, 5 active loans for 12 000 000 FCFA, 88 360 001 FCFA of principal repaid and 5 219 831 FCFA of interest.

Indicator	Value	Reading
Members	7	One member per named account
Capital / contributions	8 000 000 FCFA	Basis for allocating interest
Loans disbursed	49	100 360 001 FCFA in total
Loans settled	44	93 579 832 FCFA repaid (principal + interest)
Loans outstanding	5	12 000 000 FCFA of principal
Principal repaid	88 360 001 FCFA	Excluding interest
Interest generated	5 219 831 FCFA	Allocated pro rata to capital
2026 dividends	5 433 499 FCFA	7 shares, validated calculation basis
Loans / funds ratio	150 %	Parameter specific to HORIZON

Table 8.1 — HORIZON key figures from the actual import, verified in the application.

8.6 Loans screen

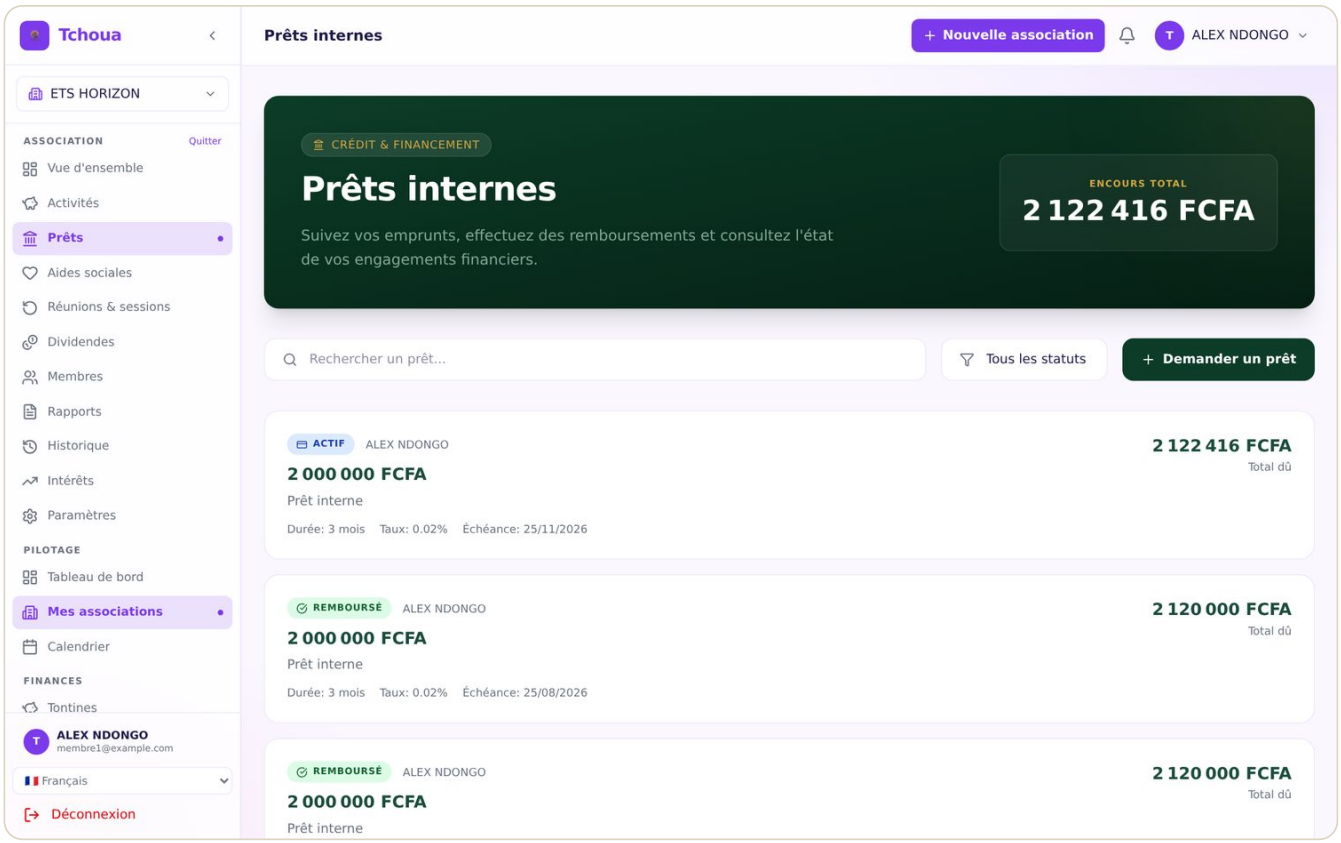


Figure 8.3 — Internal loans: available cash box, loans outstanding, history, statuses and access to repayment schedules.

8.7 Risks and controls

Risk	Control
Over-financing	Cap per member, cash ratio, max active loans
Repayment default	Guarantees, repayment schedule, reminders, penalties
Double loan	Check of active loans
Misappropriation	Dual approval, audit, bank reconciliation
Misallocated interest	Capital key on the loan date, detailed journal
Politically motivated loan	Justification, vote, traceability of the decision

WARNING

An unrepaid loan must never be deleted or hidden: it must be written off with a board decision and an adjusting entry. Traceability takes precedence over the appearance of the balance.

CHAPTER 9

Solidarity and social aid

Solidarity is the very purpose of many associations. Tchoua turns informal mutual support into a legible entitlement: a catalogue, caps, an application, an approval and a traceable disbursement.

9.1 The aid catalogue

Category	Event	Example cap
Health	Member hospitalisation	SLD 70 000; FEV 10 000/day (3–15 days)
Health	Spouse hospitalisation	SLD 50 000
Health	Serious illness	Interest-free loan up to 100 000
Family	Marriage (first)	50 000
Family	Birth	20 000 + 10 000 twins
Bereavement	Death of the member	SLD 900 000; FEV 300 000
Bereavement	Death of the spouse	405 000
Bereavement	Death of a child	50 000 (<10 years) / 100 000 (>10 years)
Bereavement	Death of the father or mother	100 000
Emergency	Disaster, fire, accident	According to budget and board decision
Education	School fees, supplies	According to cash box and internal rules

Table 9.1 — Indicative catalogue. Each association defines its own categories, caps and conditions.

9.2 Application workflow

1. **Opening** — the member selects the category, describes the event and attaches the documents (certificate, invoice, deed).
2. **Review** — the board checks eligibility, completeness and the available cap.
3. **Decision** — approval, reasoned rejection or request for further documents.
4. **Disbursement** — payment to the member or to a third party (hospital, funeral home), with a reference.
5. **Follow-up** — the aid appears in the member's history and in the association's reports.

```
fund balance = allocated contributions − aid paid − costs
available cap = annual cap − aid already paid in the period
```

9.3 Solidarity screen

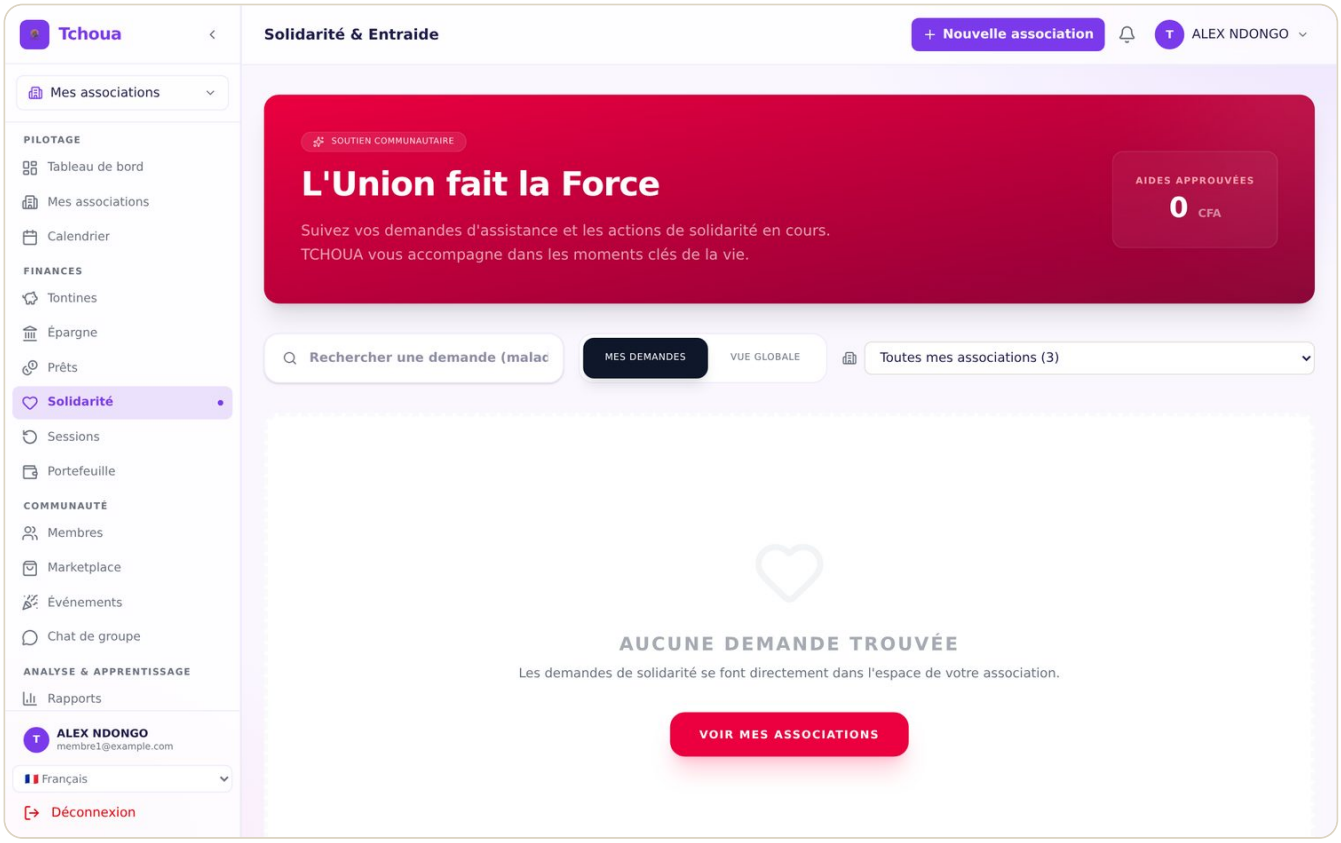


Figure 9.1 — Solidarity & Mutual Aid area: application search, filters by category and by association, queue status.

9.4 SLD rules

The SLD internal rules specify the amounts and conditions: hospitalisation of at least five days, observed by a designated member and verified with the health facility; illness aid limited to a one-year period; marriage limited to a first marriage; birth within a marriage; death of a member with a breakdown for burial, transport, catering, drinks and miscellaneous costs; a wake attendance requirement of three days in Douala.

SLD aid	Amount	Condition
Member hospitalisation	70 000	5 days minimum, verification
Spouse hospitalisation	50 000	First declared spouse
Serious illness	Loan 100 000 max	Interest-free, 3-month repayment
Marriage	50 000	First marriage only
Birth	20 000 (+10 000 twins)	Birth within a marriage
Death of member	900 000	Statutory breakdown
Death of spouse	405 000	Statutory breakdown
Death of child	50 000 / 100 000	Depending on age
Death of parent	100 000	Father or mother

9.5 FEV rules

At national level, the FEV has a provisional deposit fund of 10 000 FCFA per member. In the event of a death, 300 000 FCFA is drawn: 200 000 for the family, 50 000 for the wreath and 50 000 for the deceased's section. Illness aid is 10 000 FCFA per day of hospitalisation, with a minimum of 3 days and a maximum of 15 days per year, subject to verification by a section team and a medical certificate.

9.6 Controls and anti-abuse

- Supporting documents mandatory and archived.
- Verification by a designated member for hospitalisations.
- Annual cap per member and per category.
- One aid per event, unless by exceptional decision.
- Aid history available to the board and the auditor.
- Disbursement approved by at least two people for large amounts.

> **PHILOSOPHY**
>
> Aid is not a favour: it is a contributory entitlement. The member has contributed to the fund; they must be able to understand the amount they are entitled to, the documents to provide and the processing time.

CHAPTER 10

Meetings, discipline and catering

The meeting is where association life is decided. Tchoua turns it into a structured object: convening notice, agenda, attendance, quorum, votes, sanctions, minutes and catering.

10.1 Meeting types

Type	Purpose	Typical quorum
Ordinary meeting	Routine activities, contributions, decisions	Majority of members
Extraordinary meeting	Urgent matter, special convening	According to the bylaws
Ordinary general assembly	Accounts, budget, elections, guidelines	1/2 of members (FEV)
Extraordinary general assembly	Amendment of bylaws, dissolution	2/3 depending on the case
Board meeting	Review, arbitration, preparation	Majority of the board
Committee	Study of a technical matter	Committee members

10.2 Running a meeting

1. **Convening** — date, venue, agenda, attached documents, statutory notice period.
2. **Opening** — quorum check, appointment of the chair and the secretary.
3. **Roll call** — attendance, lateness, absences, excuses; any fines.
4. **Reading of the previous minutes** — adoption or corrections.
5. **Agenda items** — presentation, debate, vote, resolution.
6. **Contributions and catering** — collection, service, receipt.
7. **Closing** — next date, action owners, signed minutes.

```
quorum met = present ≥ required quorum
majority = votes for > votes against (or 2/3 according to the rule)
fine = scale according to the breach observed
```

10.3 Meetings screen

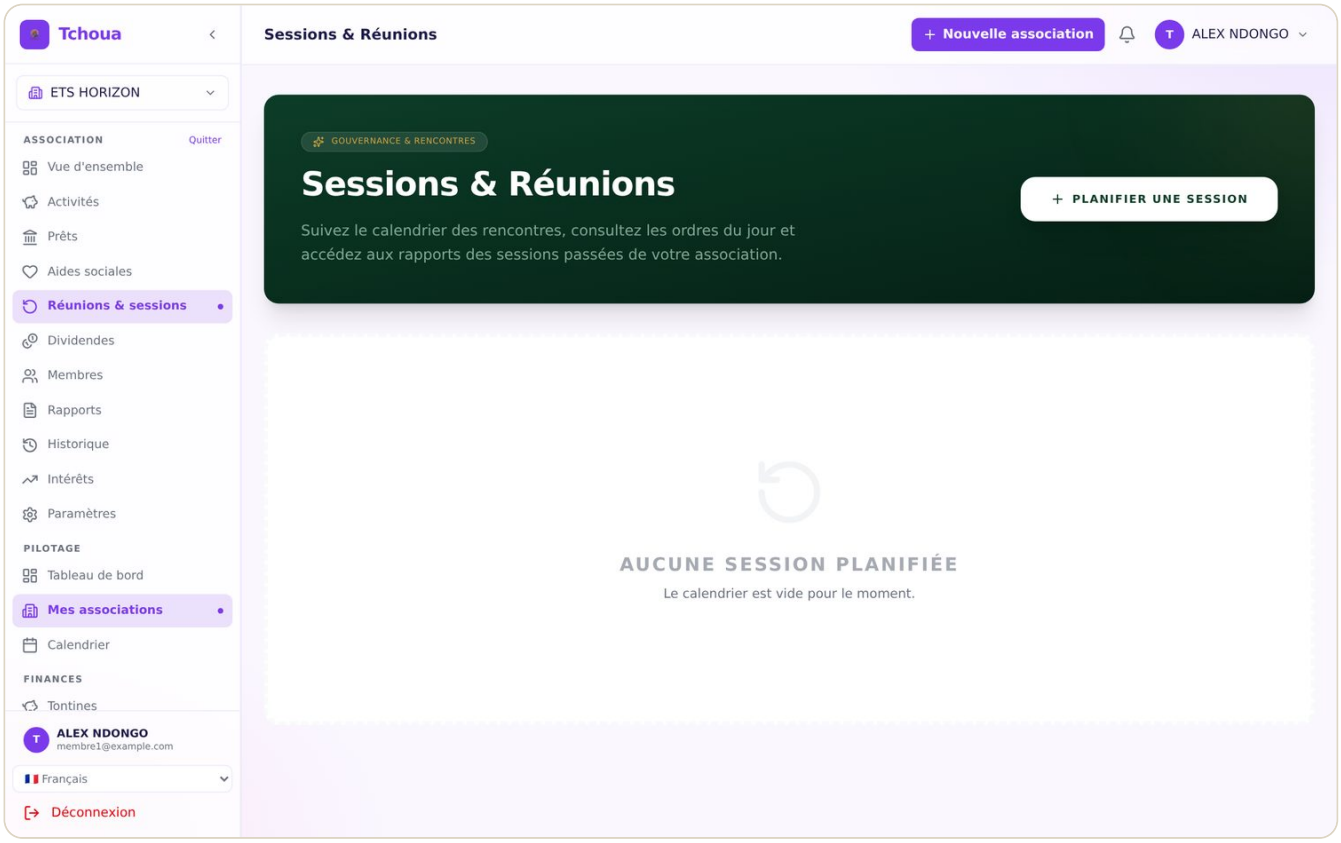


Figure 10.1 — Sessions & Meetings: search, filters by activity and by association, status of the meeting queue.

10.4 Discipline

Discipline is handled as a register, not as a mood. Each breach is linked to a session, a member, an article of the internal rules and an amount. The censor applies; the treasurer collects; the auditor verifies. In the event of a dispute, the member may ask the board for a discretionary waiver, which the board decides and justifies.

Step	Actor	Trace
Observation	Chair / censor	Attendance line, time-stamp
Qualification	Censor	Reason, article of the internal rules
Sanction	Board	Amount, due date, notification
Collection	Treasurer	Receipt, cash box, journal
Appeal	Board / GA	Reasoned decision

10.5 Catering and receptions

Catering is a contribution from each member at each statutory session. The member hosting the meeting receives a flat fee, after deduction of their arrears. At the SLD, the contribution is 3 000 FCFA and the flat fee 150 000 FCFA; the payment is handed to the recipient during the session.

- The catering rate is an association parameter.
- The recipient's flat fee is also configurable.
- Catering arrears are deducted from the flat fee.
- The catering balance is tracked in the reports.

10.6 Minutes and archives

Each meeting produces minutes: agenda, those present, those absent, resolutions, votes, actions and owners. The minutes are uploaded to the association's documents, with a dedicated type, and remain available to members. Decisions can be linked to tasks and deadlines.

> **INFORMATION**
>
> Convening notices and minutes can be exported to PDF. Documents are time-stamped and attached to the association; they are never overwritten silently.

10.7 Good practice

- ✓ Publish the agenda at least 15 days in advance (30 days for a GA).
- ✓ Check the quorum before opening the votes.
- ✓ Have the minutes signed by the president and the secretary.
- ✓ Collect fines and catering contributions on the spot.
- ✓ Circulate the minutes to members within 30 days.
- ✓ Keep the supporting documents with the minutes.

CHAPTER 11

Accounting, reports and dividends

An association without reports is an association without proof. Tchoua produces statements that the board, the auditors and the members can read: cash boxes, movements, balance sheet, dividends and exports.

11.1 The cash boxes

Each activity has its own cash box: tontine, solidarity, investment, loan, catering. The balance is calculated from validated movements. Transfers between cash boxes are explicit operations, never silent corrections.

Cash box	Funded by	Used for
Tontine	Member contributions	Distribution to beneficiaries
Solidarity	Contributions, fines, donations	Illness, bereavement and family aid
Investment	Contributions, interest, dividends	Loans, projects, reserve
Loan	Repayments, allocations	Loan disbursements
Catering	Session contributions	Recipient's flat fee

11.2 The activity report

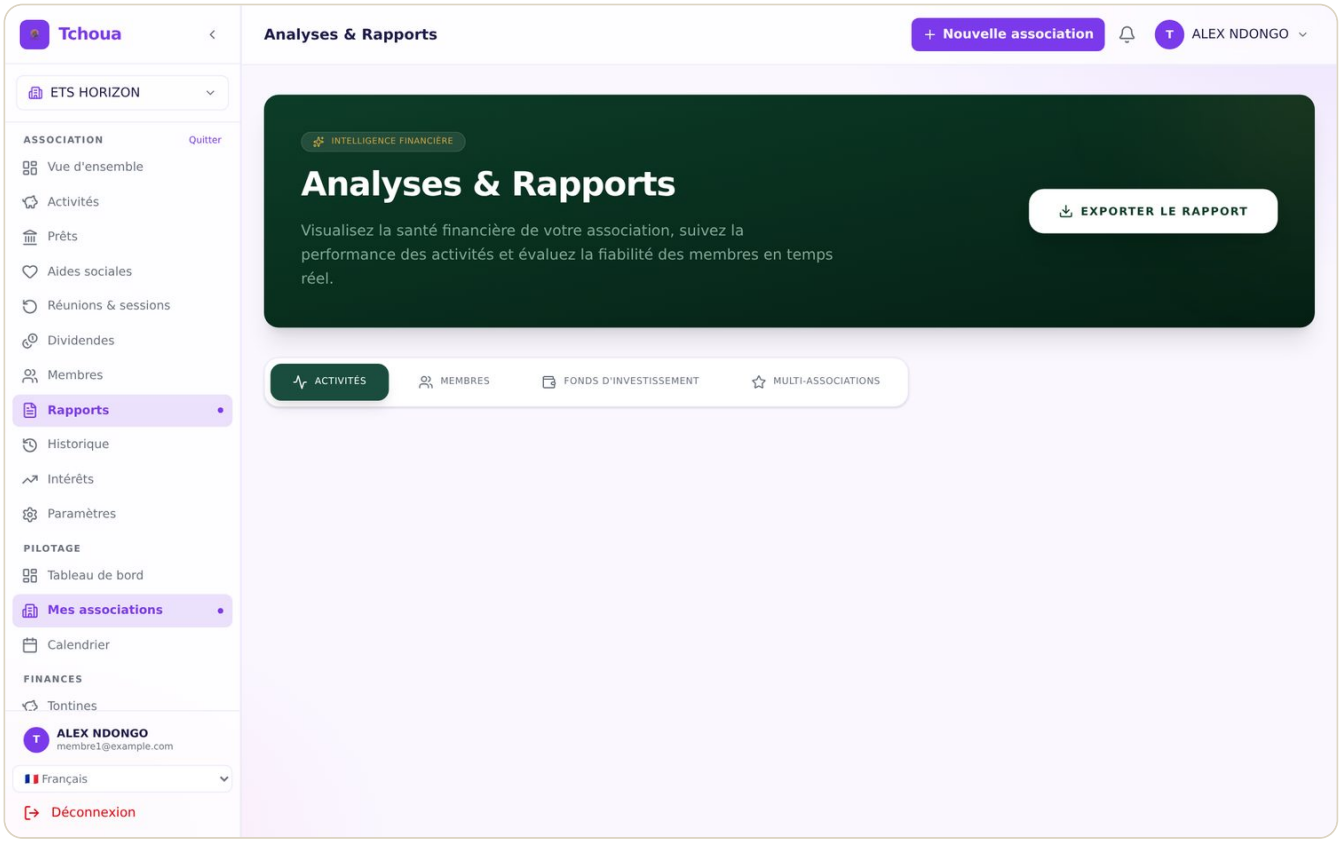


Figure 11.1 — Reports: summary by activity, cash boxes, recent contributions, exports and steering indicators.

The report presents expected and collected contributions, unpaid amounts, loans disbursed and repaid, aid paid, cash box balances, expenses and results. It can be filtered by period, activity and member, then exported to PDF or CSV.

11.3 Dividends

The dividends module calculates the allocation of interest and profits among members, using a key defined at the general assembly: number of shares, capital held, seniority or a combination. The calculation is frozen in an annual "run", with status, calculation basis and each member's share.

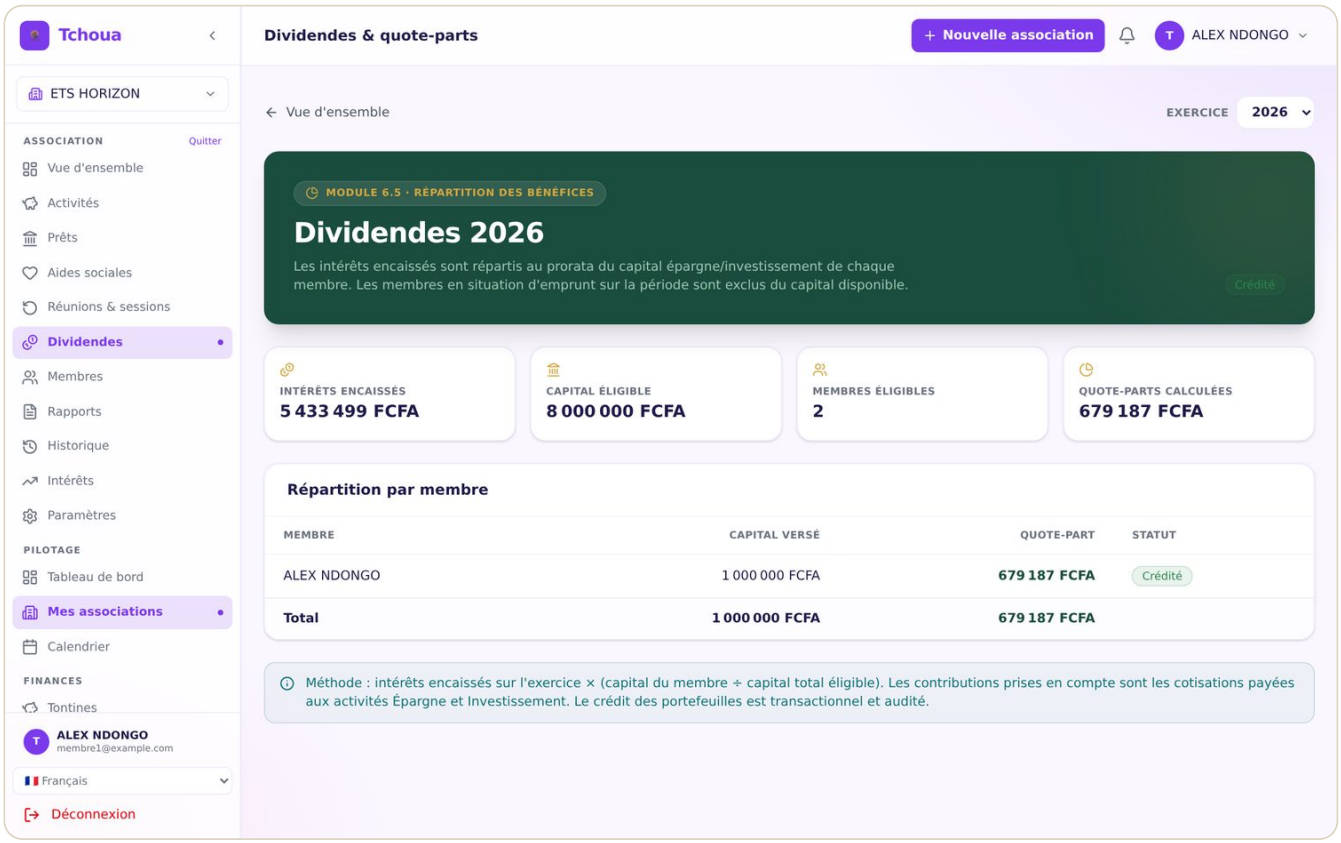


Figure 11.2 — HORIZON's 2026 dividends: base of 5 433 499 FCFA, seven shares, amount per share and allocation by name.

```
dividend per share = total amount to allocate ÷ number of shares
member dividend = member's shares × dividend per share
```

11.4 The HORIZON case

Item	Value
Financial year	2026
Allocation base	5 433 499 FCFA
Shares	7
Amount per share	776 214.14 FCFA
Status	Validated / published
Traceability	Annual run, share per member, export

11.5 Exports and documents

- **PDF** — management report, balance sheet, list of movements, balance certificate.
- **CSV** — detailed journal for a spreadsheet or an accountant.
- **Documents** — bylaws, internal rules, minutes, auditors' reports, supporting documents.

- **Archiving** — each export is time-stamped and attached to the association.

11.6 Audit and internal control

Control	Objective	Frequency
Cash box / bank reconciliation	Detect discrepancies	Monthly
Review of unpaid amounts	Send reminders and set provisions	Monthly
Loan control	Verify guarantees and due dates	Quarterly
Auditors' report	Certify the accounts	Quarterly and annual
Document inventory	Completeness of archives	Annual
System audit	Traceability of sensitive actions	Continuous

> **WARNING**
> An exported report does not replace statutory accounting. For associations subject to accounting obligations, exports must be handed to an accountant and reconciled with the official ledgers.

11.7 Steering indicators

Rate
of contribution collection

Outstanding
loans and repayment rate

Balance
per cash box and per activity

Coverage
of aid by the solidarity fund

These indicators are calculated from actual movements, never entered manually. They are available in the reports and on the board's dashboard.

CHAPTER 12

History and analytics

Every member must be able to answer three questions: what have I paid in, what have I received, and what is left for me? The history module answers them at member, activity and association level.

12.1 Three scopes

Scope	Content	Use
Association-wide	All movements across all activities	Steering, transparency, auditors
Per member	Contributions, loans, repayments, aid, interest shares	Member record, statement, claim
Per activity	Movements of the tontine, the fund or the cash box	Cycle monitoring, reports

12.2 The four views

The history screen adopts the display modes of modern ERPs: list, kanban, chart and grid. The user filters by type, member, activity, period and text, then groups by member, activity, type or month.

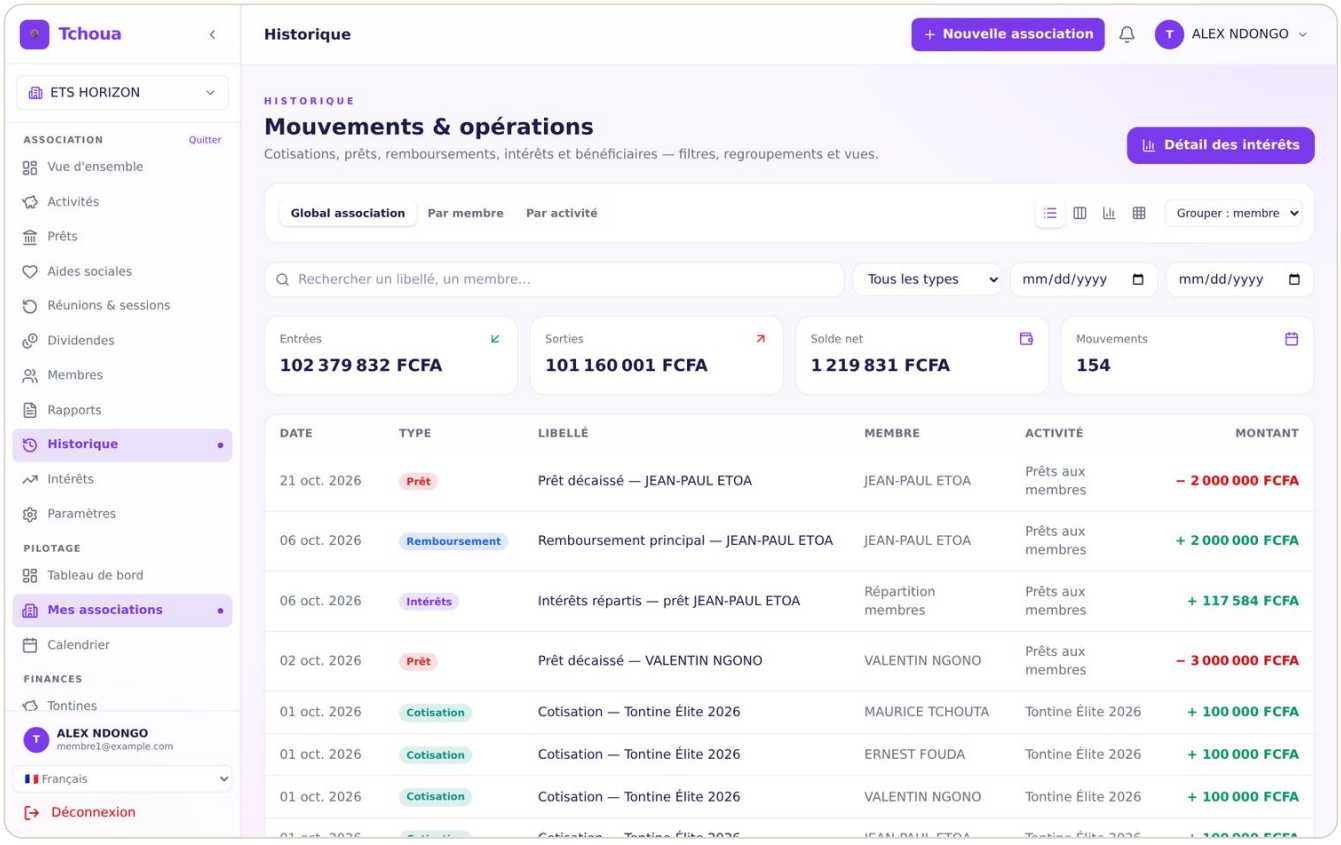


Figure 12.1 — History: inflows, outflows, net balance, paginated list, filters, groupings and four display modes.

View	What it shows	When to use it
List	Movements line by line, signed amount	Control, search, export
Kanban	Cards grouped by member, activity, type or month	Quick reading, distribution
Chart	Inflow / outflow bars by group	Comparison, trend
Grid	Cross-tab of groups × types, net	Analysis, report

12.3 Movement types

Type	Direction	Source
Contribution	Inflow	Session / activity
Loan disbursed	Outflow	Approved request
Principal repayment	Inflow	Loan instalment
Interest	Inflow	Repayment, shared pro rata
Beneficiary	Outflow	Tontine session
Social aid	Outflow	Request approved and paid

12.4 Reference figures

On HORIZON, the global history returns 144 movements: 101 579 832 FCFA of inflows (8 000 000 in capital, 88 360 001 in principal repaid, 5 219 831 in interest) and 100 360 001 FCFA of outflows (loans disbursed), i.e. a net balance of 1 219 831 FCFA. These figures are computed from the movements, not keyed in.

```
net balance = Σ inflows − Σ outflows
interest = Σ interest on repayments
member share = Σ (interest × capital share)
```

12.5 Interest detail

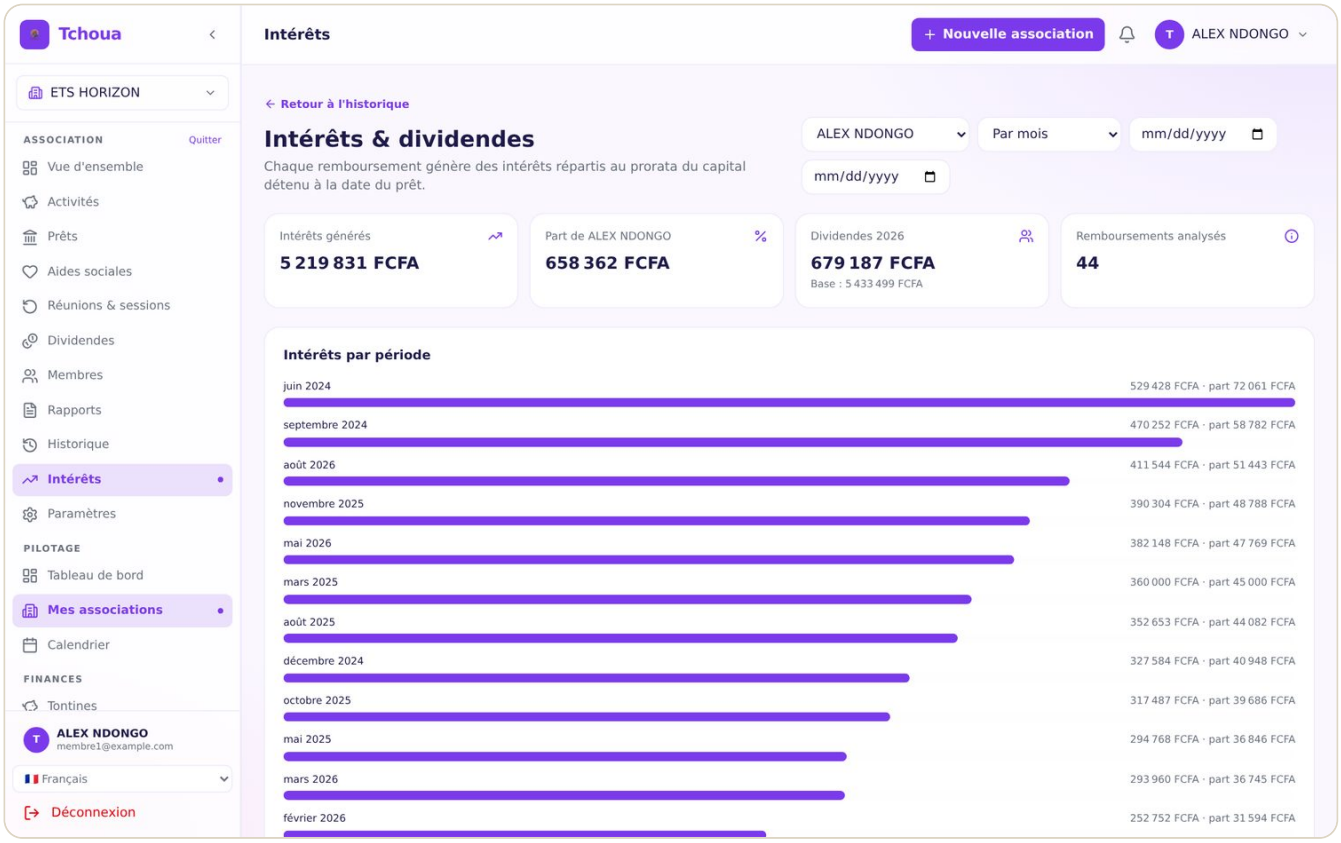


Figure 12.2 — Interest: total generated, member share, dividends, distribution by period and detail by repayment.

The board can open each line to see how it is made up: total interest on the repayment, the member's capital at the loan date, total capital, share and amount allocated. This is the proof of the distribution, not only its result.

12.6 Pagination, filters and export

- Server-side pagination (25 rows by default, up to 100).
- Combinable filters: type, member, activity, start date, end date, text search.
- Dynamic grouping: member, activity, type, month.
- Totals recalculated over the entire filtered set, not only over the page.
- CSV/PDF export for reports and auditors.

12.7 Transparency and privacy

The association's global history is visible to members, in line with the transparency promise. Personal data, medical documents and contact details, however, are not exposed. An ordinary member sees their own transactions in detail; the board sees everything within its remit.

> **INFORMATION**
>
> Visibility is computed on every request: it depends on the effective role in the association, not on a display setting. A member whose role changes immediately sees their scope change.

CHAPTER 13

Member area, themes and mobile

The member area is designed as a personal cockpit: it shows what matters, lets you act in two clicks and adapts to the chosen language and theme.

13.1 Structure of the area

Zone	Content
Top bar	Breadcrumb, notifications, user menu, association selector
Side navigation	Steering, finances, community, analysis, account; association context menu
Launchpad	12 modules with direct access + quick actions
Content	Dashboard, activities, history, reports, settings
Footer	Version, help, contact, status

13.2 The six themes

Each member chooses their theme in Settings → Appearance & language. The theme is applied immediately and saved on the account. The six themes cover different tastes and contexts of use, from the most sober to the most contrasted.

Theme	Mood	Use
Forest	Deep green, cream, gold	Historic, institutional theme
Midnight	Night blue, cyan	Night work, contrast
Aurora	Lavender, purple, white	Soft, modern
Ocean	Blue-green, sand	Balance, legibility
Sand	Beige, brown, terracotta	Warm, traditional
Neon	Black, magenta, acid green	Maximum contrast, demonstration

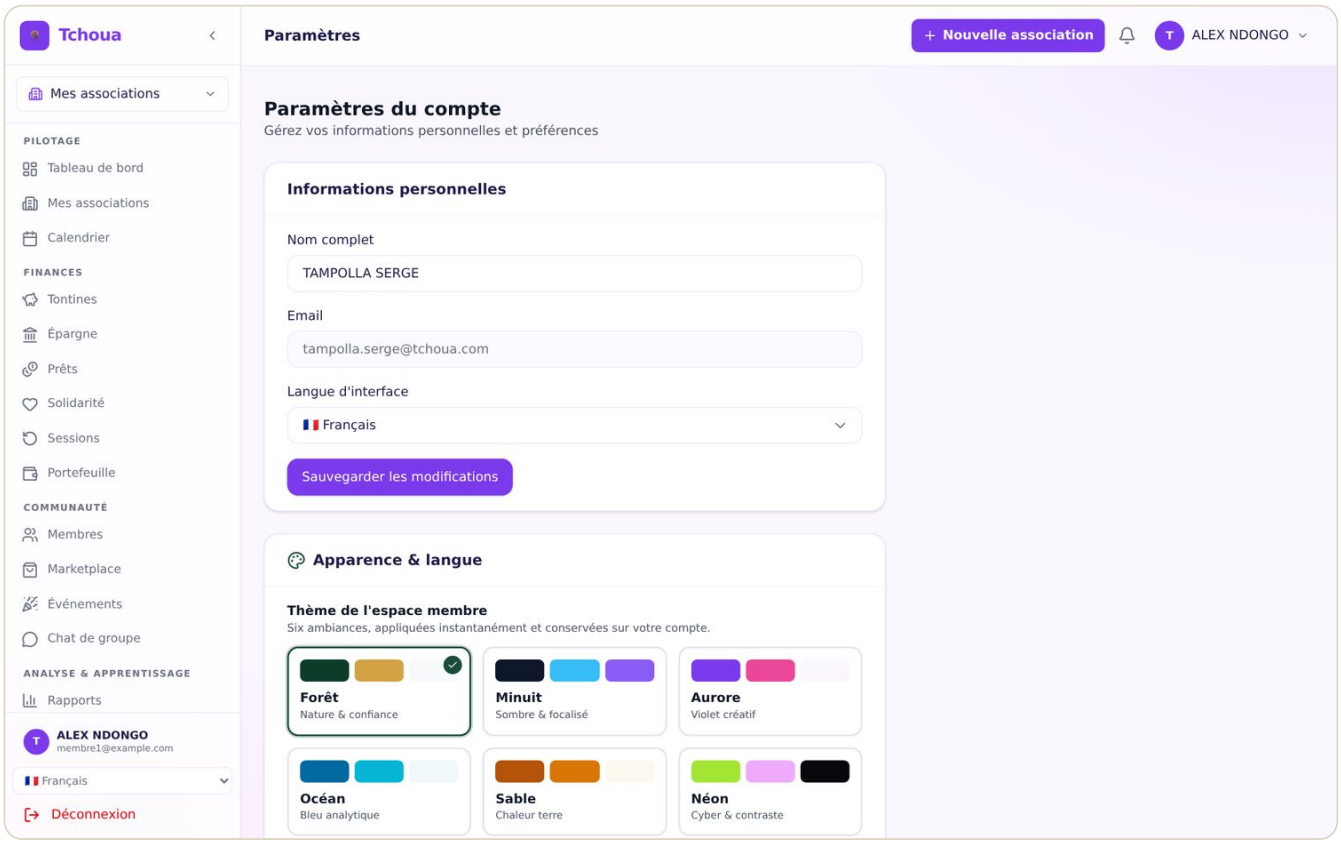


Figure 13.1 — Settings: personal information, language and theme selector with immediate preview.

13.3 Navigation and launchpad

The launchpad groups modules by use and offers three quick actions: contribute, apply for a loan, create an association. On mobile it becomes a grid of touch cards; on desktop, a grid of cards with icons and subtitles.

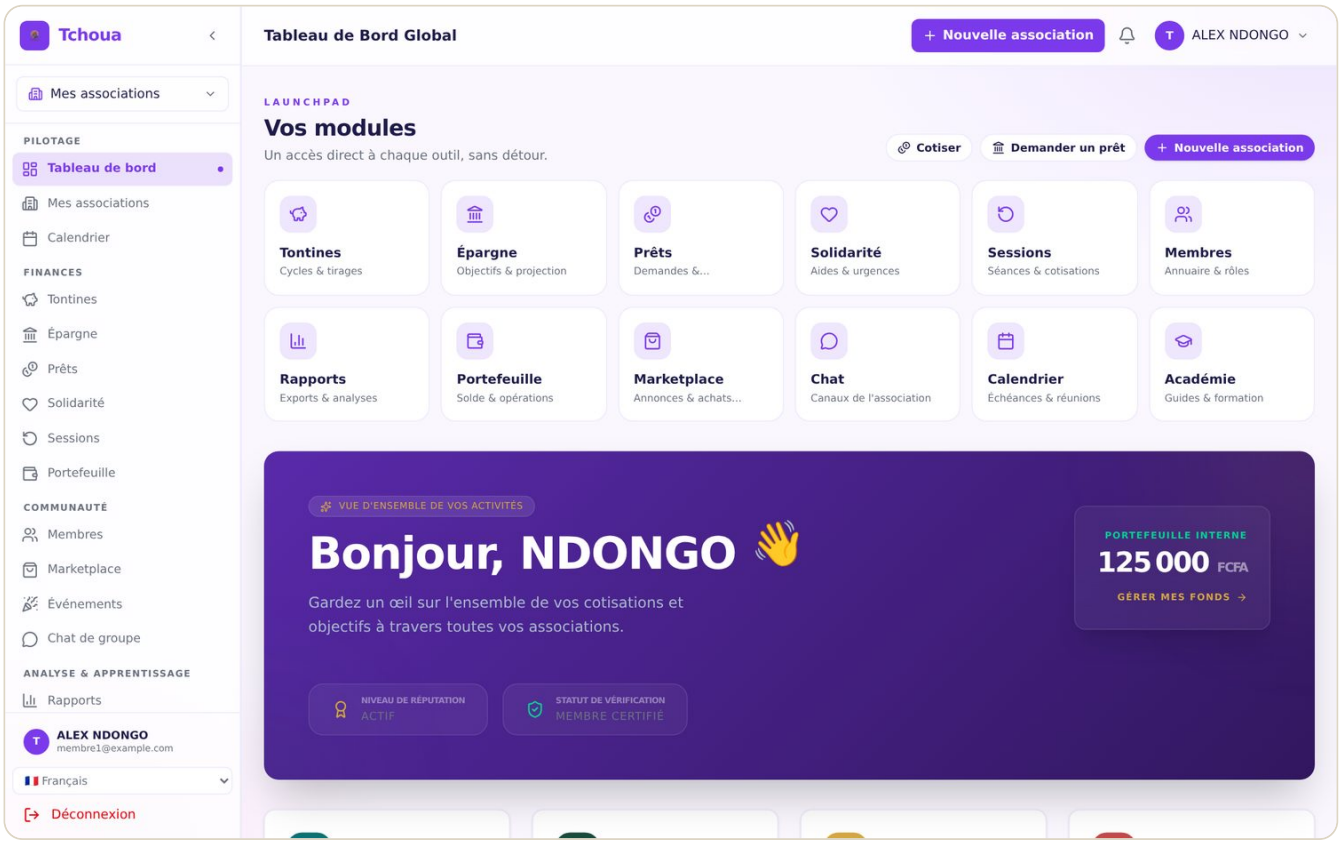


Figure 13.2 — Dashboard: personalised hero, indicators, launchpad and shortcuts. Tone and colours follow the active theme.

13.4 Mobile

The application can be used on a phone: compact navigation, stacked cards, actions within thumb reach, scrolling tables. The key screens (dashboard, tontines, loans, history) are tested at 390 × 844.

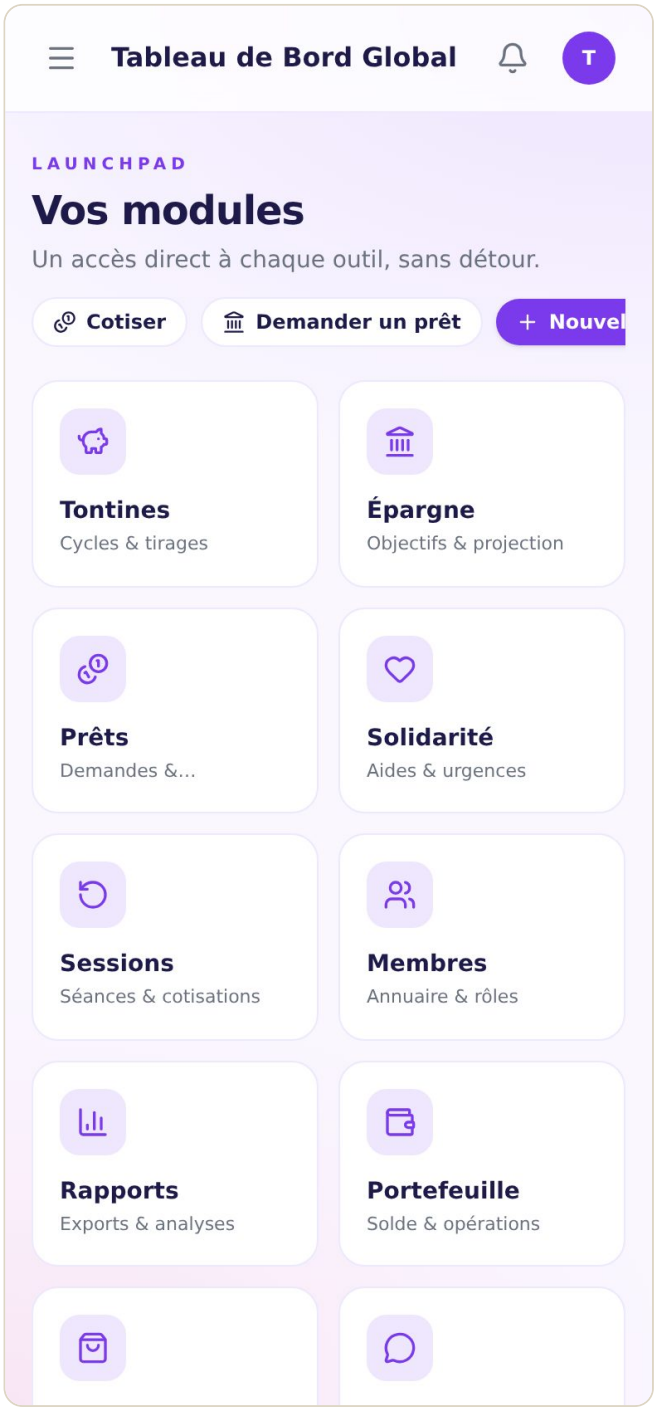


Figure 13.3 — Mobile launchpad.

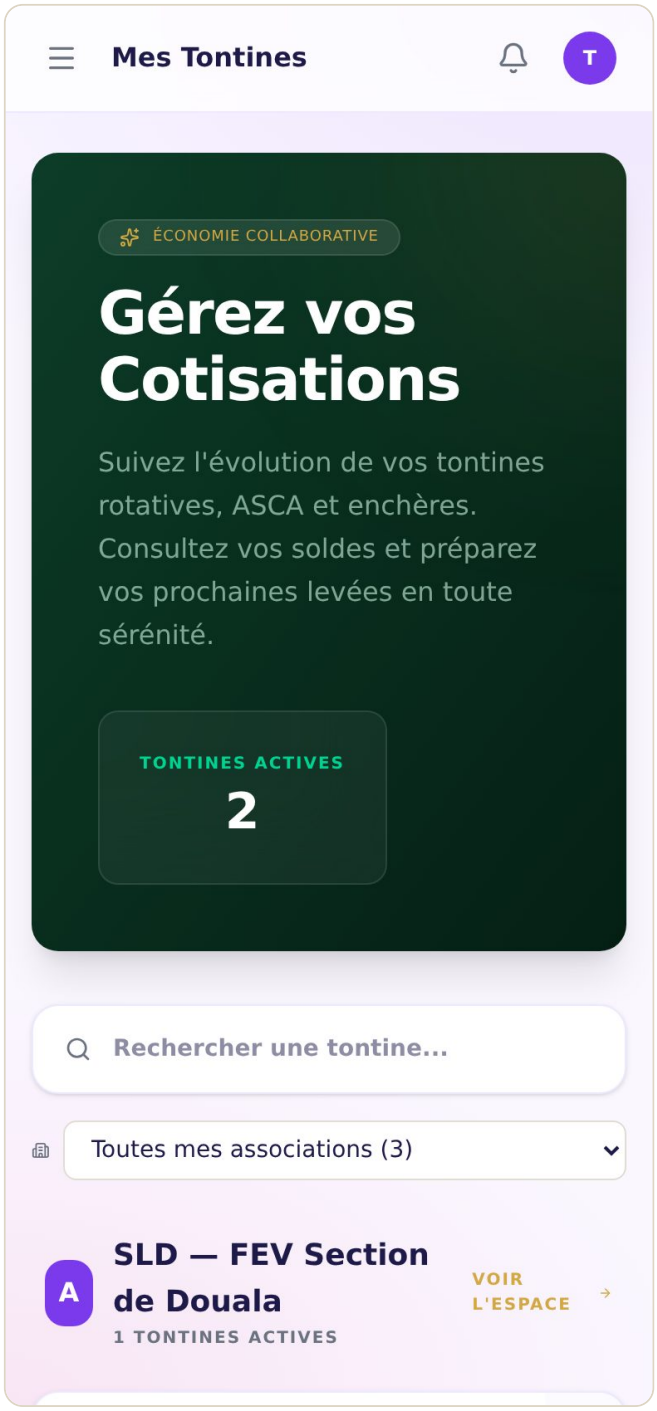


Figure 13.4 — Mobile tontine list.

13.5 Language and accessibility

The language selector is available in the user menu and in settings. Public pages accept `?lang=fr|en|es|de`. Labels, dates and amounts follow the locale; currencies are formatted with separators and symbol.

- Contrast checked on all six themes, including primary buttons.
- Font sizes legible on mobile and in print.
- Keyboard navigation on forms and buttons.

- Explicit error messages, never a technical code on its own.

13.6 Notifications

The member receives notifications for contribution due dates, upcoming sessions, decisions on their requests, repayments and board announcements. Channels (in-app, e-mail, WhatsApp, SMS) are configured by the platform and by the association.

> **PRINCIPLE**
>
> The member area must never force anyone to guess: every amount displayed points to a movement, every movement points to a session, every session points to an activity.

13.7 Support and digital inclusion

Not all members have the same digital confidence. The member area offers explicit labels, educational empty states and guided actions. Boards can organise onboarding sessions and use the screenshots in this manual as training material.

- Onboarding in under ten minutes for a member who knows how to use WhatsApp.
- Critical actions ask for confirmation and display a summary.
- Errors are explained in plain language, with the expected correction.
- Data remains viewable even when the association is suspended.
- A member can ask their board for help without leaving the application.

CHAPTER 14

Administration and templates

The system console drives the platform, not the associations. It manages reusable templates, system users, roles, global settings and maintenance.

14.1 System dashboard

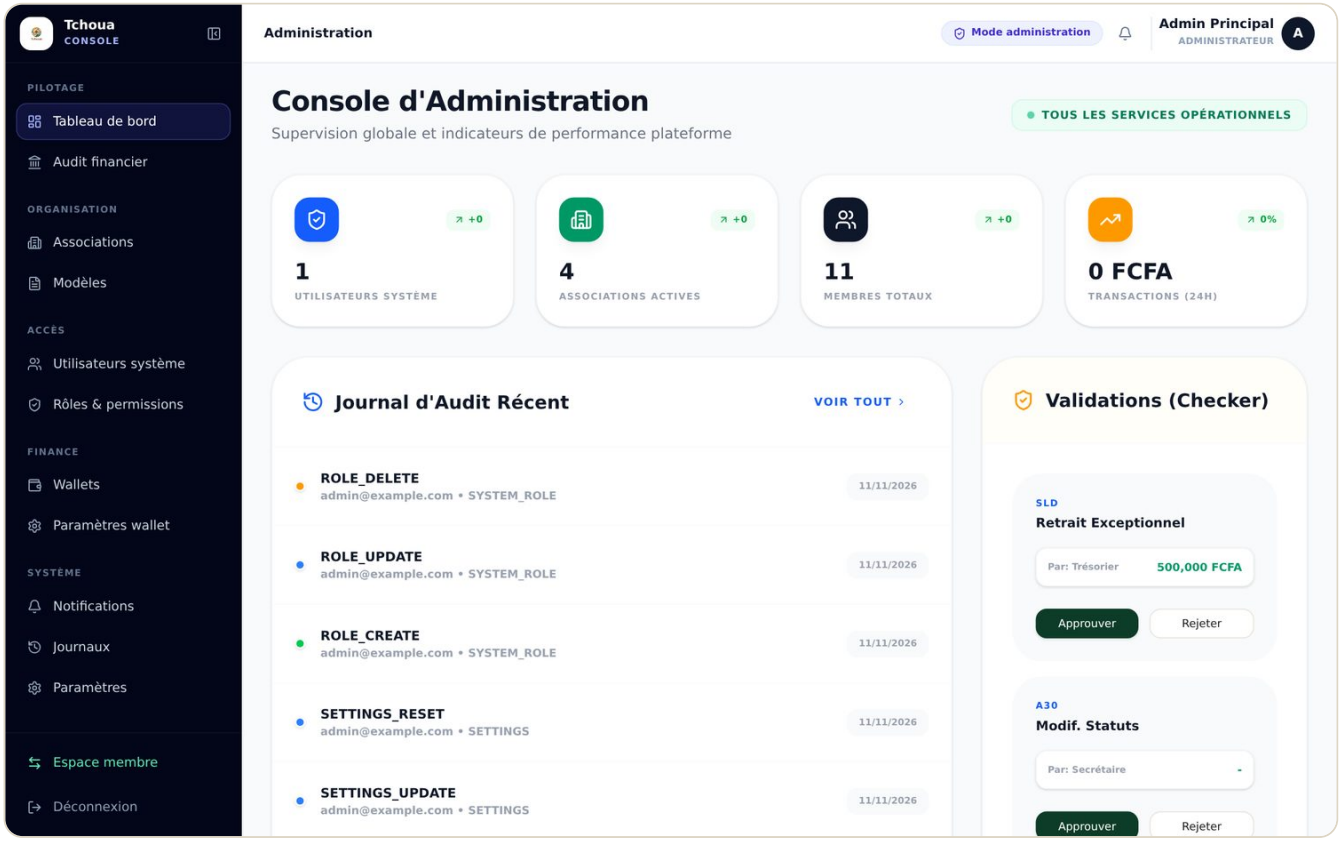


Figure 14.1 — Administration console: associations, users, templates, wallets, settings and recent audit log.

The dashboard shows the number of associations, users, templates and wallets, the latest audit actions and pending requests. It gives a quick diagnosis of the platform's state.

14.2 Association templates

A template contains a name, a category, a description, model internal rules and a list of preconfigured activities. It serves as a starting point when creating an association; the creator can then adjust each parameter.

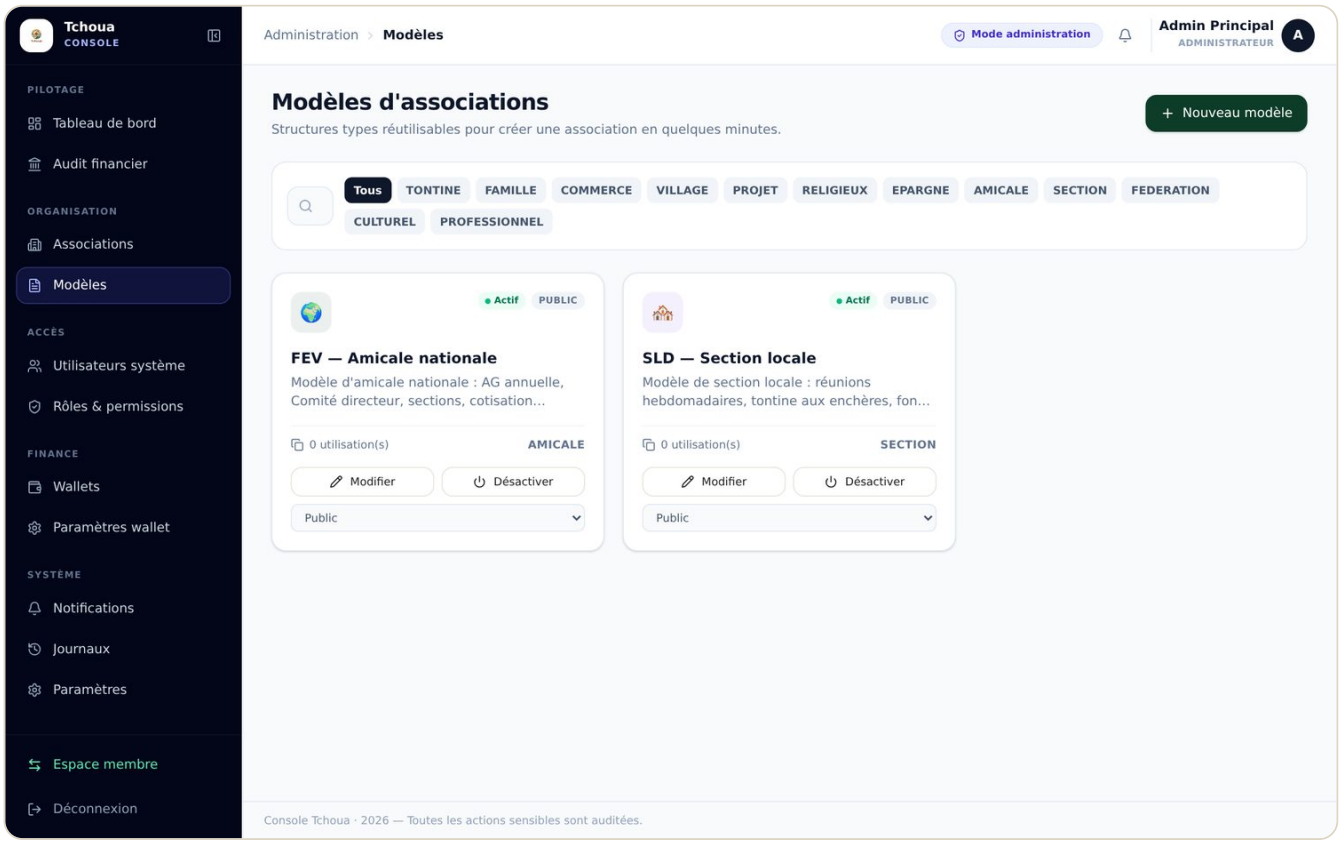


Figure 14.2 — Templates: FEV (national friendly society), SLD (local branch), categories tontine, family, business, village, project, religious, savings, friendly society, branch, federation.

Template	Category	Typical activities
FEV — National federation	AMICALE	Membership, annual contribution, national fund, illness aid, bereavement aid
SLD — Local branch	SECTION	Auction tontine, solidarity, investment, catering, three aid funds
Neighbourhood tontine	TONTINE	Rotating tontine, catering
Loan fund	ÉPARGNE	Investment fund, loans, solidarity
Village association	VILLAGE	Contribution, projects, solidarity, meetings

14.3 Users and system roles

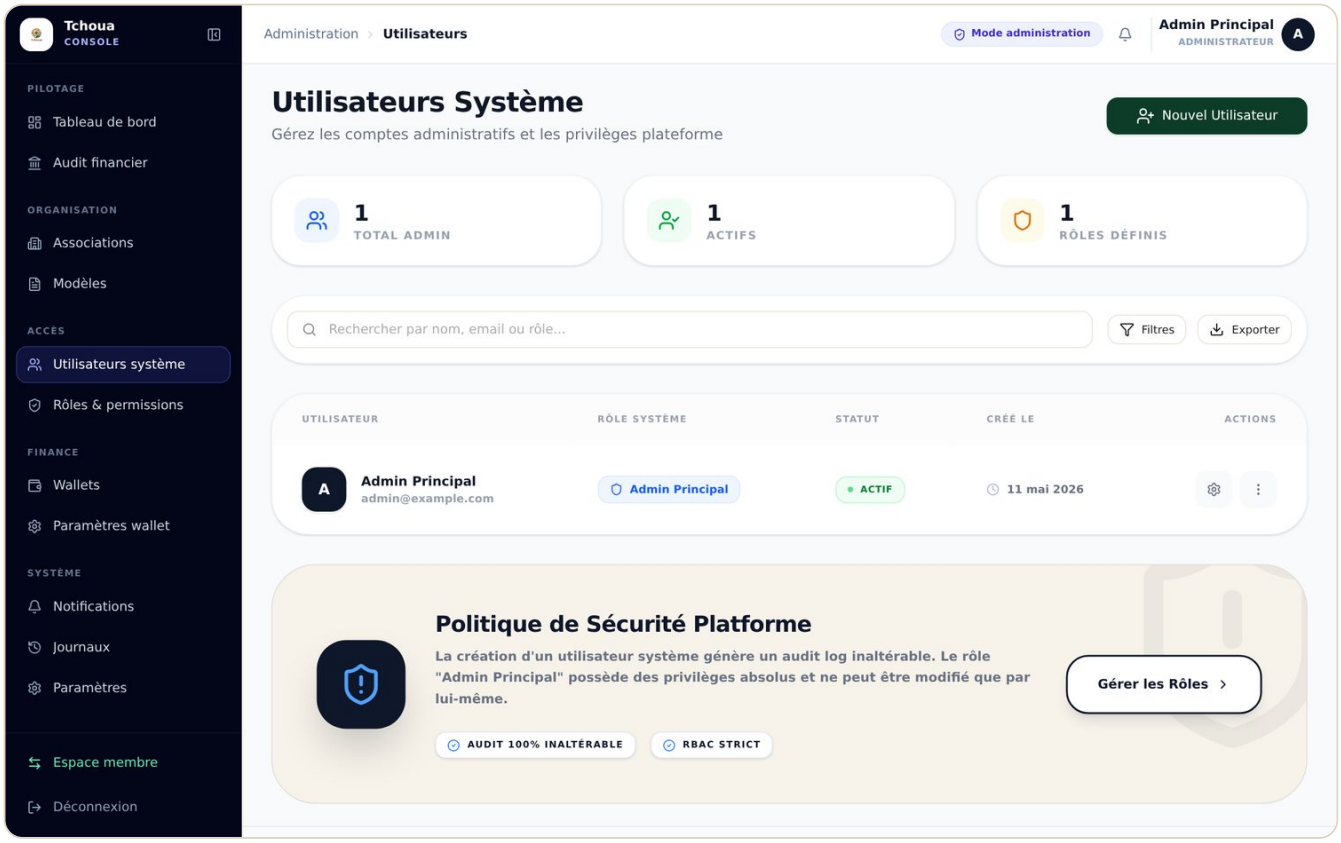


Figure 14.3 — System users: accounts, roles, status, last access and security actions.

The administrator creates accounts, assigns a system role, resets a password or disables an account. System roles combine permissions per resource and per action (CREATE, READ, UPDATE, DELETE). The "Principal Admin" role bypasses restrictions; the other roles are limited to their permissions.

14.4 Supervised associations

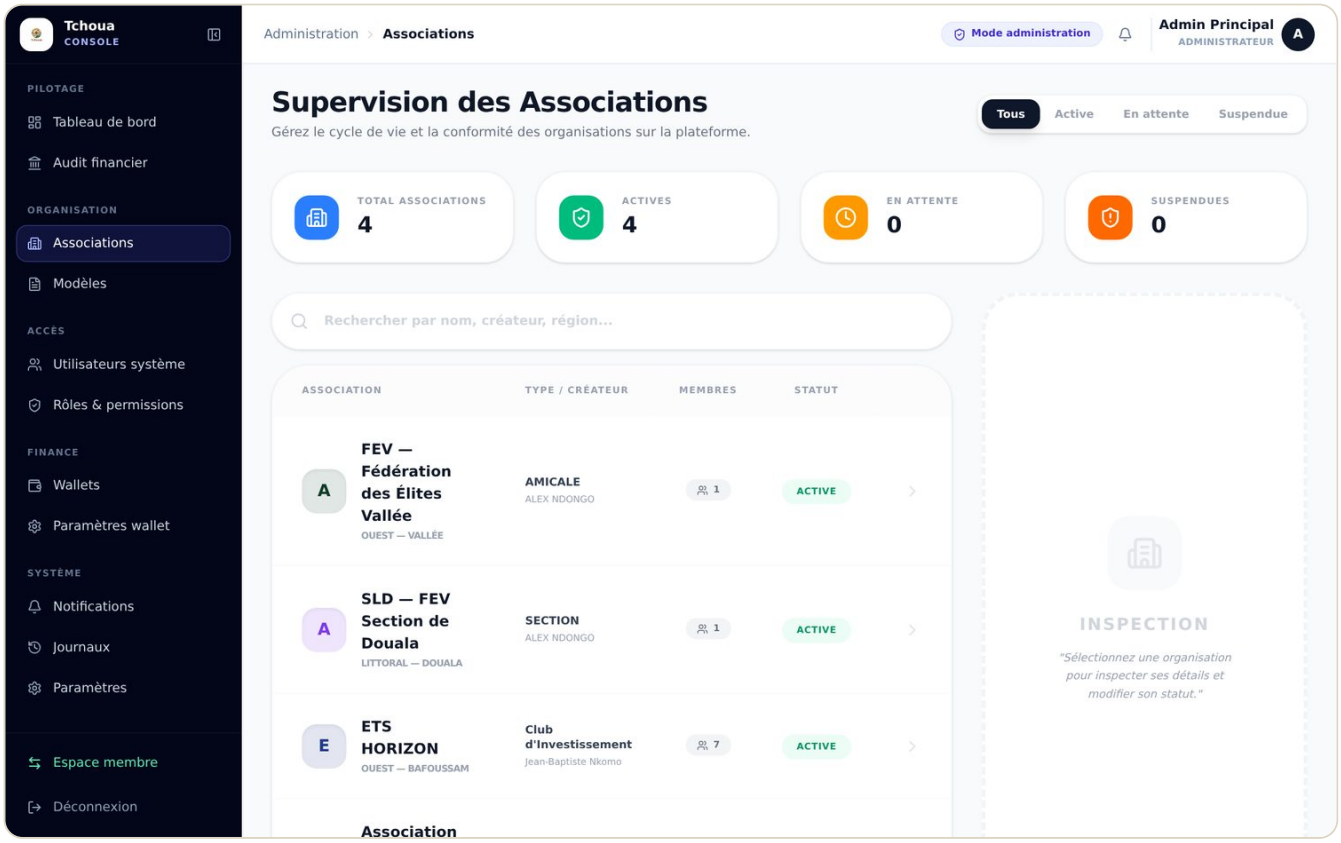


Figure 14.4 — Supervision: list of associations, type, status, members, activities and access to the record.

The console allows an association to be suspended, reactivated, its status changed or its data viewed. These actions are logged and do not alter business entries.

14.5 Platform settings

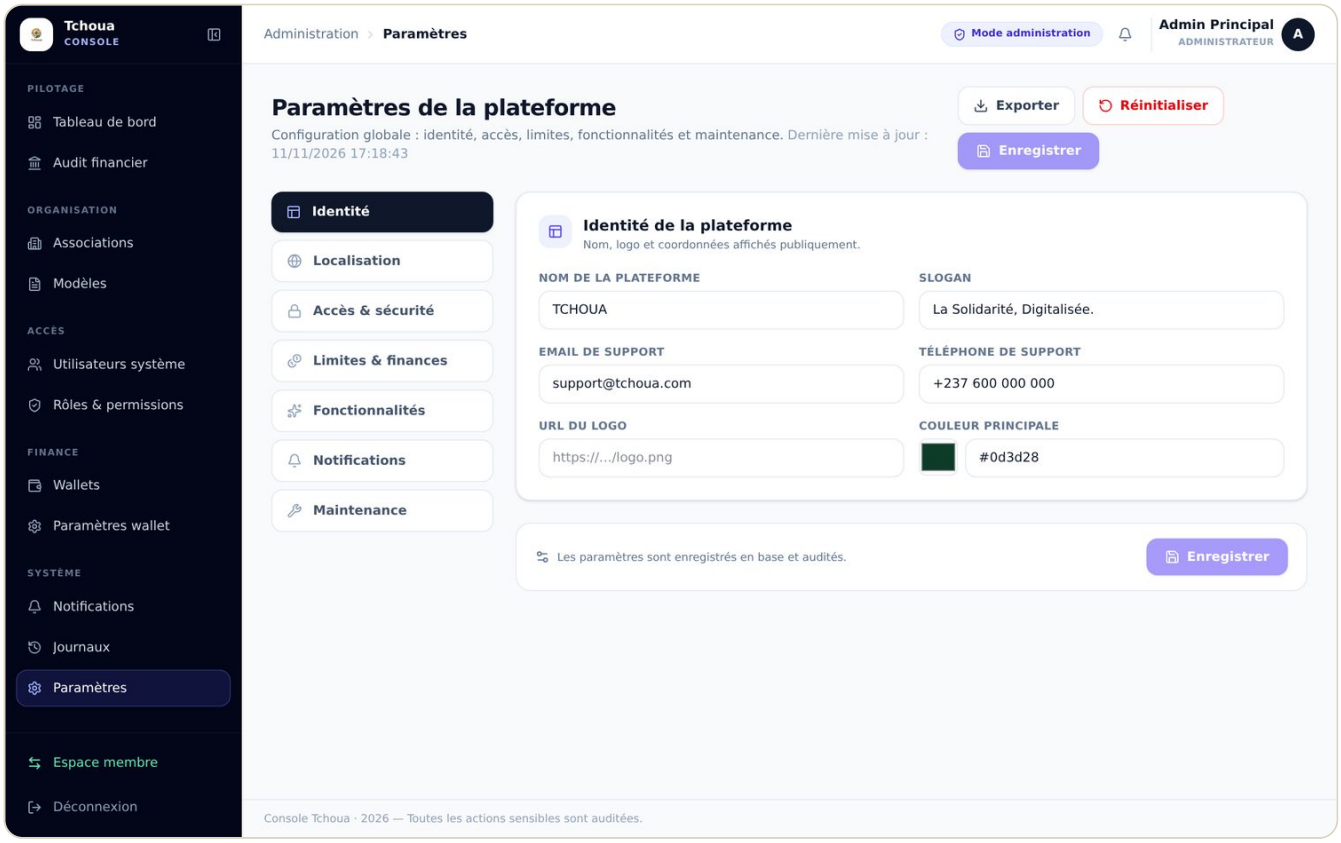


Figure 14.5 — Settings: name, currency, country, contact, limits, features, maintenance, registration.

Global settings define the platform's identity, the default currency, the limits (number of associations, members, activities), the enabled features, maintenance mode and whether registrations are open. They are stored in the database and applied immediately.

14.6 Audit log

Action	Actor	Data
Association created / modified	Administrator or creator	Before / after, date, IP
Role change	Administrator	Previous role, new role
Password reset	Administrator	Target account, date
Settings modified	Administrator	Key, old value, new value
Suspension / reactivation	Administrator	Reason, date

14.7 ARP reference framework and coverage

The **Organisation → ARP reference framework** entry gives access to the 36 domains of the ARP study, the 144 invariants, the C01–C06 conditions and the catalogue of 23 328 combinations. The same console shows each association's

coverage: framework status, origin (manual, imported, template), version, completeness, errors, alerts, active modules and number of funds.

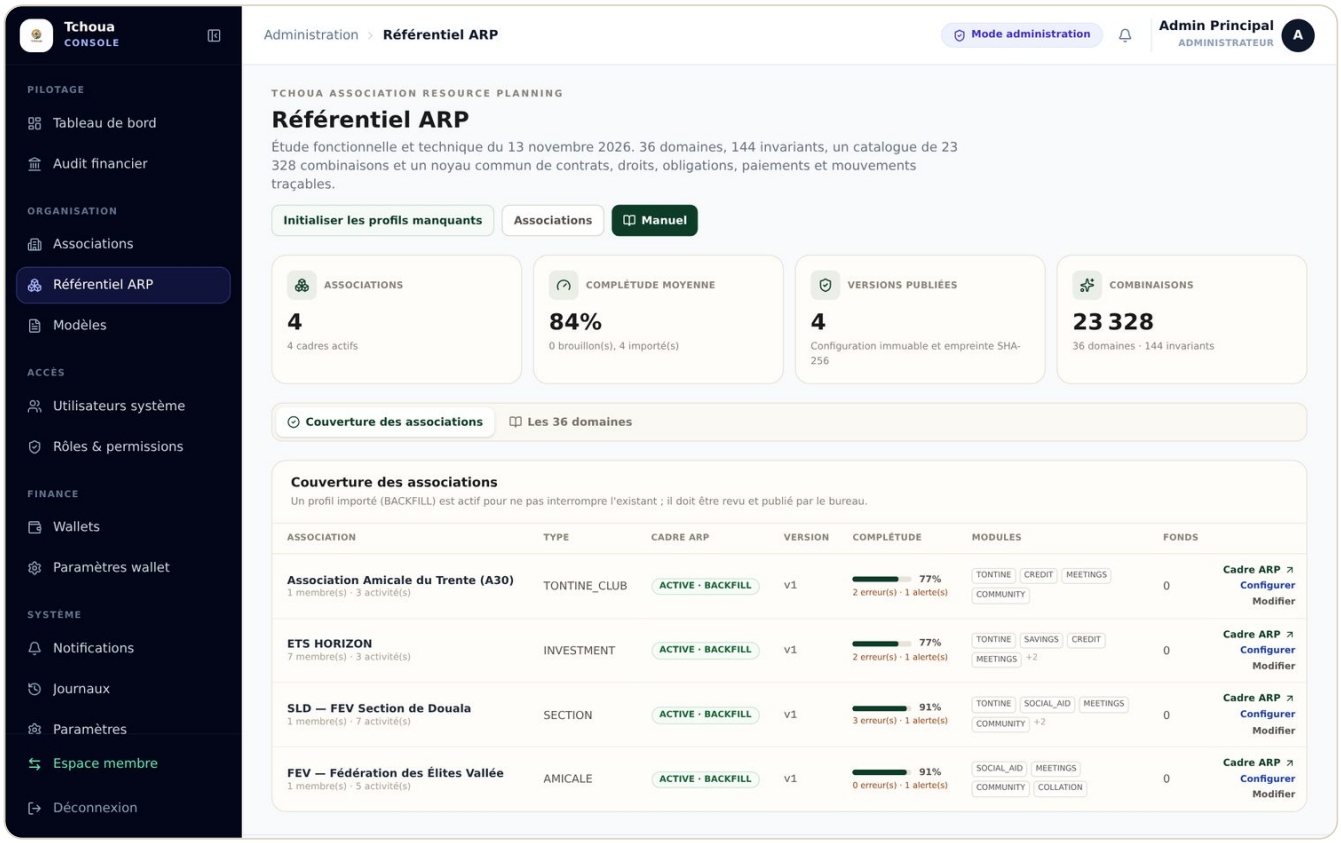


Figure 14.6 — ARP coverage: the four associations, their version, their completeness and their modules. Imported profiles remain flagged as needing validation.

The "Initialise missing profiles" button creates an `ACTIVE / BACKFILL` profile for associations that do not yet have one. This profile does not block existing operations; it must be reviewed and then published by the association's board.

14.8 Maintenance and backups

- Daily backup of the PostgreSQL database.
- Export of critical documents and reports.
- Process and TLS certificate monitoring.
- Idempotent migrations before each deployment.
- Deployment and build log.
- Rollback procedure per commit.

> **WARNING**
>
> The system console must not be used to correct business transactions. A correction to a contribution, a loan or an aid is made inside the association, with a traceable movement and the board's agreement.

CHAPTER 15

Multilingual and international

Tchoua targets French-speaking, English-speaking and other African markets. The interface, content, currencies and rules must be able to change country without rewriting the application.

15.1 Languages

Language	Code	Coverage
French	fr	Full interface, public site, documents
English	en	Public site, member interface (progressive rollout)
Spanish	es	Public site
German	de	Public site
Local languages	—	Planned extensions (Douala, Bassa, Ewondo, etc.)

The language selector is available in the user menu, in settings and on the public site. Translations are centralised in locale files; no text should be hard-coded in components.

15.2 Currencies and formats

Amounts are stored as numeric values and displayed in the association's currency. Formatting respects separators, symbol and number of decimals. Conversion rates and multiple currencies are planned for international federations.

Currency	Code	Use
CFA franc (CEMAC)	XAF	Cameroon, Gabon, Congo, Chad, CAR, Equatorial Guinea
CFA franc (UEMOA)	XOF	Senegal, Côte d'Ivoire, Benin, Togo, Mali, Burkina, Niger
Naira	NGN	Nigeria
Cedi	GHS	Ghana
Euro / Dollar	EUR / USD	Diaspora, partners

15.3 Legal adaptation

Each country has its own law on associations: in Cameroon, Law No. 90/053 of 19 December 1990; in the OHADA area, common accounting rules; elsewhere, specific regimes. Tchoua does not claim to replace a lawyer: the platform provides templates and parameters, and the association validates its articles with its advisers.

- Template articles and internal rules per country (to be expanded).
- Declaratory obligations and statutory registers.
- Holding of assemblies, quorums and archives.

- Protection of personal data.

15.4 Payments and mobile money

Payment methods are configurable per association: cash, MTN MoMo, Orange Money, Wave, bank transfer, in-kind, time. Payment integrations must comply with local operators and regulators; the platform prepares the entry points and the reconciliation.

Method	Use	Status
Cash	Meetings, rural areas	Operational
MTN MoMo	Cameroon, Ghana, Uganda…	Configurable, integration to be completed
Orange Money	French-speaking Africa	Configurable, integration to be completed
Wave	Senegal, Côte d'Ivoire	Planned
Bank transfer	Formalised associations	Operational (entry + reconciliation)
In-kind / time	Cooperatives, time banks	Operational (valuation)

15.5 International roadmap

1. Stabilise French and English across the full interface.
2. Add local currencies and formats.
3. Integrate mobile money operators country by country.
4. Build the legal library country by country.
5. Offline mode and synchronisation for areas with poor connectivity.
6. USSD and SMS for members without a smartphone.

> **INFORMATION**
>
> The public site is already available in French, English, Spanish and German. The member interface uses the same translation foundation; business screens are translated progressively.

CHAPTER 16

Real cases: HORIZON, FEV, SLD

Three associations, three different models, one platform. HORIZON illustrates investment and lending; FEV the federating national friendly society; SLD the local branch with an auction tontine.

16.1 ETS HORIZON — investment and loans

HORIZON is a seven-member association that manages capital, loans and dividends. The historical data has been imported and verified: 49 loans, 5 active, 44 repaid, 5 433 499 FCFA of 2026 dividends.

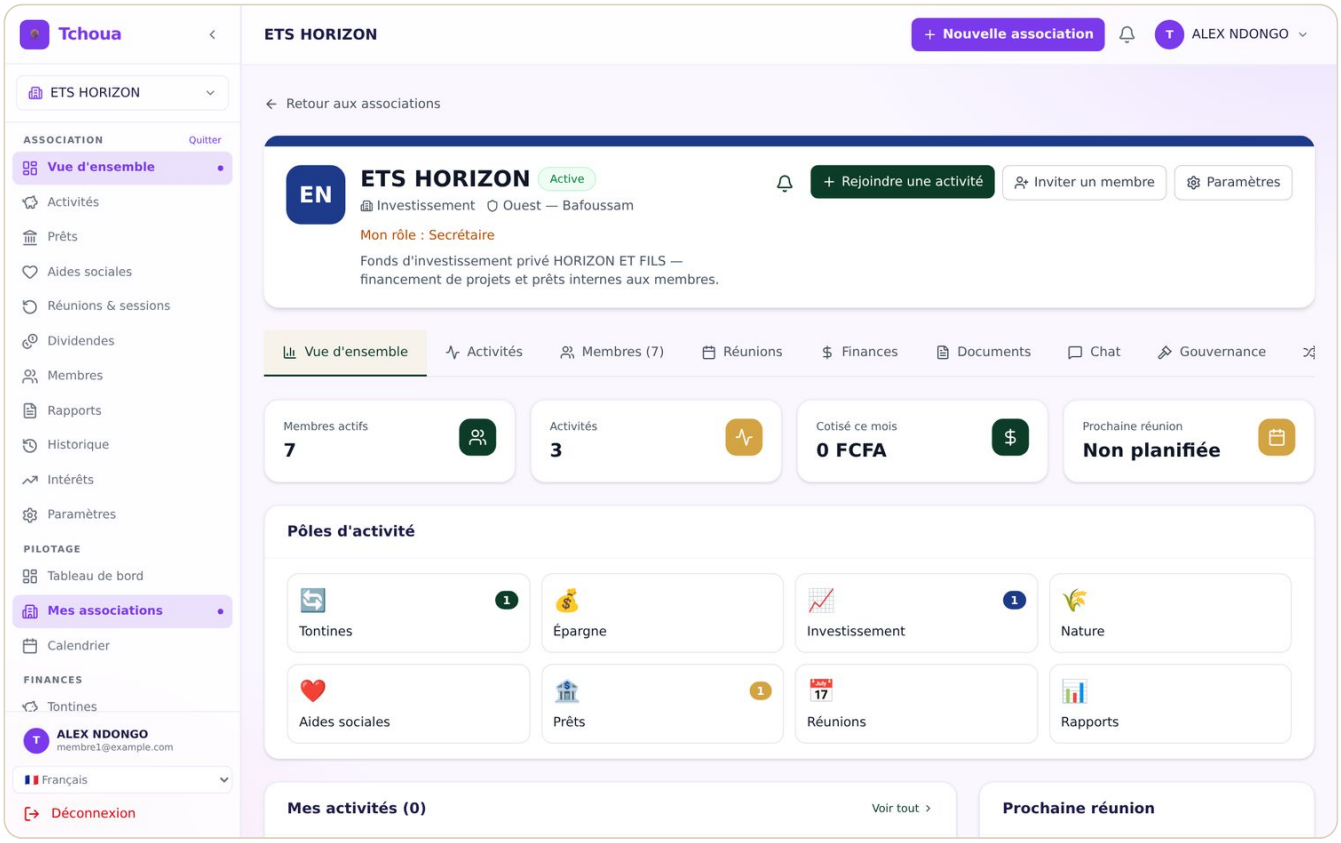


Figure 16.1 — ETS HORIZON record: 7 members, 2 activities, capital, cash box, loans, dividends and management shortcuts.

Item	Amount / number
Members	7
Capital / contributions	8 000 000 FCFA
Loans disbursed	49 — 100 360 001 FCFA
Loans outstanding	5 — 12 000 000 FCFA
Principal repaid	88 360 001 FCFA
Interest	5 219 831 FCFA
2026 dividends	5 433 499 FCFA
Loans / funds ratio	150 % (configurable)

16.2 FEV — national friendly society

FEV (Fédération des Élites de la Vallée) is an elite friendly society governed by the 1990 law. It federates branches, holds an annual GA, a half-yearly steering committee, an executive board and committees. Its resources come from membership fees (10 000 FCFA), annual contributions (15 000 FCFA), investments and fines.

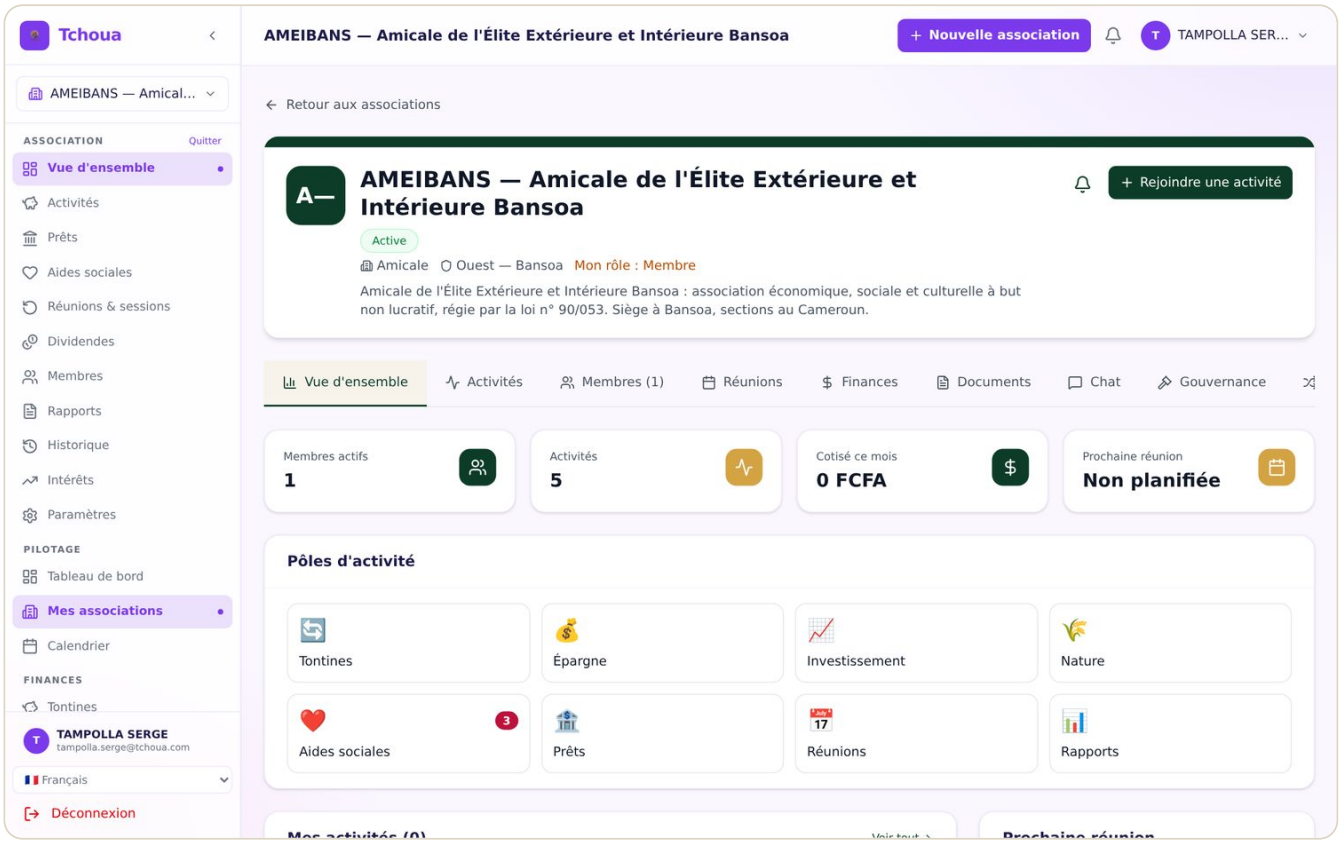


Figure 16.2 — FEV: national friendly society, 1 attached member, 5 activities, 14 articles, 3 founding documents.

Parameter	Configured value
Type	AMICALE
GA	Annual, January–February, 45 days' notice, quorum 1/2
Steering committee	Half-yearly, decisions 2/3, quorum 2/3
Board	3 years, renewable once, unpaid duties
Membership fee	10 000 FCFA
Annual contribution	15 000 FCFA
National fund	10 000 FCFA per member; 300 000 on death
Illness aid	10 000 FCFA/day, 3 to 15 days
Branches	Minimum 10 members; delegates 10 % (cap 11)

16.3 SLD — Douala branch

SLD (Section Locale de Douala) is the Douala branch of FEV. It applies the national articles and its own 2024 internal rules: meetings on the first Saturday after the 5th, quorum 3/5, auction tontine of 25 000 FCFA, solidarity fund of 72 000 FCFA/year, investment fund of 5 000 FCFA/month, catering contribution of 3 000 FCFA and family aid.

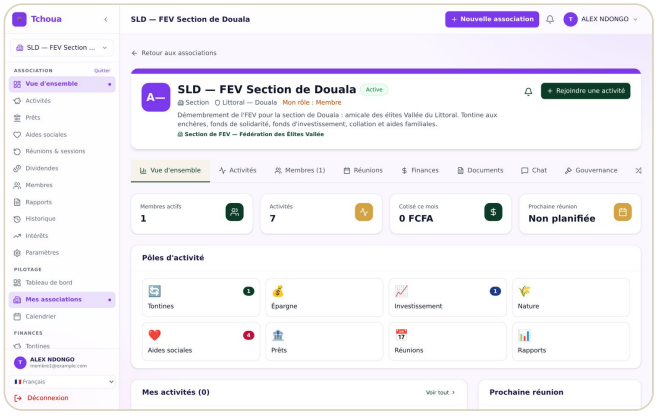


Figure 16.3 — SLD, Douala branch attached to FEV.

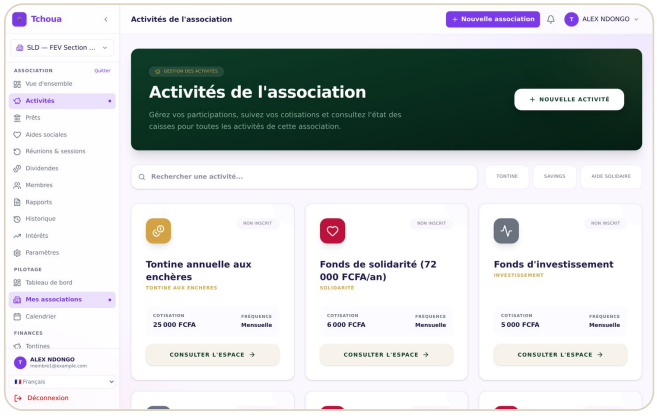


Figure 16.4 — Seven SLD activities.

Activity	Amount	Frequency / rule
Annual auction tontine	25 000 FCFA / share	Monthly, 12 sessions, reserve price 5 %
Solidarity fund	6 000 FCFA/month	72 000 FCFA/year
Investment fund	5 000 FCFA/month	Loan 3 %/month, 2 months, cap 2×
Catering	3 000 FCFA/session	Flat rate 150 000 FCFA
Illness aid	70 000 / 50 000	Member / spouse
Family aid	50 000 / 20 000	Marriage / birth
Bereavement aid	900 000 / 405 000	Member / spouse

16.4 Comparison

Dimension	HORIZON	FEV	SLD
Nature	Investment / loans	National friendly society	Local branch
Level	Primary association	Federation	Branch
Members	7	National	Douala branch
Main contribution	Capital contributions	15 000/year	25 000 tontine + 6 000 solidarity + 5 000 FI
Flagship activity	Loans with interest	National fund and aid	Auction tontine
Reports	2026 dividends	GA, committee, committees	CDS, catering, penalties

LESSON

The same engine serves three radically different models. That is the proof of generalisation: the objects (association, member, activity, session, movement) are common; the rules are parameters.

CHAPTER 17

End-to-end workflows

This chapter chains screens and rules together for the most frequent journeys. Each step states the actor, the action, the control and the trace produced.

17.1 Creating an association and its board

#	Actor	Action	Control / trace
1	Creator	Chooses the template, type, name, region, colour	Unique name, valid type
2	Creator	Optionally attaches it to a parent association	No hierarchical cycle
3	Creator	Configures membership, meetings, aid, penalties	Consistent values
4	Creator	Appoints the board and roles	One president, one treasurer
5	System	Generates the invitation code and activates the association	Lifecycle event

17.2 Enrolling a member

1. The candidate creates an account or receives an invitation.
2. They join the association by code, link or request.
3. The board checks the conditions, the sponsorship and the documents.
4. The membership is validated: role, member number, join date.
5. The member appears in the directory, the contribution calls and the reports.

> INFORMATION
>
> A refused membership is kept with its reason. Removal does not delete the history: it changes the status and closes the entitlements.

17.3 Opening a session and collecting contributions

#	Action	Result
1	Create the session (date, type, venue, agenda)	Session scheduled
2	Open the session	Expected contributions generated
3	Record attendance	Statuses and fines
4	Collect contributions	PAID / PARTIAL / LATE statuses
5	Collect the catering contribution	Catering cash box funded
6	Close the session	Frozen pot, minutes, report

17.4 Distributing a tontine

1. Check the pot and the distributable shares.
2. Apply the allocation rule (rotation, draw, auction).
3. Designate the beneficiary or beneficiaries and the number of shares.
4. Compute the amount: shares × amount per share.
5. Validate the distribution; the session balance is updated.
6. The beneficiary receives the amount; the history records it.

```
amount = beneficiary shares × amount per share
session balance = contributions − distributions
cycle balance = Σ session balances
```

17.5 Granting and tracking a loan

#	Actor	Action	Control
1	Member	Requests amount, purpose, term, guarantees	Cap, active loans
2	Board	Reviews, checks the cash position	Ratio, capacity, guarantees
3	Board / GA	Approval or reasoned refusal	Quorum, majority
4	Treasurer	Disbursement	Double validation
5	Member	Repayments of principal + interest	Schedule, receipt
6	System	Allocation of interest	Capital at the loan date

17.6 Awarding social aid

1. The member creates the request with category and supporting documents.
2. The board checks eligibility and the cap.
3. The decision is notified (approved, refused, further information required).
4. The disbursement is recorded with a reference.
5. The aid appears in the history, the report and the fund balance.

17.7 Closing a financial year

Step	Content	Owner
Inventory	Cash boxes, banks, loans, aid, contributions	Treasurer
Reconciliation	Discrepancies and reasoned corrections	Treasurer + auditors
Financial statements	Income statement, position	Board
Dividends	Allocation key, annual run	GA
Report	Activity report and financial report	President / treasurer
Archiving	Minutes, reports, exports	Secretary

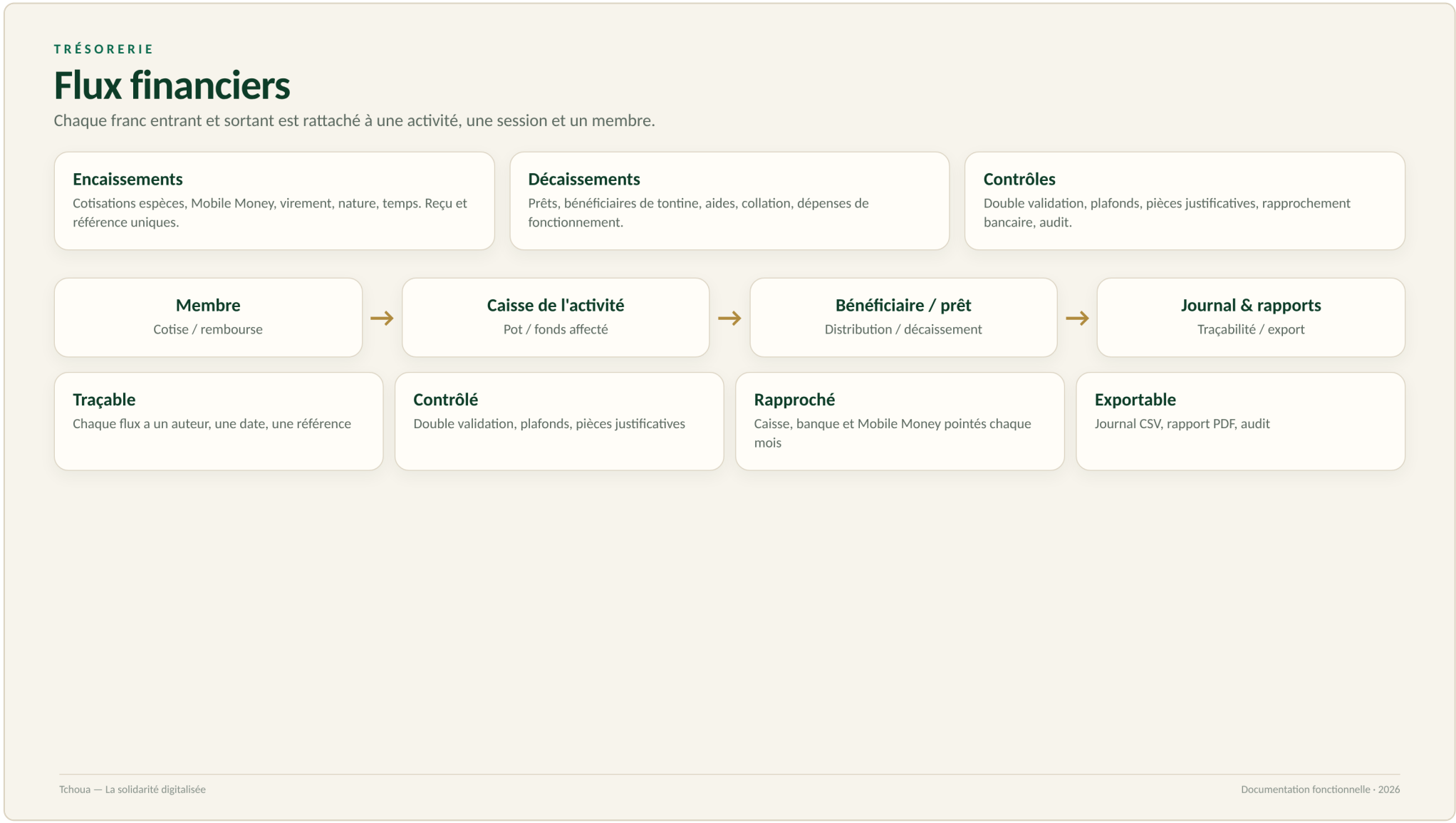


Figure 17.1 — Financial flows: collections, disbursements, controls and the member → cash box → beneficiary → journal path.

17.8 RACI matrix

Activity	Member	Board	Treasurer	Auditors	Admin
Contribute	R	A	R	I	—
Validate a membership	I	A/R	I	—	—
Disburse a loan	I	A	R	I	—
Award an aid	R	A	R	I	—
Close a cycle	I	A/R	R	C	—
Configure the platform	—	I	—	—	A/R

R = responsible for execution, A = approval authority, C = consulted, I = informed.

17.9 Handling exceptions

- **Session cancelled** — contributions remain, the distribution is cancelled by a reverse movement.
- **Data entry error** — corrected by an adjustment, never by deletion.
- **Departing member** — balance calculated, repayment or offset, LEFT status.
- **Dispute** — referred to the board, reasoned decision, appeal possible before the GA.
- **Connection loss** — validated entries remain; data entry resumes after reconnection.
- **Outage** — members can view, the board is informed; service resumes after restart.

CHAPTER 18

Roadmap, limitations and support

An honest document also states what is not yet done. This chapter presents the product trajectory, the current limitations and the support arrangements.

18.1 What is operational

- ✓ Multilingual public site and member access.
- ✓ Creation and management of associations, branches and federations.
- ✓ Membership, roles, directory, documents, internal rules.
- ✓ Tontines: shares, fractions, cycles, pot, beneficiaries, balances.
- ✓ Contributions, sessions, attendance, fines, catering.
- ✓ Loans, repayments, interest, dividends.
- ✓ Solidarity and the aid catalogue.
- ✓ History, reports, PDF/CSV exports, audit.
- ✓ Themes, language, mobile, launchpad.
- ✓ System console, templates, roles, settings.

18.2 Current limitations

Area	Limitation	Workaround / next step
Automatic designation of beneficiaries	The computation and the parameters are ready; automatic allocation based on eligibility and auctions remains a workflow to be finalised	Manual, traceable designation in the meantime
Federation consolidation	Hierarchy and branches operational; aggregated national dashboard still to come	Per-branch reports
OHADA accounting	Management exports and reports; full chart of accounts not integrated	CSV export to the accountant
Mobile Money	Payment methods configurable; operator integration to be finalised	Manual entry with reference
Offline / USSD / SMS	Not available	Responsive web application
Honorary distinctions	EPI — honorary distinction described in the texts, not modelled	Module to be added
Multi-country	Currency and language configurable; legal library by country to be expanded	Templates by country
Electronic signature	Not integrated	Minutes signed and uploaded

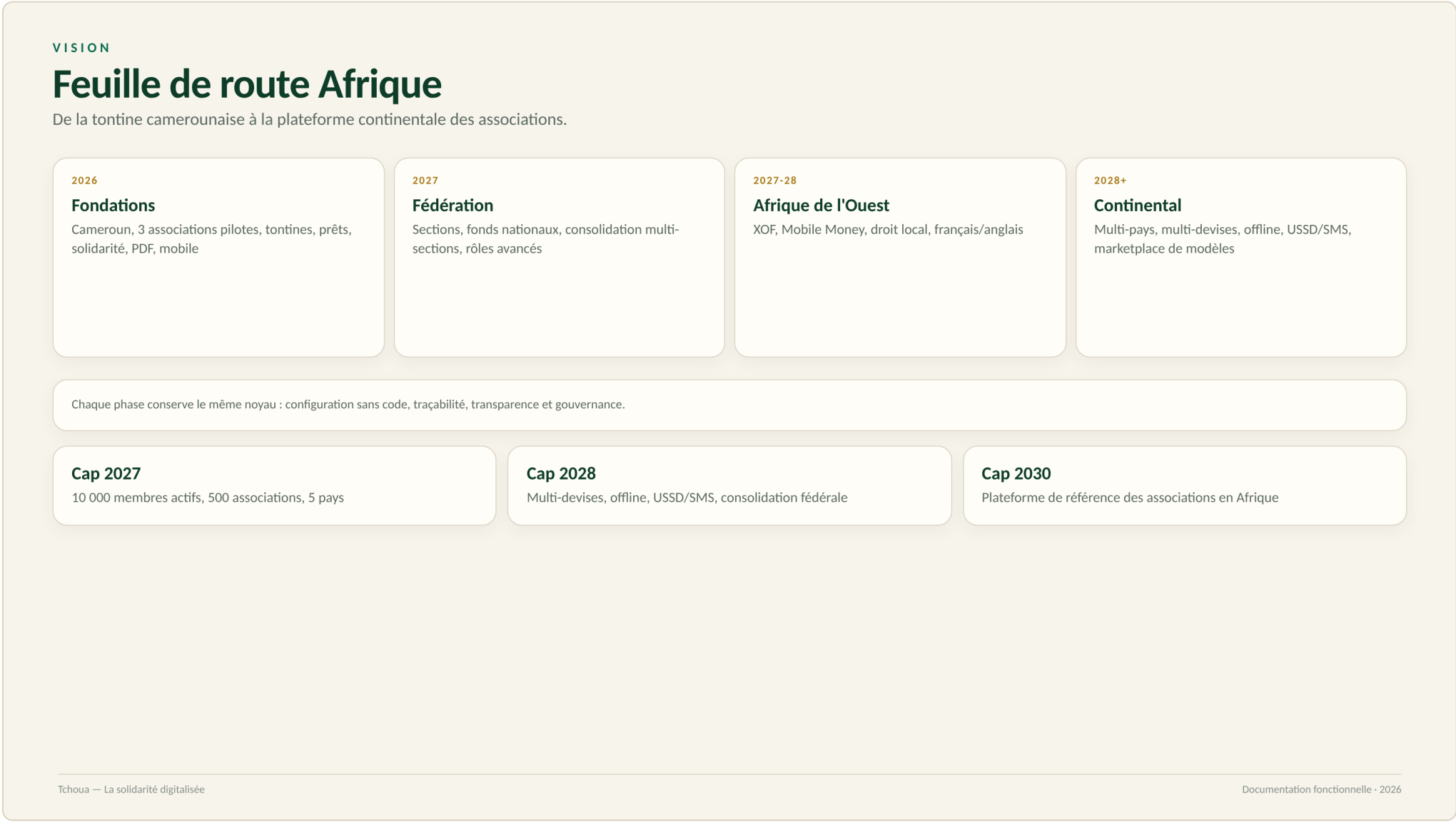


Figure 18.1 — Trajectory: Cameroon foundations, federation, West Africa, continental.

18.3 Security and compliance

The platform applies the principles of least privilege, traceability and separation of powers. Personal data is limited to what is necessary; medical documents and contact details are not exposed to other members. Backups and logs make it possible to reconstruct operations.

> **WARNING**
>
> No system replaces human vigilance: the board must check the reports, the auditors must examine the accounts, and members must verify their own movements.

18.4 Support and maintenance

Level	Channel	Target response time
Blocking incident	Priority support	4 business hours
Functional defect	Standard support	2 business days
Change request	Product committee	Monthly review
Training	Dedicated session	On schedule
Documentation	This manual + guides	Quarterly update

18.5 Document governance

This manual is the functional reference. Any product change must be reflected in the relevant chapter, with the version and date updated. Screenshots must be redone when the interface changes; diagrams can be regenerated by script.

- Version 1.0 — November 2026 — state of production.
- Quarterly review: features, figures, screenshots.
- Annual review: architecture, security, roadmap.
- Validation: product, operations, management.

> **CONCLUSION**
>
> Tchoua is not just tontine software: it is a way of making solidarity verifiable, governance readable and trust transferable. The platform is ready to serve; the documentation is there so that everyone can use it.

CHAPTER 19

ARP framework: rules, combinations and versions

Since November 2026, Tchoua no longer merely records activities: each association has an **ARP framework** (Association Resource Planning) that describes its form, its rules, its authorised modules and its compatibility conditions. This framework is versioned, checked and enforceable against new operations.

19.1 Why an ARP framework

A neighbourhood tontine, a family friendly society, a local branch and a national federation do not have the same bodies, the same funds or the same obligations. The ARP framework separates what must be separated:

- the **user account** and the **membership**;
- the **expected contribution** (receivable), the **payment** (treasury) and the **allocation** (imputation);
- the **fund** (authorised use) and the **account** (place of safekeeping);
- the **decision** to allocate and the **disbursement**;
- the **published rule** and the **historical operations**, which keep their version.

> **PRINCIPLE**
>
> The ARP framework does not replace the articles or the internal rules. It makes them **executable**: a declared rule becomes a control, a mandatory field or an explicit refusal. The software model is worth neither filed articles nor authorisation to carry on a regulated activity.

19.2 Thirty-six domains, one hundred and forty-four invariants

The ARP reference covers 36 domains: identity, federations, memberships, families, governance, meetings, discipline, documents, rules engine, contributions, payments, cash boxes, tontines, allocations, auctions, cycles, savings, loans, schedules, repayments, guarantees, solidarity, aid, reserves, investments, projects, purchases, assets, events, community, reports, security, architecture, mobile, operations and internationalisation.

36	144	23 328	6
design domains	management invariants	bounded combinations	compatibility conditions

Each domain contains its options, invariants, form, workflow, exceptions, test recipes and indicators. The administration console lets you browse this reference without leaving the application.

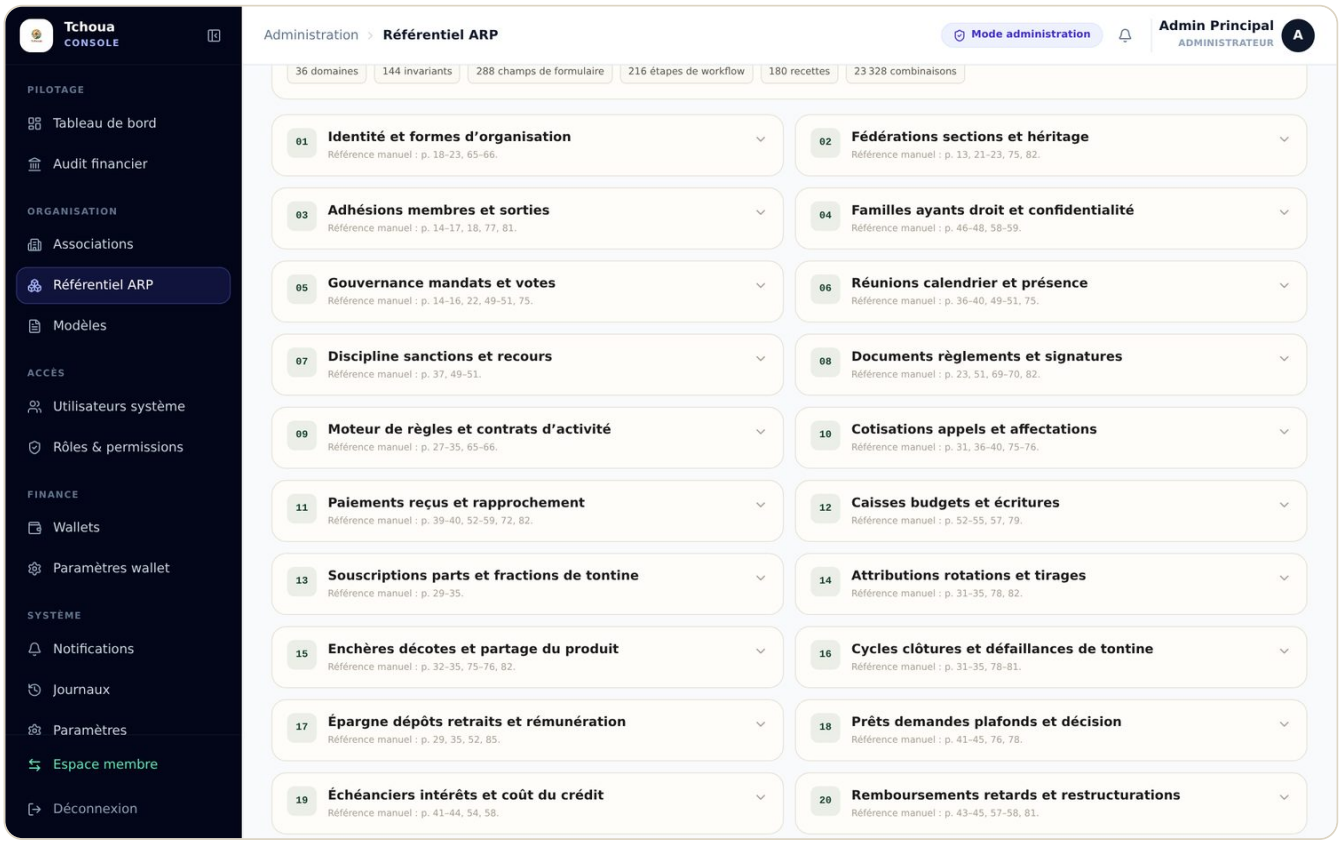


Figure 19.1 — ARP reference: 36 domains, 144 invariants, 288 form fields, 216 workflow steps and 180 test recipes available for consultation.

19.3 The ten dimensions of an association

An ARP configuration is described by ten dimensions. They determine the modules available and the conditions to be met before publication.

Dimension	Values	Main consequence
Organisational form	Autonomous · Branch · Federation	Scope of decisions, ownership of funds, inheritance
Membership mode	Individual · Family · Institutional	Unit of rights, caps and representation
Contribution basis	Fixed · Shares · In-kind	Call computation, valuation, liquidity
Tontine mode	None · Rotation · Draw · Auction	Allocation, proof of draw, auction
Credit policy	None · Zero rate · Amortising	Schedule, interest, caps
Social aid	None · Flat rate · Daily allowance	Caps, events, supporting documents
Payment channel	Cash · Bank · Mobile Money	Proof, fees, reconciliation, confirmation
Currencies	Single · Conversion	Rate, margin, presentation currency
Connection mode	Online · Offline tolerated	Draft, synchronisation, server validation
Validation level	Single · Double	Separation of duties, second approver

19.4 The six compatibility conditions

Not all combinations are equally simple. Six conditions flag the configurations that require additional parameters. They do not block creation, but they prevent publication until the required fields are filled in.

Code	Trigger	Required fields
C01	In-kind contribution + open tontine	Valuation method, liquidity rule (sale or funding)
C02	Auction + offline connection	Discount base, closing rule, offline bidding policy
C03	Mobile Money + offline connection	Provider, payment policy, mandatory external confirmation
C04	Currency with conversion	Rate source, date, margin, presentation currency
C05	Family membership + social aid	Cap unit (member, household, dependant), household definition
C06	Credit + single validation	Decision threshold, recusal rule for interested persons

> CONCRETE EXAMPLE
>
> An association that enables loans without defining a threshold or a recusal rule displays **C06_THRESHOLD** and **C06_CONFLICT**. The framework remains active for existing operations, but publication of a new version is refused until the board decides.

19.5 Twenty-three thousand combinations, a single logic

The ten dimensions produce 23 328 combinations. The engine does not store these rows: it computes the conditions from the values and checks each configuration with a test recipe. This approach avoids frozen catalogues and guarantees that a new value remains compatible with the existing controls.

```
conditions = C01(dimension nature + tontine) ∪ C02(auction + offline) ∪ C03(mobile money +
offline) ∪ C04(conversion) ∪ C05(family + aid) ∪ C06(credit + single validation)
qualification = "conditional" if conditions ≠ ∅ "no additional condition" otherwise
```

19.6 Profile, draft, publication and versions

Each association has a single ARP profile but several versions. The active profile carries the configuration applied to new operations; the draft allows a change to be prepared without interrupting the existing one.

State	Meaning	Effect
DRAFT	Unpublished profile	New financial subscriptions are blocked until publication
ACTIVE	Published version	Modules and conditions are applied
BACKFILL	Profile imported for an existing association	Continues to operate; to be reviewed and published by the board
SUSPENDED / ARCHIVED	Framework suspended or archived	Viewing and liquidation possible, new subscriptions closed

A publication creates an immutable version with a SHA-256 fingerprint. Old operations remain attached to their rule version; a retroactive correction never rewrites the history, it creates a linked operation in an open period.

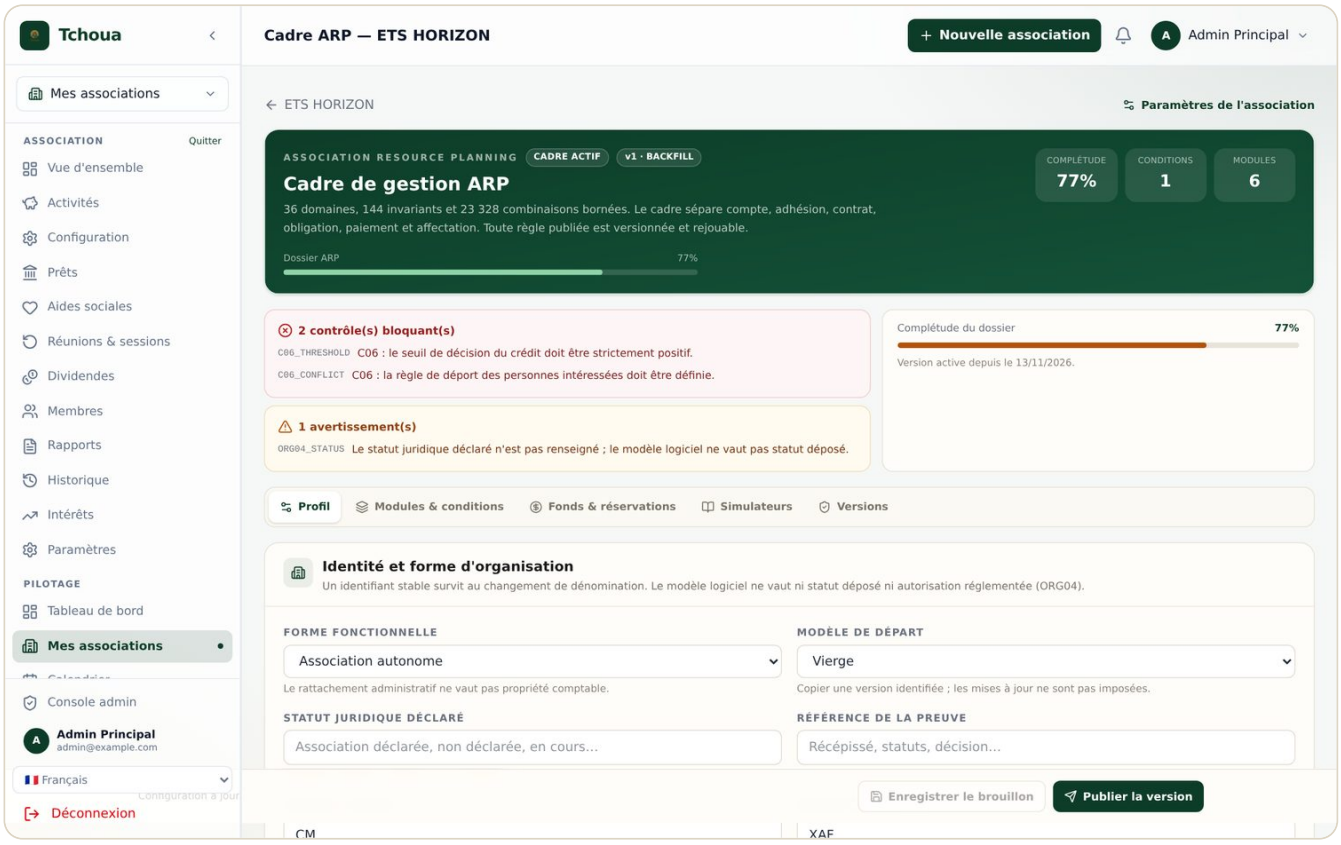


Figure 19.2 — ARP framework of an association: status, version, completeness, active conditions and blocking controls.

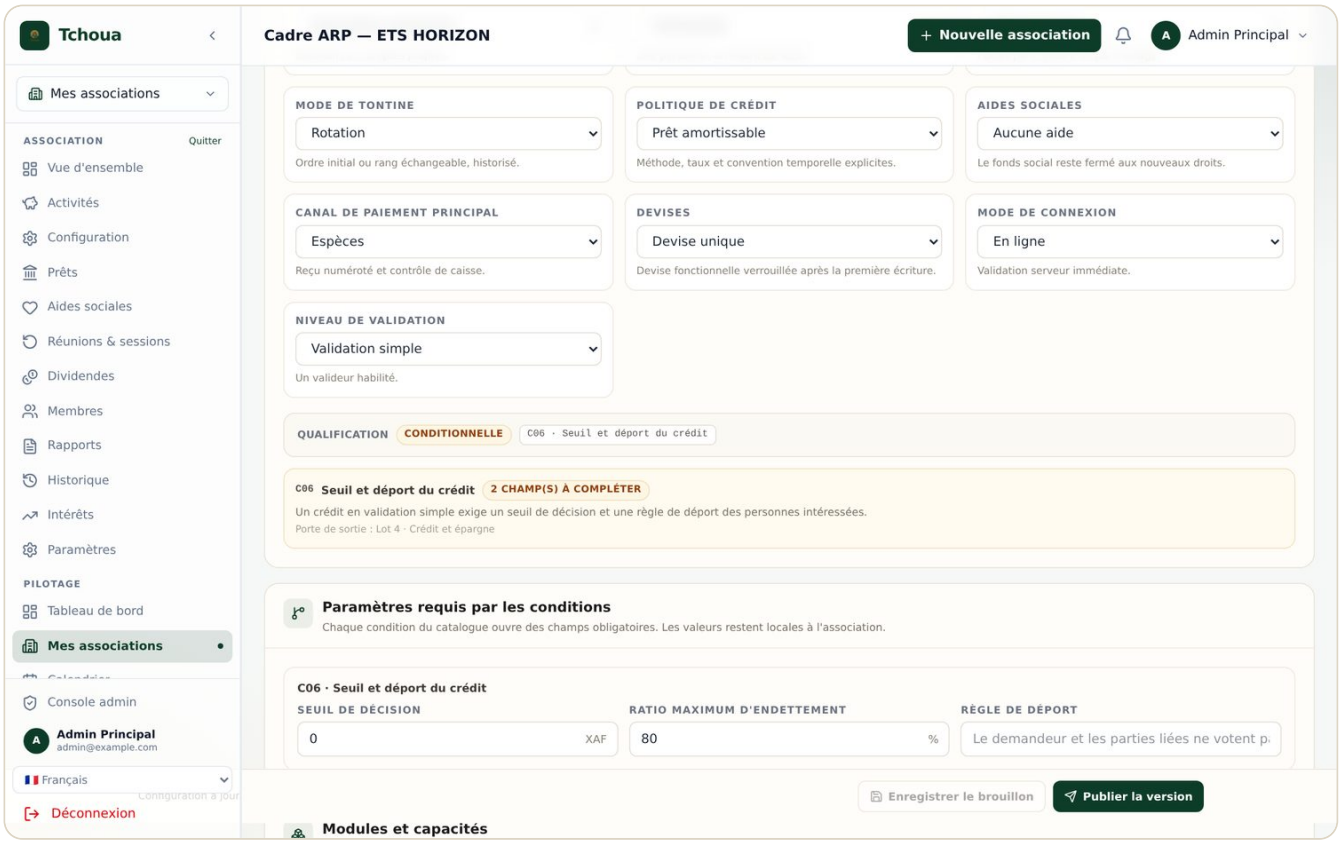


Figure 19.3 — Modules & conditions tab: ten dimensions, computed qualification, required fields and guardrails that cannot be disabled.

19.7 Funds, movements and reservations

The financial core distinguishes the **fund** (economic allocation), the **movement** (confirmed inflow or outflow) and the **reservation** (a commitment that protects the available balance without changing the balance). The available balance is computed as follows:

```
available = balance (confirmed inflows − confirmed outflows) − active reservations
```

- An **outflow** can never exceed the available balance; the check is performed under a concurrent lock.
- A **reservation** can be released (returned to the available balance) or consumed (creating a linked disbursement).
- Amounts are kept in **whole minor units**: XAF without decimals, EUR in cents. No financial calculation uses binary floating point.
- **Disabling a module** hides new subscriptions but preserves the viewing, corrections and liquidations required by existing obligations.

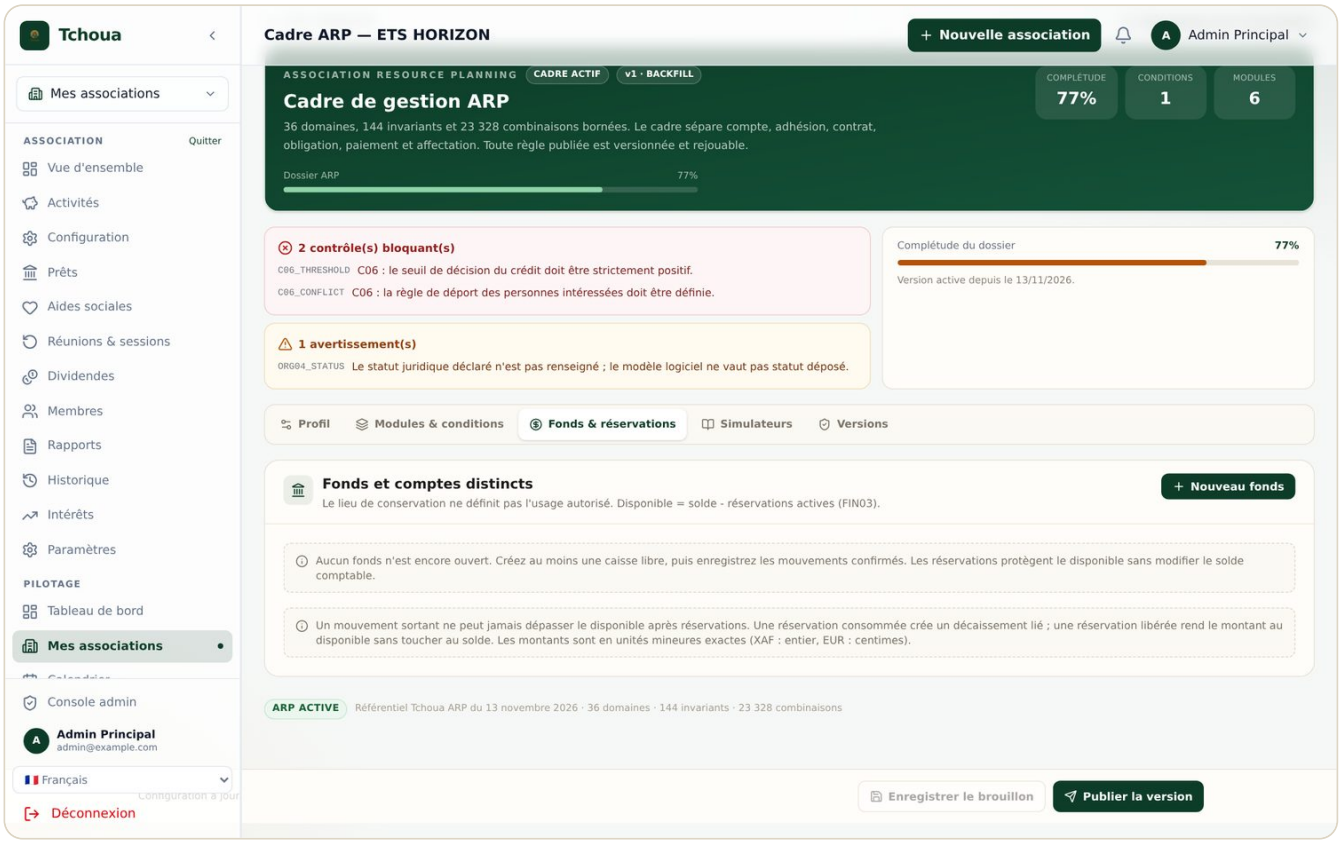


Figure 19.4 — Funds and reservations: balance, reserved amount and available balance kept distinct, with creation of movements and reservations.

19.8 Simulators and proof of calculations

Three simulators make the rules verifiable before any entry is posted: loan schedule (constant principal, constant annuity, bullet), tontine allocation (pot, session cap, remaining entitlements, minimum fraction) and social fund projection (headcount, contribution, collection rate).

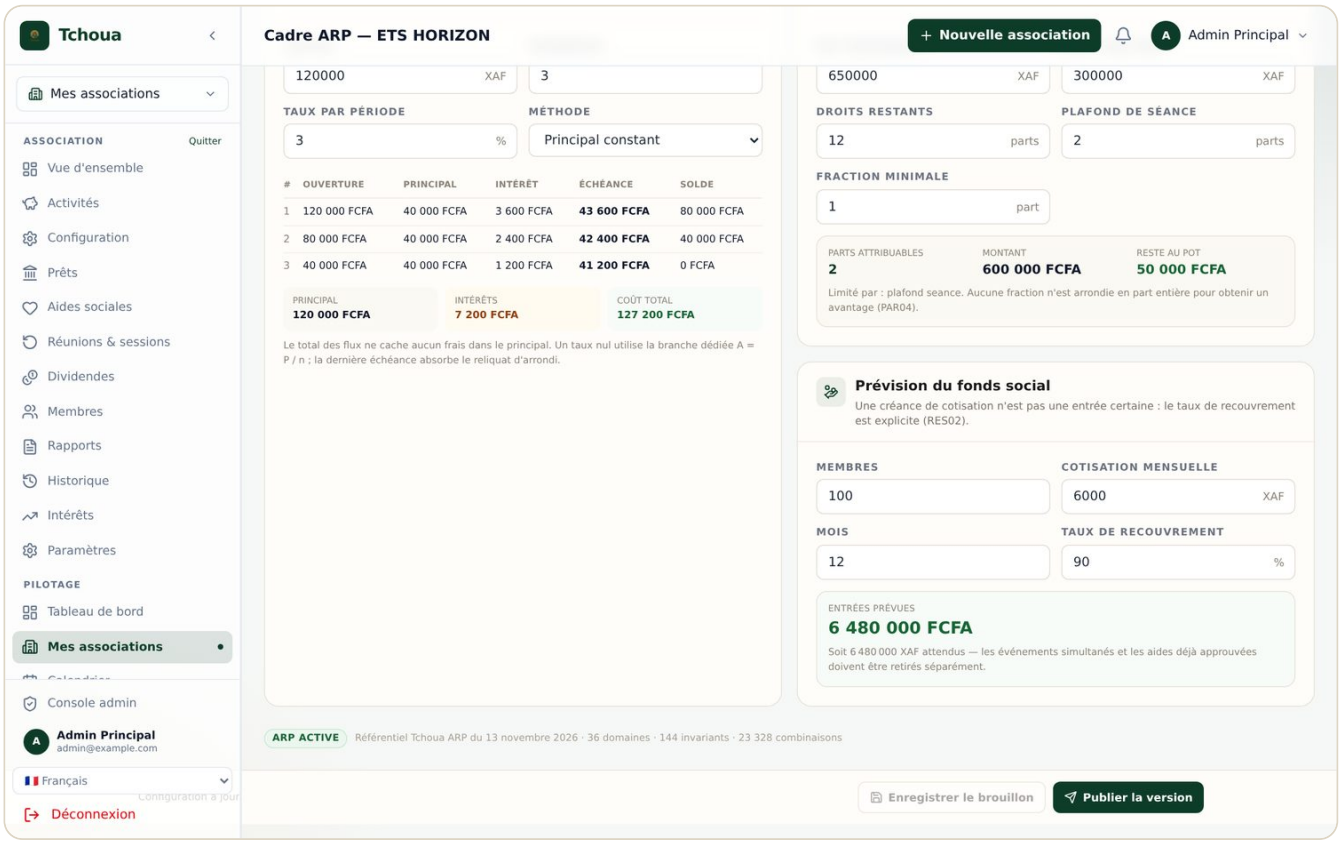


Figure 19.5 — Schedule simulator: 120 000 XAF over three months at 3 % per month, constant principal — 43 600, 42 400 and 41 200 XAF, interest 7 200 XAF.

19.9 Administration and coverage

The administration console shows the ARP coverage of all associations: status, origin, version, completeness, errors, alerts, active modules and number of funds. It can initialise missing profiles from existing activities, without interrupting production.

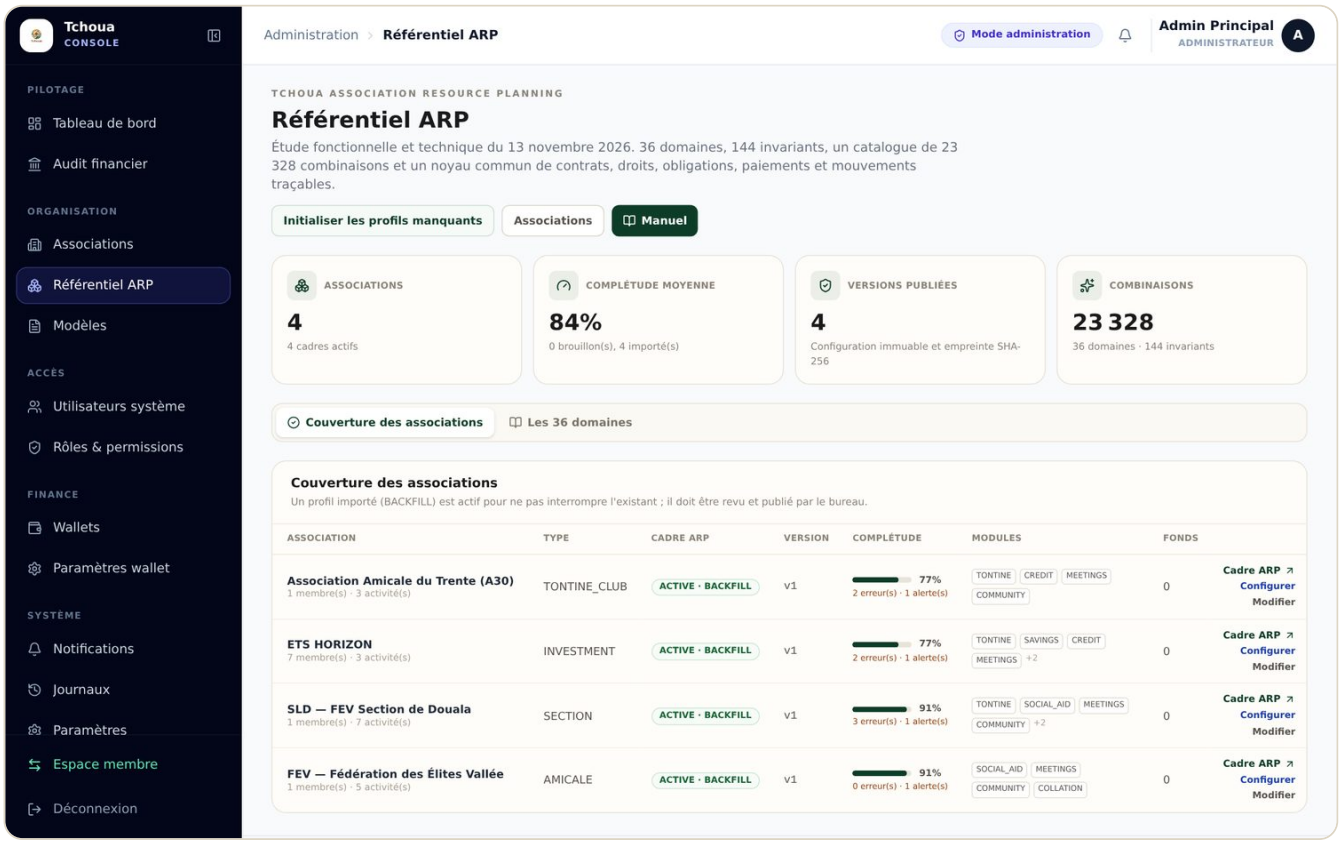


Figure 19.6 — ARP coverage: each association, its framework, its completeness and its modules; imported profiles remain flagged as needing validation.

> **ACCEPTED LIMITATIONS**
>
> The ARP framework is the anchor point for future developments. The modules not yet built — double-entry accounting journal, guarantees register, payment connectors, projects and inventory, real-time auctions, federation consolidation — remain identified workstreams and are not presented as finished.

CHAPTER 20

Creating or editing an association with tabs

The former seven-step wizard has been replaced by a single form organised in tabs. You can go back, correct a field, consult the ARP framework or save at any time: no linear progression is imposed.

20.1 A single form, two modes

The same component is used for **creation** and for **editing**. In creation it starts from default values and offers the modules to activate. In editing it reloads the association, its ARP framework, its meetings, its caps, its internal rules and its existing activities.

Mode	Address	Saving
Creation	`/dashboard/associations/create`	Creates the association, its initial activities, its internal rules and its ARP profile
Editing	`/associations/[id]/modifier`	Updates the record; the ARP framework moves to draft if it has changed

20.2 The eight tabs

Tab	Main fields	Checks
Identity	Name, type, description, parent, legal status, country, currency, time zone, purposes, template	Name ≥ 3 characters; ISO currency code; ISO country; time zone required
Contacts	Region, email, phone, website, logo, colour, public visibility	Valid email; public page with no financial data
ARP framework	Ten dimensions, C01–C06 conditions, modules, required parameters	Conditions computed; mandatory fields; invariants that cannot be disabled
Activities	Modules to activate, initial tontine parameters, session frequency	Module / ARP capacity consistency; positive amounts; valid cycle
Meetings & catering	Frequency, day, time, quorum, contribution, host flat rate	Quorum 0–100; positive amounts
Membership & finances	Entry fee, age, sponsors, aid caps, penalties, bank account	Positive amounts; consistent scales
Internal rules	Titled articles and content	Atomic update; approvals recreated
Summary	Overview, completeness, conditions, errors and warnings	Final check before creation or saving

20.3 Navigating and saving at any time

The tab bar stays visible at the top of the form. The save button stays visible at the bottom, whichever tab is open. Switching tabs loses no input: the state is kept in the component until saving or cancelling.

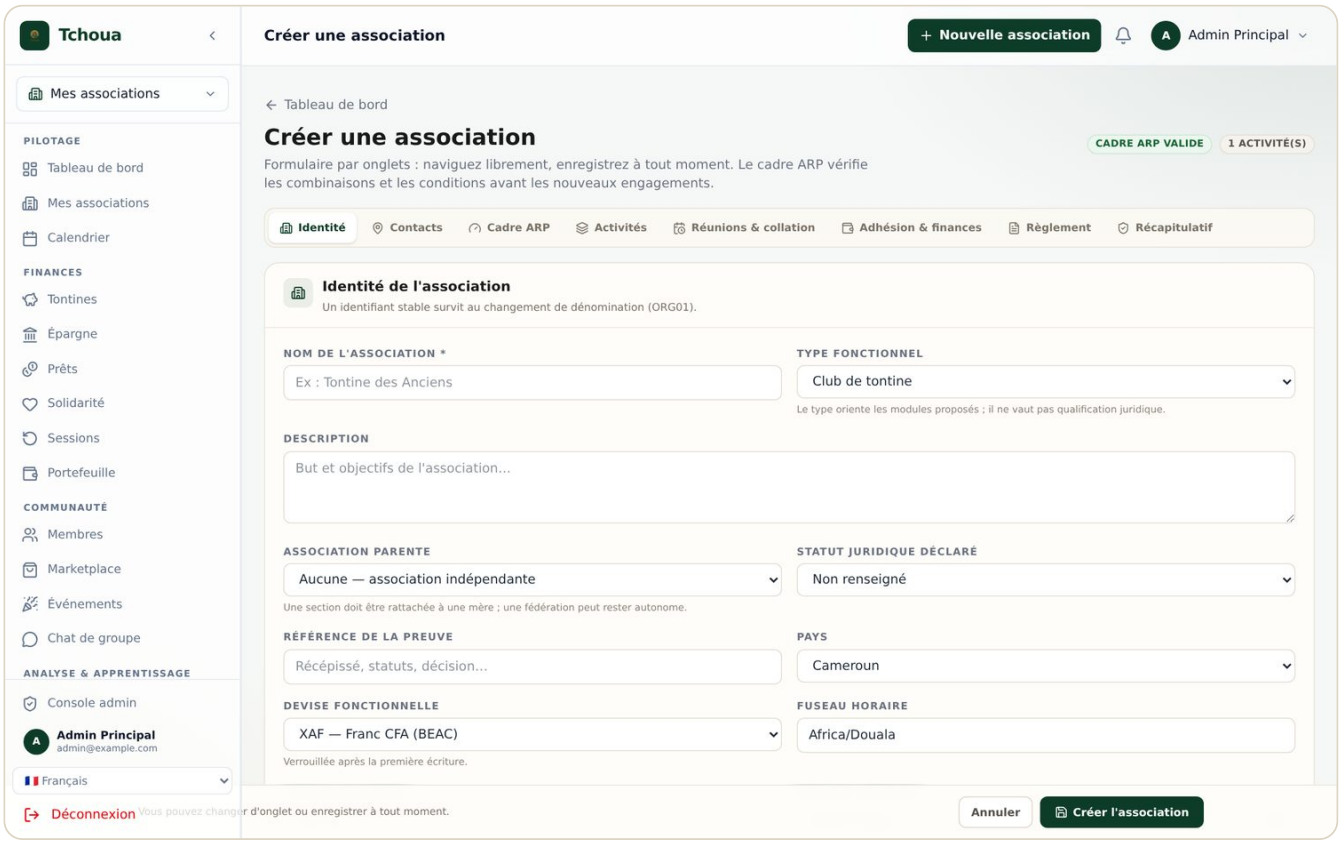


Figure 20.1 — Creation by tabs: all eight entries are directly accessible, with no "Next" button. The "Create the association" button remains available.

> **TIP**
>
> Fill in the name and type first, then open the **ARP framework** tab: the "Apply the standard profile" button suggests a consistent combination (neighbourhood tontine, friendly society, federation, branch, cooperative…). You can then adjust each dimension.

20.4 ARP framework and module consistency

The **Activities** tab does not allow a module that the ARP framework does not authorise to be activated. If you select "Loans" without a credit policy, the form sends you back to the ARP framework tab to choose the credit mode and fill in the conditions. This rule avoids creating an activity that the controls would block on the very first request.

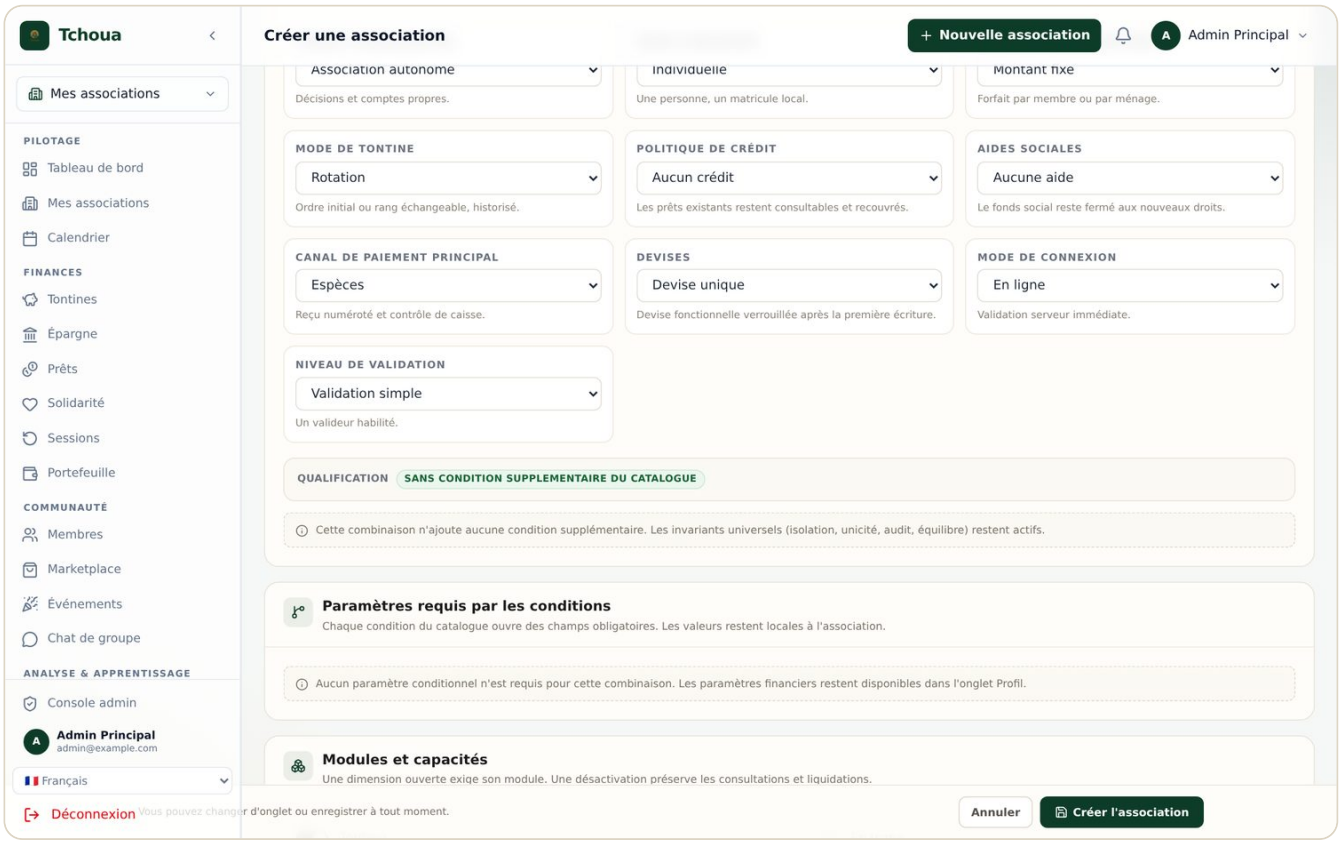


Figure 20.2 — ARP framework tab: ten dimensions, qualification, conditions, modules and required parameters. The standard profile can be applied and then customised.

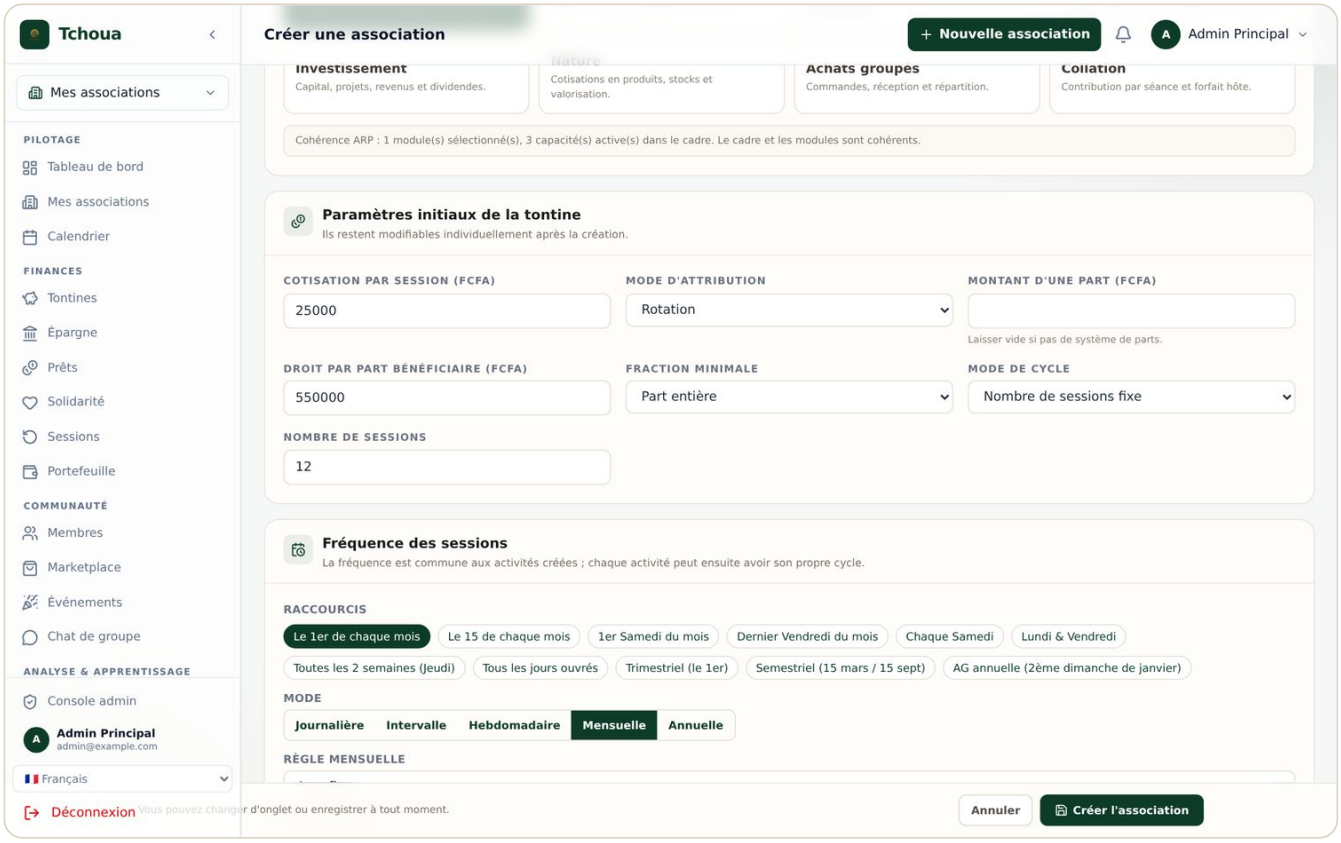


Figure 20.3 — Activities tab: modules, ARP consistency, initial tontine parameters and session frequency.

20.5 Editing an existing association

In editing mode, the data is reloaded from the database. The **Edit** button is visible on the association record for authorised board members. The principal administrator can also open the direct address.

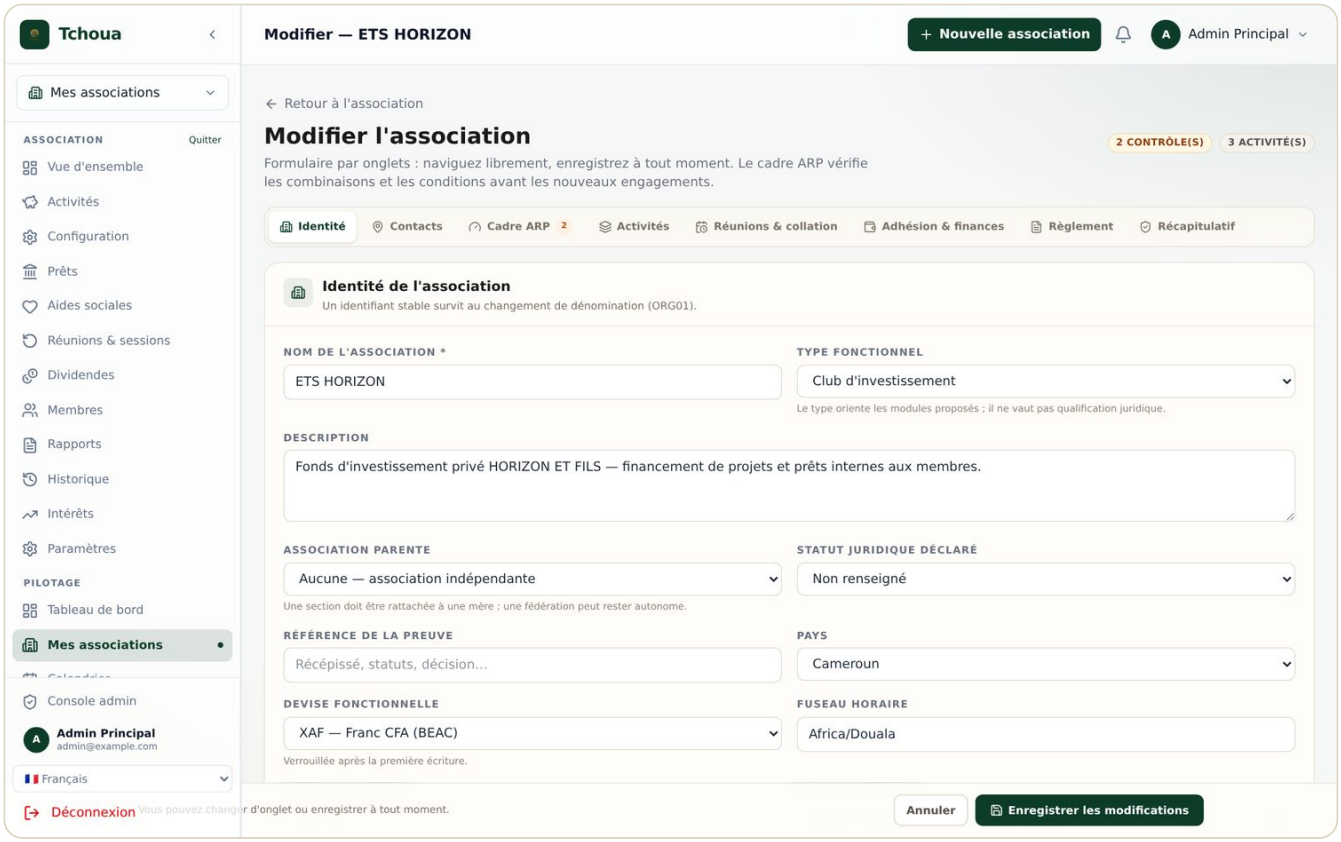


Figure 20.4 — Editing: the same tabs, pre-filled with the existing identity, ARP framework, meetings, caps and internal rules.

Three behaviours are worth knowing:

- the record fields (name, contacts, meetings, membership, internal rules) are applied immediately;
- if the ARP framework has changed, it is saved as a **draft** and the ARP page opens so that it can be published;
- if only the rest of the record has changed, no ARP draft is created.

> **WARNING**
>
> Editing the internal rules deletes the previous approvals and recreates the articles in a single transaction. Members must approve the new version; the history of the previous approvals is not kept as proof of the new version.

20.6 Who can create, edit and configure

Action	Authorised roles	Permission
Create an association	Any authenticated user	Platform limits
Edit the association record	Founder, president, vice-president, principal administrator	`EDIT_SETTINGS`
Configure activities	Founder, president, vice-president, secretary, treasurer, principal administrator	`MANAGE_ACTIVITIES`
Publish the ARP framework	Founder, president, principal administrator	Publication decision
View the ARP framework	All members of the association	Read only

> **SEPARATION OF DUTIES**
>
> A member can view the framework and the activities but cannot modify them or publish a version. Creating a financial activity is also controlled by the corresponding ARP capacity.

CHAPTER 21

Configuring each activity and managing funds

Once the association is created, each activity is configured separately. The **Configuration** page groups tontines, loans, savings, contributions, solidarity, in-kind activities, investments, purchases and catering, with a form suited to each family.

21.1 Accessing the configuration

Two paths lead to the same page:

- from the association record, the **Configure** button;
- from the side menu, section **Association → Configuration**, once the association is selected.

Access is restricted to authorised board members and to the principal administrator. Other members can view the activities but cannot change their parameters.

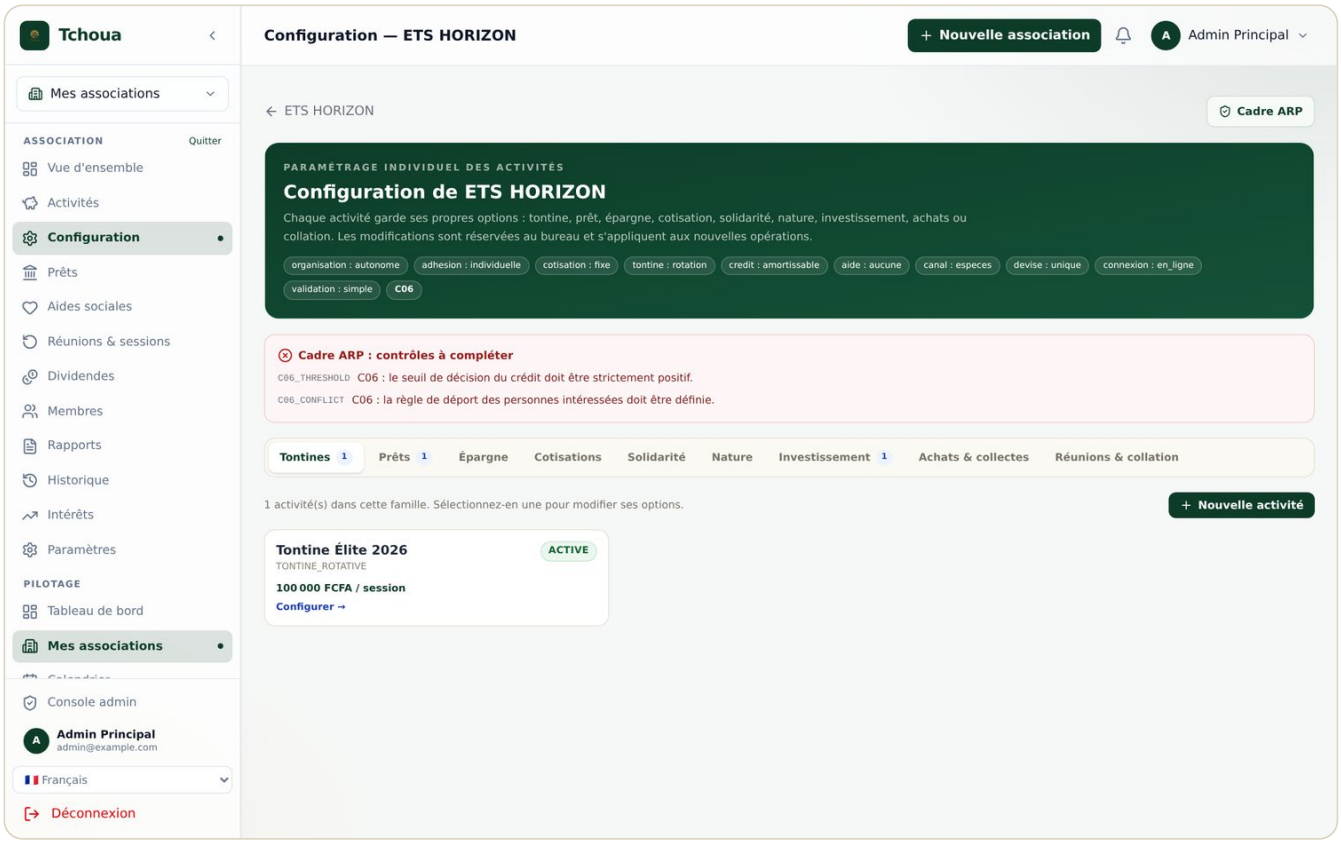


Figure 21.1 — Association configuration: ARP dimensions, active conditions, activity families and creation button.

21.2 The activity families

Family	Activities concerned	Specific parameters
Tontines	Rotating tontine, ASCA, auctions	Contribution, shares, beneficiary entitlement, fractions, cycle, reserve price
Loans	Loan fund	Short/long rate, cap, fund ratio, validation, active loans
Savings	Savings / investment	Contribution, frequency, payment method
Contributions	Cross-cutting rules	Basis, entry fee, admission, family cap, in-kind
Solidarity	Solidarity fund	Contribution, caps, events, validation
In-kind	Contribution in goods	Unit, valuation, liquidity, catalogue
Investment	Collective investment	Capital, income, dividends
Purchases & collections	Group purchases, one-off collection	Quantities, costs, allocation, collection amount
Meetings & catering	Cross-cutting parameters	Frequency, quorum, contribution, host flat rate

21.3 The activity form

The form adapts to the selected type. It keeps an identity, a status (active, paused, closed), a participation mode (mandatory, optional, conditional), a contribution unit and a frequency. The following blocks appear only when relevant:

- **Tontine**: amount per share, beneficiary entitlement, minimum fraction, caps, cycle mode, auctions;
- **Loan**: month 1–3 rate, rate beyond, cap per member, fund ratio, validation, number of active loans;
- **Collection**: collection amount;
- **All**: late penalties, payment methods, preview of the next twelve sessions.

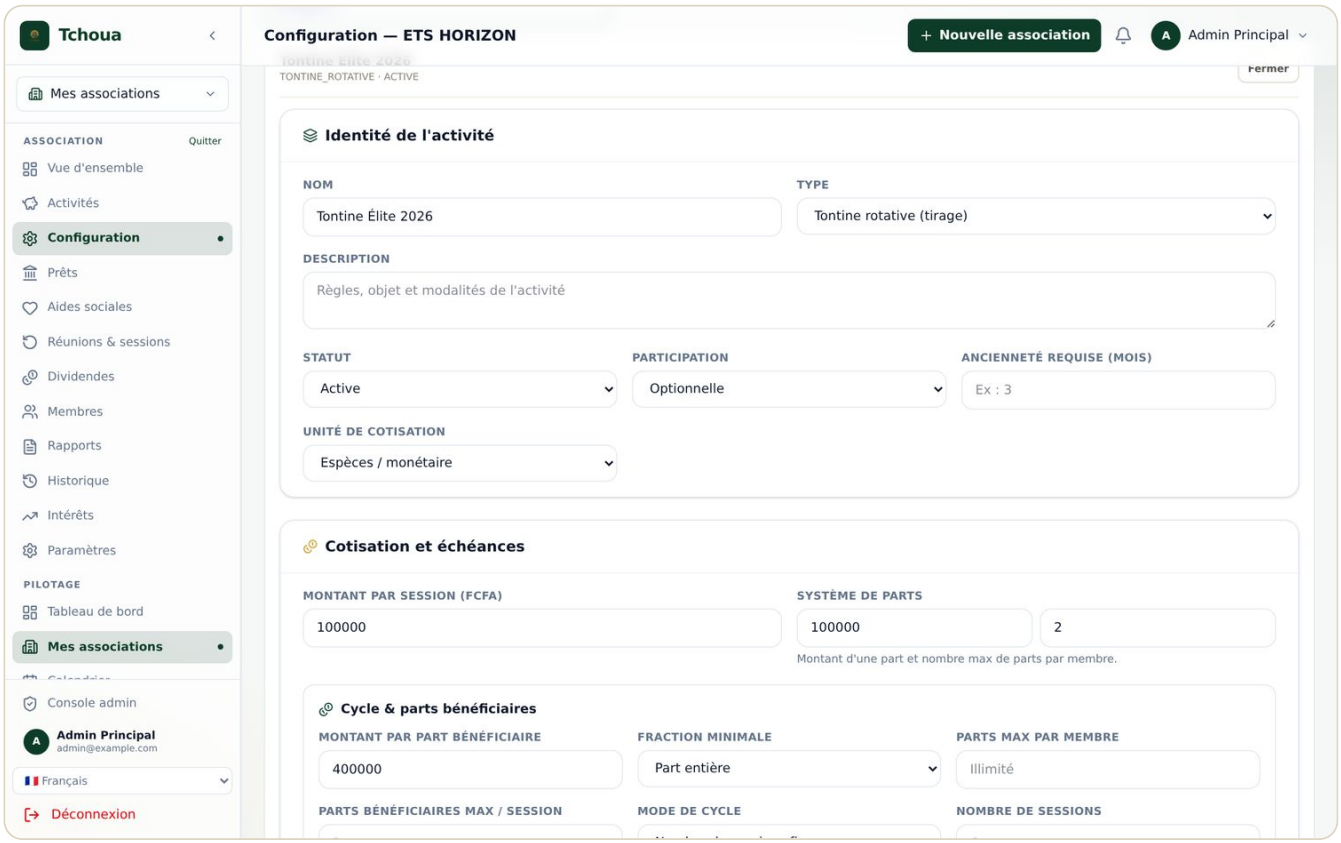


Figure 21.2 — Activity form: identity, status, participation, contribution, cycle, beneficiary shares and frequency.

> **TIP**
>
> A change of frequency does not rewrite past sessions: the new due dates apply to upcoming sessions. For an activity already under way, check the cycle and the remaining entitlements before changing a contribution.

21.4 Contribution and membership rules

The **Contributions** tab does not create an activity: it sets the cross-cutting parameters that feed the calls. It holds the basis (fixed, shares, in-kind), the entry fee, the minimum age, the admission mode, the sponsors, the waiting period, the family cap unit and the household definition.

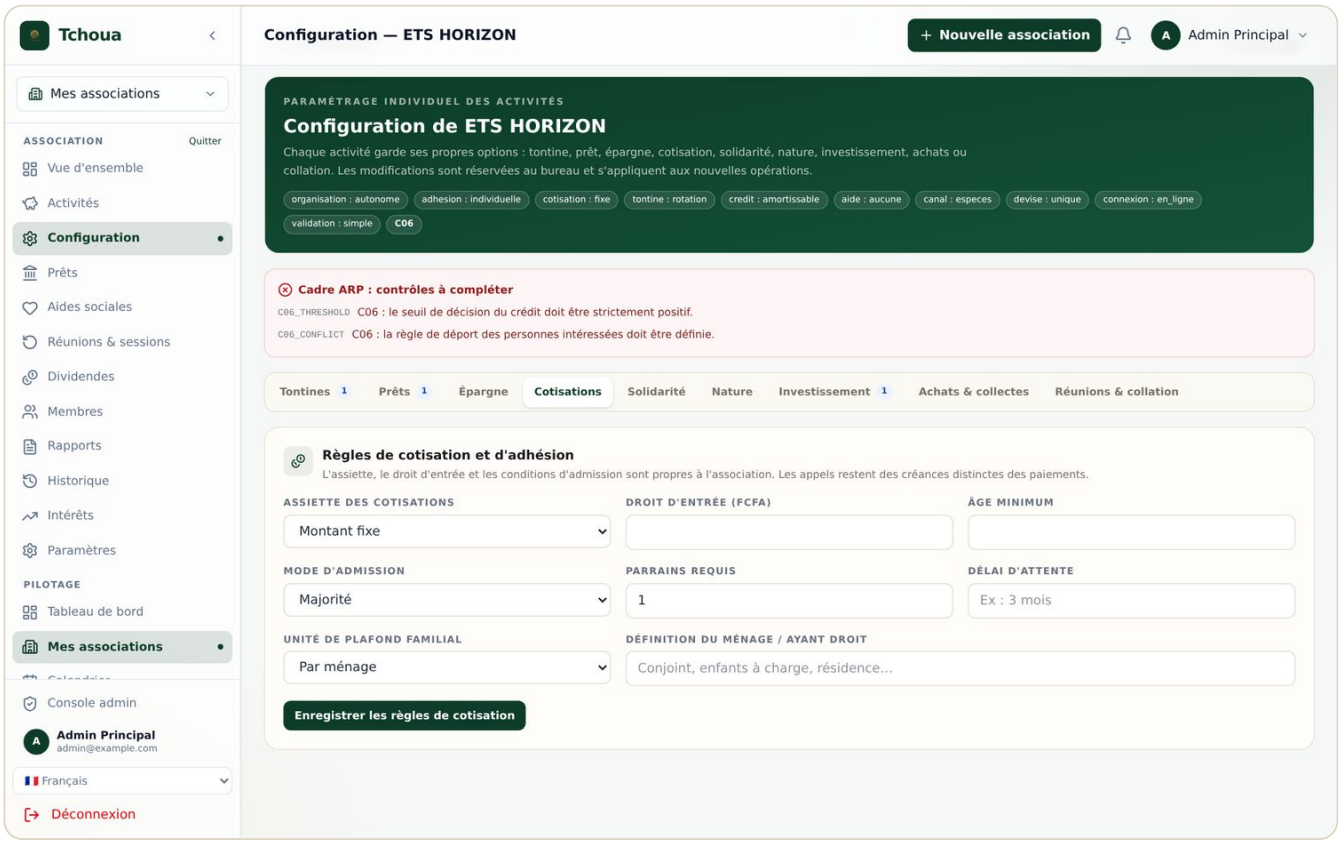


Figure 21.3 — Contribution and membership rules: basis, entry fee, admission, family cap and in-kind valuation.

When the basis is "In-kind", the C01 fields appear: valuation method and liquidity rule. An in-kind contribution does not automatically become cash; it must be sold or funded before distribution.

21.5 Meetings and catering

The **Meetings & catering** tab sets the frequency, day, time, quorum, contribution per member and the flat rate paid back to the host. These parameters are distinct from financial sessions: a meeting can cover several activities, and an activity can run without a physical meeting.

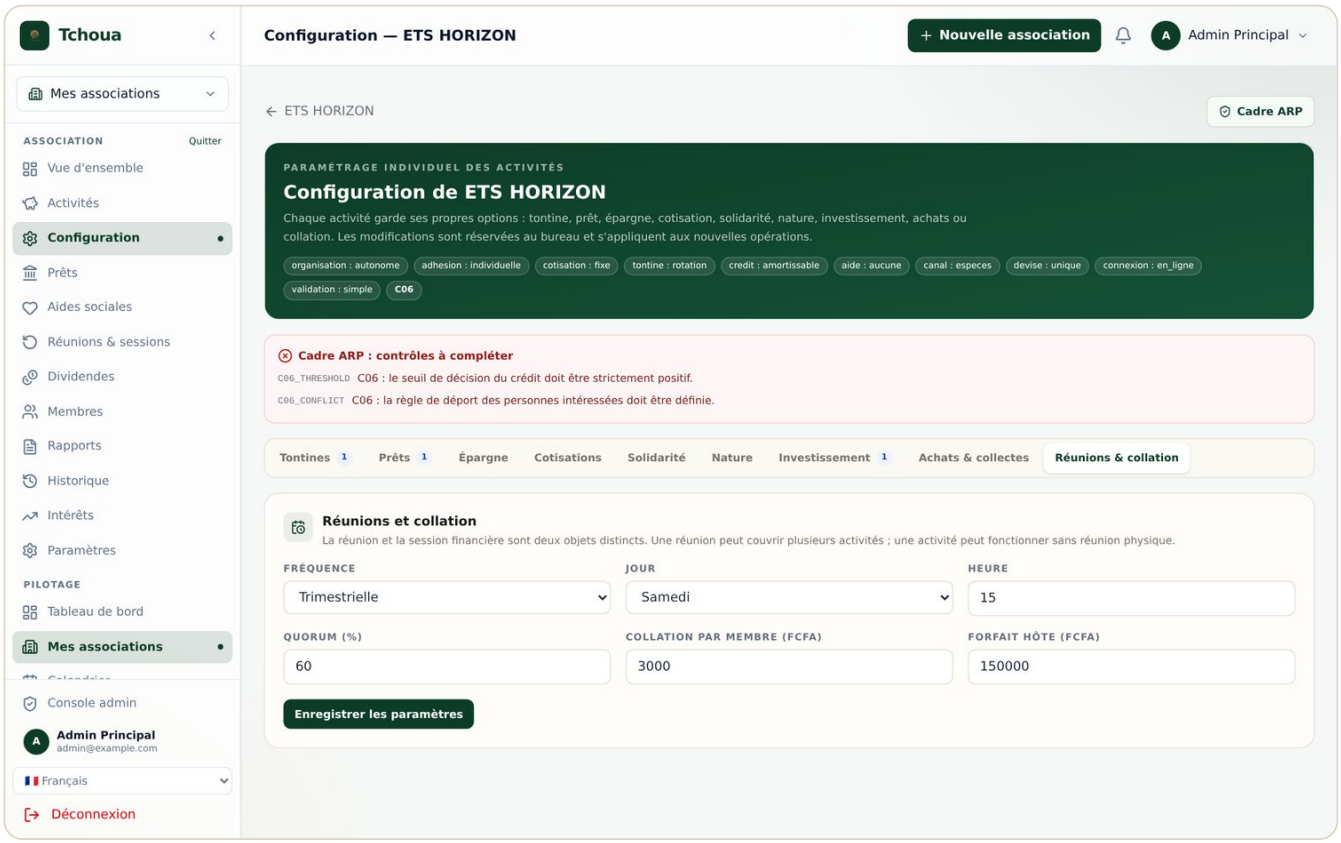


Figure 21.4 — Meetings and catering: frequency, quorum, contribution and host flat rate, with no confusion with activity contributions.

21.6 Funds and reservations

The ARP page contains the **Funds & reservations** tab. It allows several funds to be created (unrestricted, restricted, reserve, refundable deposit), inflows and outflows to be recorded, and an amount to be reserved for an aid, an allocation or a guarantee.

Concept	Definition	Impact on the available balance
Balance	Confirmed inflows minus confirmed outflows	Accounting measure
Active reservation	Commitment on an amount, with purpose and expiry	Reduces the available balance without changing the balance
Released reservation	Commitment cancelled	Returns the amount to the available balance
Consumed reservation	Commitment converted into a disbursement	Creates a linked outflow; the available balance stays consistent

> **RULE**
>
> An outflow can never exceed the available balance after reservations. The check is performed under a concurrent lock: two treasurers cannot consume the same amount twice.

21.7 Versions, audit and control

Each save of the ARP framework creates a draft. Publication turns that draft into an active, immutable version, with a SHA-256 fingerprint. Versions can be viewed in the **Versions** tab; old operations keep the rule that applied on their date.

- Activity changes are logged with the author and the date.
- Financial parameters are never corrected by deletion: a linked operation keeps the trace.
- Disabling a module closes new subscriptions but lets repayments and liquidations continue.
- The board can return to an earlier version by publishing a new version, never by overwriting the history.

21.8 Configuration checklist

- ✓ The name, type and country are filled in.
- ✓ The functional currency is correct and the time zone matches the head office.
- ✓ The ARP dimensions match the activities actually open.
- ✓ Conditions C01–C06 are complete or explicitly accepted as still to be completed.
- ✓ Each activity has a consistent status, participation mode and frequency.
- ✓ Caps, rates and penalties have been reviewed by the treasurer.
- ✓ The internal rules are up to date and members can approve them.
- ✓ Funds and reservations are created before the first disbursements.

ANNEXES

Glossary, API and checklist

The references used day to day: vocabulary, endpoints, configuration checklist and index of figures.

A. Glossary

Term	Definition
Activity	Collective contract managed by the association: tontine, savings, loan, solidarity, catering.
ARP	Association Resource Planning: versioned framework describing an association's dimensions, rules, modules and conditions.
ARP framework	Configuration profile of an association, with draft, published version and compatibility checks.
Condition C01–C06	Compatibility requirement that blocks publication until the required parameters are filled in.
Membership	Link between a user and an association, with role, status and join date.
Beneficiary	Member designated to receive one or more shares during a session.
Board	Executive body of the association: president, secretary, treasurer, etc.
Cash box	Financial envelope allocated to an activity or a fund.
Cycle	Complete period of an activity, from opening to closing.
Dividend	Share of the profits or interest distributed among the members.
Fraction	Fraction of a share: 1, 1/2 or 1/4.
Fund	Envelope allocated to a use (unrestricted, restricted, reserve, refundable deposit); distinct from the account that holds it.
Membership	Record of a user's belonging to an association.
Share	Unit of participation in a tontine.
Pot	Amount available for a session after contributions and carry-overs.
Quorum	Minimum number of people present in order to deliberate validly.
Reservation	Commitment on an amount that reduces a fund's available balance without changing its balance.
Session	Dated meeting of an activity.
Subscription	A member's commitment to a number of shares.
Solidarity	Non-repayable funds and aid intended for members.
ASCA tontine	Internal savings and credit tontine.
Rotating tontine	Tontine in which each member receives the pot in turn.
Template	Reusable association model with activities and internal rules.
Minor unit	Smallest whole monetary unit (XAF, euro cent) used for all financial calculations.
Wallet	Internal platform wallet for financial flows.

B. Main endpoints

Method	Endpoint	Purpose
GET	/api/associations	List of the user's associations
POST	/api/associations	Create an association
GET	/api/associations/{id}	Details, members, activities, documents
PATCH	/api/associations/{id}	Update the settings
GET	/api/associations/{id}/history	Paginated history, filters, groupings
GET	/api/associations/{id}/arp	ARP profile, draft, versions, checks
PUT	/api/associations/{id}/arp	Save the draft or publish a version
GET	/api/associations/{id}/arp/funds	Funds, balances, reservations and movements
POST	/api/associations/{id}/arp/funds/{fundId}/movements	Fund inflow or outflow
POST	/api/associations/{id}/arp/funds/{fundId}/reservations	Reserve, release or consume an amount
GET	/api/admin/arp	ARP coverage of all associations
POST	/api/admin/arp/backfill	Initialise missing profiles
GET	/api/arp/referential	36 domains, 144 invariants, conditions
GET	/api/associations/{id}/interests	Interest by member, repayment, period
GET	/api/associations/{id}/activities/{actId}/cycle	Tontine cycle summary
POST	/api/associations/{id}/activities	Create an activity
PATCH	/api/associations/{id}/activities/{actId}	Update an activity
GET	/api/associations/{id}/documents	Association documents
POST	/api/associations/{id}/documents	Upload a document
GET	/api/admin/settings	Platform settings (admin)
GET	/api/admin/templates	Association templates (admin)
GET	/api/me/aggregate	Personal multi-association summary

Table B.1 — Main endpoints. All of them check the session and the membership of the association.

C. Configuration checklist

Step	To check	Done
Identity	Name, type, region, logo, colour, description	☐
Attachment	Parent association, level, invitation code	☐
Board	President, secretary, treasurer, auditors	☐
Membership	Conditions, sponsorship, fee, contribution	☐
Meetings	Day, time, venue, quorum, notice	☐
Aid	Categories, caps, documents, deadlines	☐
Penalties	Scale, grounds, appeals	☐
Activities	Type, share, frequency, cycle, beneficiaries	☐
Loans	Rate, term, cap, guarantees, approval	☐
Cash boxes	Bank accounts, cash boxes, responsible persons	☐
Documents	Articles, internal rules, minutes, reports	☐
Tests	Membership, contribution, loan, aid, report	☐

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E. References

- FEV articles revised 2018; steering committee internal rules 2013; compilation of texts 2023.
- SLD internal rules 2024.
- ETS HORIZON data (import workbook and internal rules).
- Functional specifications of the repository: tontine, association, wallet, savings, marketplace, vote.